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**FINAL GEOLOGICAL REPORT ON
GENERAL EXPLORATION FOR COAL IN
MARWATOLA SECTOR-I & II BLOCK
SOHAGPUR COALFIELD
DISTRICT - SHAHDOL (MADHYA PRADESH)**

**VOLUME I
TEXT
MAY, 2023**



**EXPLORATION DEPARTMENT
REGIONAL INSTITUTE - V**

CMPDI

**(A Subsidiary of Coal India Ltd.)
BILASPUR, CHHATTISGARH**

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LIST OF ABBREVIATION

AFT	- Ash Fusion Temperature Range
Ash%, A%	- Ash percent
BH No.	- Borehole Number
Bh. Intersec.	- Borehole Intersection
B	- Bottom
Cal	- Calculated
CIMFR	- Central Institute of Mining and Fuel Research
CD	- Closing Depth
CVAR	- Gross Calorific Value, as received basis
CVDF	- DMMF Gross CV or Unit Gross CV
Det.	- Determined
Dist	- District
DMMF	- Dry Mineral Matter Free
F/Faulted, FF	- Floor Faulted
F	- Faulted
FC%	- Fixed Carbon percent
FRL / RRL	- Floor / Roof Reduced Level
Gross CV	- Gross Calorific Value
GSI	- Geological Survey of India
H%	- Hydrogen percent
HGI	- Hard Grove Grindability Index
HQ	- Head Quarters
Kcal/kg	- Kilocalories per kilogram
Km	- Kilometre
m	- Metre
M%	- Moisture percent
MCM	- Million Cubic Metres
Mt/mt	- Million tonnes
SECL	- South Eastern Coalfields Limited
ND	- Not Drilled
N Dev/NDV	- Not Developed

LIST OF ABBREVIATION

M	- Middle
MTY	- Million Tonnes per Year
NA	- Not Available
N%	- Nitrogen percent
O%	- Oxygen percent
OB	- Overburden
°C	- Degree Celsius
OC	- Opencast
P/Faulted, PF	- Parting Faulted
P ₆₀	- Proximate Analysis on 60% RH and 40°C
R/Faulted, R/F	- Roof Faulted
RF	- Representative Fraction
RH	- Relative Humidity
RL	- Reduced Level
S%	- Sulphur percent
S/WM	- Soil / Weathered mantle
Sl.No.	- Serial Number
sq.km.	- Square kilometre
T	- Top
TD	- Total Depth
UA	- Ultimate Analysis
UG	- Underground
UHV	- Useful Heat Value
UG	- Ungraded
UVM%	- Unit Volatile Matter percent
VM%	- Volatile Matter percent
WM	- Weathered Mantle
L	- Local
L1, L2, L3, L4	- Local1, Local2, Local3, Local4.

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SALIENT FEATURES

1. LOCATION OF THE BLOCK

- A. Located in the west-central part of the Sohagpur coalfield
- B. Latitude - 23° 17' 44.991" N – 23° 20' 38.858" N
- C. Longitude - 81° 12' 24.273" E – 81° 14' 51.807" E
- D. The block is covered in Survey of India old Toposheet No. 64 E/3 & E/7 (RF1:50,000)
- E. The bounding co-ordinates of the cardinal points on the block boundary are given in Table SF.1 A
- F. The block is located in 12 kms North-West of Shahdol town Shahdol district of M.P State.

Table–SF.1 A: Bounding Co-ordinates of the Cardinal Points of the Block Boundary of Marwatola Sector I&II Block, Sohagpur Coalfield in (Coal grid and UTM)

CARDINAL POINTS	UTM COORDINATE SYSTEM (WGS 1984 UTM ZONE 44N)		COAL GRID COORDINATE	
SNo	EASTING	NORTHING	X	Y
1	525316.549	2581637.035	2718942.695	1040098.846
2	525331.130	2576290.313	2718847.772	1034754.006
3	525096.743	2576307.516	2718613.833	1034776.001
4	524221.384	2576373.162	2717740.177	1034859.546
5	523523.379	2576565.217	2717046.391	1035065.815
6	522576.445	2576729.579	2716103.209	1035249.499
7	521933.799	2576782.859	2715461.916	1035315.916
8	521174.687	2576838.767	2714704.258	1035387.344
9	521163.071	2578026.391	2714716.963	1036574.723
10	521156.016	2578026.270	2714709.909	1036574.746
11	521154.760	2578618.761	2714720.785	1037167.023
12	521134.090	2581063.874	2714750.190	1039611.569
13	521135.210	2581138.048	2714752.828	1039685.691
14	521132.991	2581472.141	2714757.451	1040019.694
15	521135.001	2581610.436	2714762.293	1040157.892
16	521235.834	2581611.789	2714863.113	1040157.180
17	524038.593	2581631.913	2717665.152	1040119.899

2. ACCESSABILITY / COMMUNICATION

The block is well connected by metal road approachable all through the year. The road distance from the block to various destinations is listed below.

- | | | | |
|----|---------|---|-------|
| A. | Shahdol | - | 12 km |
| B. | Burhar | - | 7 km |

The nearest rail head is Badhwabara located just near to the south-eastern part of the block.

3. OBJECTIVE

The G2 stage exploration is proposed to fulfil following objectives:

- A. To prove the existence and continuity of coal seams occurring in the Marwatola Sec- I & II Block.
- B. To establish the lay, disposition and potentiality of coal seams.
- C. To assess the coal resources in G2 stage of exploration.

4. STATUS OF MINING

The present block of investigation is virgin from the view point of mining activity.

5. AREA & LIMITATIONS

The block measures 21.03 sq.km which includes forest area of about 15.45 sq.km. The block is bounded by:

North : The block is bounded on north side by Pathora East, Arjuni West block

South : The south side bounded by Marwatola sector VII block.

East : The block is bounded on East side by Marwatola Sector-III & IV block.

West : On the west the Ghunghuti block is located.

6. DETAILS OF DRILLING

Table–SF.6 A: Details of boreholes drilled in Marwatola Sector I & II Block, Sohagpur Coalfield.

S.No.	Series of Boreholes	Nos. of boreholes drilled	Period
1	MWS	20	April.2004 to May.2007
2	CSMI&II	15	Dec.2021 to June2022

7. BOREHOLE DENSITY

The overall borehole density in the block is around 1.66 bhs/sq.km

8. CHEMICAL ANALYSIS

The laboratory-wise quantum of chemical analyses undertaken is tabulated below: -

Table SF.8 A: Details of coal analysis in Marwatola Sector I&II Block, Sohagpur Coalfield

Seam	Number of intersections	Band by band	Overall proximate analysis on 60%RH & 40°C (>0.90M)		GCVK. cal./kg	Ultimate analysis	AFT	HGI
			Determined	Calculated				
IV	27	20	10	-	10	-	-	-
IIIA	21	15	03	02	03	1	1	1
III Top	04	02	02	-	02	-	-	-
III Middle	04	02	01	-	01	-	-	-
III Bottom	04	02	03	-	03	-	-	-
III COMB	26	22	15	04	15	3	-	-
II Top	15	13	12	-	12	-	-	-
II Bottom	15	13	08	01	08	-	-	-
II Comb.	14	12	09	01	10	3	1	1
IB	28	26	22	03	22	2	1	1
IA	29	25	10	02	10	3	1	1

9. OTHER INVESTIGATION

A. Geological mapping - Geological mapping has been carried out in the block however, most of the area is covered with weathered mantle, soil and forest cover there are scanty or lack of exposures. The area is mostly covered by Barakar and Raniganj formation.

- | | | |
|------------------------------------|---|---|
| B. Remote Sensing | - | Not done. |
| C. Surface geophysical | - | Not Done |
| D. Geo-physical logging | - | 5869.86m. |
| E. Physico-mechanical study | - | Tests carried out for core
Samples of 2 boreholes
MSM-20 & CSMI&II-12 |
| F. Petrographic Test
and MSM-19 | - | Two borehole CSMIII&IV-04 |
| G. Hydrogeology | - | Data since 2000. Data
acquisition is continuing |
| H. Washability test | - | Not done. |

10. GEOLOGICAL FORMATIONS

The area is covered by barren measure and lower and middle Barakar Formations. 11 coal seams viz. Seams IV, IIIA, II Top, III Middle, III Bottom, III Comb., II Top, II Bottom, II Comb., IB and IA occur in Barakar Formation.

11. GEOLOGICAL STRUCTURE

A. Strike & Dip: The strata including coal seams exhibit a broad NE-SW trend in the south-western part of the block exhibits a rolling strike. The strata including coal seams show a broad NE-SW trend.

The strata in general show a northwesterly dip. The general dip of the strata varies from 2°-3° but near and in between the faults, the dip of the strata varies from 6°-8°.

B. Faults - Six faults have been interpreted in the block. These faults are numbered as F1-F1 to F6-F6 from north to south and fault strike is almost E-W. All the faults F1-F1to F6-F6in the Marwatola Sector-I&II block is continuing in the adjacent Marwatola Sector-III&IV block.

12. DESCRIPTION OF FAULTS

Description of all the faults deciphered in the block is given in Table SF. 12 A below:-

Table SF.12. A: Description of Faults, Marwatola Sector I&II Block, Sohagpur Coalfield

Sl.No.	Fault No.	Trend	Throw		Evidences
			Direction	Amount(m)	
1	F ₁ -F ₁	W-NNE	Southerly throw	Varies from 10-30m	i) Difference in FRL on either side of the fault.
2	F ₂ -F ₂	E-W	Southerly throw	0 to 2m	i) Difference in FRL on either side of the fault.
3	F ₃ -F ₃	E-W to SE	South-easterly throw	30 to 60m	i) Difference in FRL on either side of the fault.
4	F ₄ -F ₄	NW-SE	Southerly throw	50 to 100m	i) Difference in FRL on either side of the fault.
5	F ₅ -F ₅	NW-SE	Southerly throw	350 to 370m	i) Difference in FRL on either side of the fault.
6	F ₆ -F ₆	Almost E-W	Northerly throw	290 to 660m	i) Difference in FRL on either side of the fault. ii) The Boreholes CSM I&II-3, 9 have been faulted.

13. INTRUSIVE

The area is free from any igneous activity and as such the seams are free from pyrolitisation.

14. COAL SEAMS

14.1. SEQUENCE OF COAL SEAMS

The sequence of coal seams and intervening partings found to occur within the block, as per borehole data, are given below:

Table SF.14.1. A: Succession of Coal Seam Marwatola Sector I&II Block, Sohagpur Coalfield

Seams / Parting	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		Nos. of Bh
	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	
SOIL	0.00	0.00	0.20	6.00	0.20	6.00	470.08	542.58			35
			CSM I&II-06	MSM-23	CSM I&II-06	MSM-23		MSM-27			
WM	0.20	6.00	5.20	46.00	5.20	46.00					35
	CSM I&II-06	MSM-23	CSM I&II-01	MSM-55	CSM I&II-01	MSM-55					
Parting	5.20	46.00	120.00	201.73	107.56	179.17					03
	CSM I&II-01	MSM-55	CSM I&II-02	MSM-23	CSM I&II-02	MSM-55					
V	120.00	201.73	120.10	201.83	0.05	0.10	316.56	378.64	316.46	378.54	03
	CSM I&II-02	MSM-23	CSM I&II-02	MSM-23	CSM I&II-13	CSM I&II-02	MSM-23	CSM I&II-02	MSM-23	CSM I&II-02	
Parting	120.00	201.73	136.20	217.81	15.98	58.49					03
	CSM I&II-02	MSM-23	CSM I&II-02	MSM-23	MSM-23	CSM I&II-13					
IV	99.64	480	100.24	480.65	0.10	1.80	43.28	405.79	42.63	405.19	27
	CSM I&II-10	CSM I&II-03	CSM I&II-10	CSM I&II-03	MSM-51	MSM-24	CSM I&II-03	CSM I&II-10	CSM I&II-03	CSM I&II-10	
Parting	100.24	226.73	135.84	258.50	24.73	52.20					17
	CSM I&II-10	MSM-28	CSM I&II-10	MSM-24	CSM I&II-12	MSM-52					
L2	135.84	258.50	135.94	258.50	0.05	1.35	249.57	369.59	249.13	369.49	19
	CSM I&II-10	MSM-24	CSM I&II-10	MSM-24	CSM I&II-06	CSM I&II-05	MSM-54	CSM I&II-10	MSM-54	CSM I&II-10	
Parting	135.94	258.50	144.60	595.50	6.51	20.10					13
	CSM I&II-10	MSM-24	CSM I&II-10	CSM I&II-15	CSM I&II-02	MSM-28					
IIIA	144.60	595.50	147.00	597.00	0.10	2.70	-82.86	360.83	-84.36	358.43	21
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	MSM-18	MSM-28	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-10	

Seams / Parting	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		Nos. of Bh
	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	
Parting	147.00	597.00	164.40	619.10	6.17	25.73	-	-	-	-	21
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-04	CSM I&II-03					
III TOP	251.45	272.30	252.93	273.60	0.30	1.48	227.94	257.96	227.64	256.82	04
	CSM-I&II-07	MSM-55	CSM-I&II-07	MSM-55	MSM-51	CSM-I&II-07	MSM-51	CSM-I&II-12	MSM-51	CSM-I&II-12	
Parting	252.93	273.60	253.93	277.00	0.62	3.40					04
	CSM-I&II-07	MSM-55	CSM-I&II-07	MSM-55	MSM-51	MSM-55					
III MIDDLE	253.93	277.00	254.23	278.15	0.30	3.22	227.02	255.10	226.72	251.88	04
	CSM-I&II-07	MSM-55	CSM-I&II-07	MSM-55	CSM-I&II-07	CSM-I&II-12	MSM-51	CSM-I&II-12	MSM-51	CSM-I&II-12	
Parting	254.23	278.15	256.78	280.45	1.08	2.55					04
	CSM-I&II-07	MSM-55	CSM I&II-07	MSM-55	MSM-51	CSM I&II-07					
III BOTTOM	256.78	280.45	258.66	281.90	1.30	3.42	225.64	249.82	224.34	246.40	04
	CSM I&II-07	MSM-55	CSM I&II-07	MSM-55	MSM-51	CSM I&II-12	MSM-51	CSM I&II-12	MSM-51	CSM I&II-12	
III COMB	164.40	619.10	166.30	619.20	0.10	5.06	-106.46	341.03	-106.56	339.13	26
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-15	CSM I&II-05	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-10	
Parting	166.30	619.20	199.30	650.37	18.67	60.96					28
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	MSM-27	MSM-26					
II TOP	237.85	328.65	238.75	330.13	0.40	4.80	188.53	269.52	188.13	268.99	15
	CSM I&II-02	MSM-24	CSM I&II-02	MSM-24	MSM-52	MSM-55	MSM-52	MSM-27	MSM-52	MSM-27	
Parting	238.75	330.13	239.22	332.15	0.47	4.10					15
	CSM I&II-02	MSM-24	CSM I&II-02	MSM-24	CSM I&II-02	MSM-21					
II BOTTOM	239.22	332.15	241.15	334.40	0.36	2.30	186.88	268.30	185.38	267.94	15
	CSM I&II-02	MSM-24	CSM I&II-02	MSM-24	MSM-27	MSM-21	MSM-52	MSM-27	MSM-52	MSM-27	

Seams / Parting	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		Nos. of Bh
	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	
II COMB	199.30	650.37	202.30	650.87	0.03	5.60	-137.73	306.13	-138.23	303.13	14
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-13	MSM-17	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-10	
Parting	202.30	650.87	215.90	684.70	7.99	33.83					26
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-04	CSM I&II-15					
I B	20	684.7	20.98	685.3	0.05	2.47	-172.06	467.99	-172.66	467.01	28
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	MSM-27	CSM I&II-10	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	
Parting	20.98	685.30	29.72	691.98	3.90	17.25					28
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	CSM I&II-12	CSM I&II-08					
I A	29.72	691.98	29.83	693.88	0.11	2.11	-179.34	458.27	-181.24	458.16	29
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	MSM-20	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	
Parting	233.10	362.85	243.38	369.40	4.11	11.77					14
	CSM I&II-10	MSM-23	MSM I&II-10	MSM-23	MSM-19	MSM-34					
I TOP	243.38	369.40	243.58	369.70	0.10	0.75	148.20	262.05	147.70	261.85	14
	CSM I&II-10	MSM-23	CSM I&II-10	MSM-23	MSM-19	CSM I&II-02	MSM-34	CSM I&II-10	MSM-34	CSM I&II-10	
Parting	243.58	369.70	246.24	371.70	0.40	6.15					14
	CSM I&II-10	MSM-23	CSM I&II-10	MSM-23	CSM I&II-13	MSM-21					
I BOT	246.24	371.70	247.60	372.25	0.20	1.36	144	259.19	143.80	257.83	14
	CSM I&II-10	MSM-23	CSM I&II-10	MSM-23	CSM I&II-13,MSM-34	CSM I&II-10	MSM-34	CSM I&II-10	MSM-34	CSM I&II-10	
Parting	247.60	372.25	-	-	-	-					14
	CSM I&II-10	MSM-23									
I COMB	51.80	700.05	52.00	702.10	0.20	2.05	-187.41	436.19	-189.46	435.99	13
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	MSM-17, 22	CSM I&II-15	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	

14.2. QUALITY OF COAL SEAMS

Quality of coal in this block is generally of high moisture high ash power grade. The details of quality of coal on I₃₀ basis in this block are tabulated below:

Table SF.14.2.A: Qualitative Description of Coal Seams, Marwatola Sector I&II Block, Sohagpur Coalfield

Seam Name	Range of Seam Thickness (m)		Range of Effective Thickness (m)		Range of Proximate Analysis on 60% RH & 40°C								Range of GCV in k. cal/kg		Grade Range	
					M%		Ash%		VM%		FC%					
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
IV	0.95	1.8	0.95	1.8	4.7	11.6	12.6	25.9	25.8	31.8	40.9	52.5	5088	6368	G-8	G-4
	MSM-34	MSM-24	MSM-34	MSM-24	MSM-34	MSM-17	MSM-31	CSM I&II-04	CSM I&II-04	CSM I&II-08	CSM I&II-04	MSM-19	CSM I&II-04	MSM-31		
IIIA	0.96	2.7	0.91	2.4	2.3	4.8	20.1	55.1	19.5	28.1	31.2	49.5	2934	6439	G-15	G-3
	MSM-17	MSM-28	MSM-17	CSM I&II-10	CSM I&II-15	MSM-28	CSM I&II-15	MSM-17	CSM I&II-13	CSM I&II-15	CSM I&II-13	CSM I&II-15	MSM-17	CSM I&II-15		
III top	1.3	1.48	1.3	1.38	3.7	5.6	15.4	33.7	23.4	30.8	37.3	50.2	4715	6571	G-9	G-3
	MSM-55	CSM I&II-07	MSM-55	CSM I&II-07	MSM-55	CSM I&II-07	MSM-55	CSM I&II-07	CSM I&II-07	MSM-55	CSM I&II-07	MSM-55	CSM I&II-07	MSM-55		
III Middle	1.15	1.15	1.15	1.15	3.4	3.4	24	24	28.7	28.7	43.9	43.9	5805	5805	G-5	G-5
	MSM-55	MSM-55	MSM-55	MSM-55	MSM-55	MSM-55	MSM-55	MSM-55	MSM-55	MSM-55	MSM-55	MSM-55	MSM-55	MSM-55		
III Bottom	1.3	1.88	1.3	1.75	3.2	6.7	15.2	32.8	23.8	30.7	39.7	47.4	5005	6192	G-8	G-4
	MSM-51	CSM I&II-07	MSM-51	CSM I&II-07	MSM-55	CSM I&II-07	CSM I&II-07	MSM-55	MSM-51	CSM I&II-07	MSM-55	CSM I&II-07	MSM-55	CSM I&II-07		
III Combined	1.1	5.06	1.02	4.81	2	10	13.4	49.6	21.2	31.1	35.2	48.2	3394	6366	G-14	G-4
	CSM I&II-14	CSM I&II-05	CSM I&II-14	CSM I&II-05	CSM I&II-01	MSM-17	MSM-28	MSM-19	CSM I&II-01	CSM I&II-11	CSM I&II-13	CSM I&II-01	MSM-19	MSM-28		

CMPDI

Seam Name	Range of Seam Thickness (m)		Range of Effective Thickness (m)		Range of Proximate Analysis on 60% RH & 40°C								Range of GCV in k. cal/kg		Grade Range	
					M%		Ash%		VM%		FC%					
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
II top	0.9	4.8	0.9	4.8	2.8	7.6	11.6	30.2	16.4	30.9	42.5	55.5	5192	6637	G-8	G-3
	CSM I&II-02	MSM-55	CSM I&II-02	MSM-55	CSM I&II-01	CSM I&II-11	MSM-24	CSM I&II-01	CSM I&II-01	MSM-28	CSM I&II-05	MSM-24	CSM I&II-05	MSM-24		
II Bottom	0.94	2.3	0.94	2.3	2.5	6.7	13.4	35.1	14.2	31.2	38.1	51.5	4643	6402	G-9	G-3
	CSM I&II-14	MSM-21	CSM I&II-14	MSM-21	CSM I&II-01	CSM I&II-11	MSM-24	MSM-52	CSM I&II-01	CSM I&II-11	CSM I&II-14	MSM-24	MSM-52	MSM-28		
II Combined	1.39	5.6	1.39	5.08	1.9	7.4	12	33.83	22.09	28.5	38.6	57.5	4961	7025	G-8	G-1
	MSM-23	MSM-17	MSM-23	MSM-19	CSM I&II-03	MSM-18	MSM-23	CSM I&II-10	CSM I&II-10	MSM-23	MSM-26	CSM I&II-03	MSM-26	CSM I&II-03		
IB	0.98	2.47	0.98	2.1	1.5	7.9	9.6	32.8	13.9	31.2	38.2	69.5	4918	6910	G-8	G-2
	MSM-22	CSM I&II-10	MSM-22	CSM I&II-10	MSM-22	MSM-17	MSM-31	CSM I&II-01	MSM-22	CSM I&II-11	CSM I&II-14	MSM-22	CSM I&II-05	MSM-22		
IA	0.9	2	0.9	2	2.7	7.6	11.3	42.2	9.8	29.6	45.2	54	4324	6737	G-10	G-2
	CSM I&II-02	MSM-31	CSM I&II-02	MSM-31	CSM I&II-01	MSM-17	MSM-17	CSM I&II-01	CSM I&II-01	MSM-19	CSM I&II-11	MSM-17	CSM I&II-01	CSM I&II-02		
I Bottom	1.06	1.06	1.06	1.06	3.4	3.4	35.81	35.81	22.38	22.38	38.4	38.4	4892	4892	G-8	G-8
	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10	CSM I&II-10		

15. GEOLOGICAL RESOURCES

The Seam-wise and category-wise Net Geological Proved Reserves are provided below:

**Table–SF.15 A: Seam-wise and category-wise ‘Gross’ Indicated Geological resources In Marwatola Sector I&II Block,
Sohagpur Coalfield.
(Reserves in million tonnes)**

Seam Name	Indicated Resources (mt)
IV	14.581
IIIA	8.229
IIIT	4.970
IIIM	3.601
IIIB	8.299
IIIC	31.583
IIT	32.314
IIB	15.635
II	22.321
IB	34.772
IA	21.460
Grand Total	197.767

The seam-wise grade-wise Depth-wise Net Proved Geological reserves are provided below:

**Table–SF.15 B: Seam-Wise, Grade-Wise and Depth-Wise ‘Gross’ Indicated Geological Resources In Marwatola Sector I & II Block, Sohagpur Coalfield.
(Reserves in million tonnes)**

Seam Name/ Grade Range	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14	G15	G16	G17	Grand Total
	INDICATED RESOURCE (in Million tonnes)																	
SEAM - IV																		
0-300	-	-	-	4.627	4.754	1.725	1.876	0.609	-	-	-	-	-	-	-	-	-	13.591
300-600	-	-	-	0.624	0.366	-	-	-	-	-	-	-	-	-	-	-	-	0.990
600 and above	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total	-	-	-	5.250	5.121	1.725	1.876	0.609	-	-	-	-	-	-	-	-	-	14.581
SEAM - IIIA																		
0-300	-	-	-	0.176	0.084	0.114	0.355	0.757	2.581	0.990	0.494	0.252	0.297	0.321	0.045	-	-	6.463
300-600	-	-	0.114	0.527	0.472	0.160	0.117	0.086	-	-	-	-	-	-	-	-	-	1.477
600 and above	-	-	-	0.267	0.022	-	-	-	-	-	-	-	-	-	-	-	-	0.289
Sub-Total	-	-	0.114	0.970	0.578	0.274	0.472	0.843	2.581	0.990	0.494	0.252	0.297	0.321	0.045	-	-	8.229
SEAM - IIIT																		
0-300	-	0.294	0.600	0.808	0.994	0.518	0.276	0.672	0.807	-	-	-	-	-	-	-	-	4.970
300-600	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
600 and above	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total	-	0.294	0.600	0.808	0.994	0.518	0.276	0.672	0.807	-	-	-	-	-	-	-	-	4.970
SEAM - IIIM																		
0-300	-	-	-	-	2.691	0.910	-	-	-	-	-	-	-	-	-	-	-	3.601
300-600	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
600 and above	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total	-	-	-	-	2.691	0.910	-	-	-	-	-	-	-	-	-	-	-	3.601
SEAM - IIIB																		
0-300	-	-	-	1.021	2.256	1.617	1.457	1.949	-	-	-	-	-	-	-	-	-	8.299
300-600	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
600 and above	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total	-	-	-	1.021	2.256	1.617	1.457	1.949	-	-	-	-	-	-	-	-	-	8.299

CMPDI

Seam Name/ Grade Range	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14	G15	G16	G17	Grand Total
SEAM - IIIC																		
0-300	-	-	0.270	1.502	3.683	4.854	3.431	6.354	5.783	2.258	0.796	0.766	1.378	0.240	0.012	-	-	31.327
300-600	-	-	-	0.005	0.156	0.079	0.002	-	0.014	0.001	-	-	-	-	-	-	-	0.256
600 and above	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total	-	-	0.270	1.507	3.839	4.933	3.432	6.354	5.797	2.259	0.796	0.766	1.378	0.240	0.012	-	-	31.583
SEAM - IIT																		
0-300	-	-	0.746	1.842	1.446	2.160	5.466	0.022	-	-	-	-	-	-	-	-	-	11.681
300-600	-	0.181	2.939	4.077	8.674	2.957	1.681	0.124	-	-	-	-	-	-	-	-	-	20.633
600 and above	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total	-	0.181	3.684	5.918	10.120	5.117	7.147	0.146	-	-	-	-	-	-	-	-	-	32.314
SEAM - IIB																		
0-300	-	-	-	0.003	2.189	0.741	0.342	0.376	0.077	-	-	-	-	-	-	-	-	3.729
300-600	-	-	1.365	3.097	5.208	1.035	0.339	0.194	0.630	0.039	-	-	-	-	-	-	-	11.906
600 and above	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total	-	-	1.365	3.100	7.397	1.776	0.681	0.570	0.707	0.039	-	-	-	-	-	-	-	15.635
SEAM - II																		
0-300	-	-	0.015	0.085	2.782	4.638	1.517	2.486	0.021	0.002	-	-	-	-	-	-	-	11.547
300-600	0.087	0.543	0.765	0.353	1.232	3.062	2.389	0.950	0.034	0.004	-	-	-	-	-	-	-	9.419
600 and above	-	0.362	0.349	0.168	0.137	0.126	0.096	0.079	0.040	-	-	-	-	-	-	-	-	1.355
Sub-Total	0.087	0.905	1.129	0.606	4.150	7.826	4.002	3.515	0.094	0.006	-	-	-	-	-	-	-	22.321
SEAM - IB																		
0-300	-	0.360	0.541	1.378	1.510	1.579	0.465	-	-	-	-	-	-	-	-	-	-	5.833
300-600	-	0.602	0.626	4.726	6.638	8.244	5.174	1.364	-	-	-	-	-	-	-	-	-	27.375
600 and above	-	-	0.017	0.178	0.430	0.939	-	-	-	-	-	-	-	-	-	-	-	1.563
Sub-Total	-	0.962	1.184	6.282	8.578	10.762	5.639	1.364	-	-	-	-	-	-	-	-	-	34.772
SEAM - IA																		
0-300	-	0.014	0.134	1.712	1.428	0.460	0.381	0.804	0.237	0.093	0.096	0.065	-	-	-	-	-	5.424
300-600	-	0.073	1.551	4.354	4.364	1.353	0.696	0.311	0.254	0.236	0.176	0.071	0.003	-	-	-	-	13.443
600 and above	-	-	-	-	-	0.224	1.030	1.024	0.316	-	-	-	-	-	-	-	-	2.594
Sub-Total	-	0.087	1.686	6.067	5.792	2.036	2.107	2.139	0.807	0.329	0.272	0.137	0.003	-	-	-	-	21.460
Grand Total	0.087	2.430	10.032	31.529	51.516	37.495	27.090	18.161	10.794	3.621	1.562	1.155	1.678	0.560	0.057	-	-	197.767

16. UNITED NATIONS FRAMEWORK CLASSIFICATION

The UNFC consists of three dimensional system for assessing the reliability and degree of assurance, which is measured in terms of codes, viz. 1, 2 and 3. The digit '1' indicates highest degree of assurance whereas the digit '3' indicates least degree of assurance. The reliability and degree of assurance of these codes are assessed with respect to three criteria. They are Economic viability, Feasibility Assessment like project report and Geological Assessment. With respect to the current report the Economic viability and Feasibility Assessment for the block is yet to be undertaken. Hence both these criteria are given least degree of assurance i.e. the digit '3'. This report has dealt with the geological aspect completely. Thus the code '2' is given for Geological Assessment.

Thus the UNFC Code given for the resources 197.767 MT of Marwatola Sector-I&II block is 332.

Chapter - 1

INTRODUCTION

1.1 INTRODUCTION

The Sohagpur coalfield, covering an area of around 3100sq km, in the south central part of the Rewa Gondwana basin of the peninsular Gondwana belt of India. Marwatola Sector I&II block is located in the northern west part of Sohagpur coalfield. The exploration of G.S.I. has proved the occurrence of Raniganj, Barren Measure, Barakar and Talchir Formations in the area and has established the presence of five regionally persistent coal seams (Seam I to V in ascending order) within the Barakar Formation in the area. Over the last two decades, G.S.I. is exploring for high rank coking coal of Barakar Formation under the cover of Raniganj and Barren Measures, occurring all along the down thrown northern block of the Bamni-Chilpa fault, roughly E-W trending intra-basinal major fault running along the middle part of the Sohagpur Coalfield.

The Sohagpur Coalfield can be broadly divided into two parts, i.e. the North and South Sohagpur, along the east-west trending intra-basinal Bahmni-Chilpi fault. The North Sohagpur represents the down thrown block of the Sohagpur basin. In South Sohagpur three sub-basins namely, Jhagrakhand, Jamuna-Kotma and Amlai-Burhar, from East to West have been recognised. G.S.I. has carried out the regional exploration 1994 to 1998 in Shahdol and Patasi areas in the western part of this coalfield.

The regional exploration in Marwatola block, Sohagpur Coalfield has been done initially by MECL covering an area of around 78 sq.km. Later the block has been subdivided into Marwatola Sector I&II, Marwatola Sector III&IV, Marwatola Sector VI and Marwatola Sector VII respectively. The Marwatola Sector VI and Marwatola Sector VII block have been explored in detail by CMPDIL (excluding forest area). In order to increase the borehole density to 2bhs/sq.km the Marwatola Sector I&II and Marwatola Sector III&IV have been proposed for G2 stage of exploration through NMET. In the present Marwatola Sector I&II block, the drilling was undertaken, to prove the structure

and existence of the coal seams V to Seam I during the period from 22-12-2021 to 17-06-2022. All the drilled boreholes are DGPS surveyed.

1.2 LOCATION

Marwatola Sector I&II block is located in the district of Umaria, Madhya Pradesh state. It lies in the western part of Sohagpur coalfield and is about 12 Km. North-West of Shahdol town, the district headquarter of Shahdol, Madhya Pradesh. The block is located at a distance of 75 Km. from Umaria, the district headquarter of Umaria, Madhya Pradesh. In its Northern Boundary Pathora East and Arjuni West Block is located. In its western boundary Ghunghuti and Panwari block is located whereas at southern boundary Marwatola Sector III&IV block is located. Marwatola Sector I&II block covers an area of about 21.03 Sq. Km. The area of block is covered by the Survey of India Toposheet No.64 E/3 & E/7 (RF1:50,000) and block boundary, coordinates and cardinal points, given by Geosystem CMPDIL, RI -V are mentioned in table 1.2.2.

TABLE-1.2.2

Limits of Marwatola Sector I&II block, Sohagpur coalfield, based on coordinates

LATITUDE	COAL GRID	UTM
Minimum	1034754.006	2576290.313
Maximum	1040157.892	2581637.035
LONGITUDE	COAL GRID	UTM
MINIMUM	2714704.258	521132.991
MAXIMUM	2718942.695	525331.130

1.3 ACCESSIBILITY & COMMUNICATION

The Sohagpur coalfield is well connected by rail and road. Marwatola Sector I&II block is about 12km from Shahdol District headquarter and about 7 km from Burhar tehsil. The nearest rail head is Badhwabara located just near to the southeastern and Ghunghuti located near the northwestern side of Marwatola Sector I&II Block. The

Bilaspur-Katni section Line of South East Central Railway passes through the southeast to northwest part of block. The Shahdol-Pali road (NH-43) passes through the northern part of the block. The block can be approached from this by an all-weather feeder road passing all along the northern boundary of the Block.

1.4 AREA OF THE BLOCK

The block measures 21.03 sq.km which includes forest area of about 15.45 sq.km. The block is bounded by:

North : The block is bounded on north side by Pathora East, Arjuni West block

South : The south side bounded by Marwatola sector VII block.

East : The block is bounded on East side by Marwatola Sector-III&IV block.

West : On the west the Ghunghuti block is located.

Table: 1.4 A

Cardinal Points of Block Boundary of Marwatola Sector I&II Block, Sohagpur Coalfield in (Coal grid and UTM).

CARDINAL POINTS	UTM COORDINATE SYSTEM		COAL GRID COORDINATE	
S No	EASTING	NORTHING	X	Y
1	525316.549	2581637.035	2718942.695	1040098.846
2	525331.130	2576290.313	2718847.772	1034754.006
3	525096.743	2576307.516	2718613.833	1034776.001
4	524221.384	2576373.162	2717740.177	1034859.546
5	523523.379	2576565.217	2717046.391	1035065.815
6	522576.445	2576729.579	2716103.209	1035249.499
7	521933.799	2576782.859	2715461.916	1035315.916

CARDINAL POINTS	UTM COORDINATE SYSTEM		COAL GRID COORDINATE	
S No	EASTING	NORTHING	X	Y
8	521174.687	2576838.767	2714704.258	1035387.344
9	521163.071	2578026.391	2714716.963	1036574.723
10	521156.016	2578026.270	2714709.909	1036574.746
11	521154.760	2578618.761	2714720.785	1037167.023
12	521134.090	2581063.874	2714750.190	1039611.569
13	521135.210	2581138.048	2714752.828	1039685.691
14	521132.991	2581472.141	2714757.451	1040019.694
15	521135.001	2581610.436	2714762.293	1040157.892
16	521235.834	2581611.789	2714863.113	1040157.180
17	524038.593	2581631.913	2717665.152	1040119.899

1.5 PHYSIOGRAPHY & DRAINAGE

The block is characterized by barakar strata undulatory to low dips have given rise to a feature less rolling topography with elevation between 542.58m and 470.08m above MSL. However, the Barakar sandstones stand out prominently against the surrounding plains. The highest elevation 542.58m (MSM-28) occurs in the northwestern part of the block whereas the lowest surface value of 470.08m (CSM I&II) is near northeastern part of the block near Majhigawan village.

The Son River with a north-westerly flow is the main drainage channel of Sohagpur coalfield. Several streams originate in the high trappean region in the southern area and join the Son River. These north flowing streams include Nargara, Sarpa, Bagiha, Bakan, Suthna, Tipon etc. Basar nala across the block in southern part of block and discharged into sarpa nala.

1.6 CLIMATE, FLORA & FAUNA

The area is characterised by tropical climate with well-defined summer from April to June. Monsoon from July to September and winter from November to February. May is the hottest month when the temperature varies from 25°C to 48°C

and at night, it is between 18°C to 24°C. The relative humidity during monsoon ranges from 76% to 90% and in summer 26% to 42%. The average rainfall in the area varies from 1000mm to 1700mm annually. The maximum annual rainfall recorded was 1894.10mm (1994). Out of which 1756.70mm was monsoonic rainfall (July to September). The nearly half of the Block area is covered by Reserved Forest. Protected forest cover is marked in the eastern, western and northern parts. The common species being Sal (*Shorea Robusta*) Mahua, (*Madhuka Latifolia*), Palas (*Butea Mansperma*) etc. The remaining part of the area is under cultivation. Pockets of cultivated land can be found around the villages within the block. Paddy is the main annual crop of the area. In the forest wild animals like bear, leopard, hyena etc.

1.7 PREVIOUS WORK

The first systematic geological survey of Sohagpur Coalfield and adjoining areas was carried out by T.W.H Hughes during 1879-1883 and he is credited with producing the first geological map of the area. As a result of the exploration, Hughes came to the conclusion that this extensive area of coal measures with only a few exposures of attractive coal seams was rather disappointing. Later survey by Reader in 1889-90 near Burhar-Amlai, however, proved the existence of thick coal seams. A.L. Coulson who traversed the field in 1922-23, in connection with the proposed railway alignment for the Central Indian coalfields, recorded several other exposures of coal. Exploration was carried out by private mine owners like Dalchand Bahadur Singh and this resulted in the opening of Jhagrakhand and Rajnagar collieries. Since 1958 the Indian Bureau of Mines carried out detailed proving operations in Bijuri and Kotma blocks. N. Venkatappayya, in company with K. Ramanna, G.P. Deshmukh and A.K. Srivastava, resurveyed the coalfield between 1958-63. They made a re-appraisal of the coal potentiality also. Of late an intensive study of the nature of the lithic fill, stratigraphy and coal potentiality of the eastern part of the coalfield has been made (Chowdhury 1974, 1977, 1979). G. Mukhopadhyay, U. Chakraborty and V. D. Borkar also carried out systematic study of the areas. N.C. D. C. Ltd. and later Coal India Ltd. carried out studies on potential mining areas like Dhanpuri, Amlai, Bhadra, Jamuna, Rungta Collieries etc.

Geological Survey of India has carried out regional exploration in Bijuri-Rajnagar-Malga area in Jhagrakhand sub-basin. They also undertook regional assessment and large scale mapping of a number of virgin tracts between, December' 1979 to April' 1986. Recently, G.S.I. has conducted regional exploration of coal in Shahdol, Kelmania-Mithauri, Patasi, Diapipar areas (1994-99) in the western part of Sohagpur coalfield to delineate potential area of high rank coal and also to decipher structure and tectonic set up of the area.

MECL undertook the detailed exploration of coal in Urtan Block in the eastern part of Sohagpur coalfield in July 1991 in an area of about 4 Sq. Km. to assess the coal resources. The data are documented in the Geological Report entitled "Geological Report on Coal Exploration Urtan Block Sohagpur Coalfield District: Shahdol M.P." (Oct'1994).

In Marwatola block, MECL has carried out regional exploration for the period of 09/04/2005 to 13/05/2007 and published the geological report named "Geological Report on Coal Exploration Marwatola Block Sohagpur Coalfield District: Shahdol M.P."

1.8 MINING HISTORY OF THE BLOCK

The area in and around Marwatola Sector I&II Block is virgin with no mining activities. Dhanpuri UG and burdhar Mines of SECL is about 40 km (Aerial distance) and by road 45km. southeast of the block (by road). Nowrozabad Mine of SECL is about 24 km south west of the block. For Malachua mine PR is under developing stages.

1.9 EXPLORATION DESIGN

In the south-east of Marwatola Sector I&II Block, the Bicharpur North Block, was proved by CMPDI and detailed drilling completed in December 2015 by APC and the Shahdol block detail explored in January 2016 drilling by MAPL. In northern side Pathora east block was explored by MECL. Later to know the continuity of the lay and disposition of the coal horizons and to find out their working potentialities further south

of the Pathora East & western side of Shahdol, Bicharpur north & Shahpur Block. In the block regional exploration was initiated by MECL during 09/04/2004 to 13/05/2007 and CMPDI through outsourcing drilling agency 15 boreholes drilled during 22/12/2021 to 17/06/2022 in the Marwatola Sector I&II Block. However, for establishing the structure and for proving the coal seam incrops these boreholes were drilled in the block. A total of 35 No. of boreholes have been drilled in the area. Central Mine Planning and Design Institute (CMPDI) is proposing to carry out General Exploration (G2 stage) exploration for coal in Marwatola Sec- I & II Block, Sohagpur Coalfield, through NMET funds comprising 6680m of drilling in **15** boreholes. The boreholes have been proposed at 800 m × 800 m grid. The details of drilling are as follows:

S. No.	Series of Boreholes	Nos. of boreholes drilled	Period
1	MWS	20	April.2004 to May.2007
2	CSM I&II	15	Dec.2021 to June 2022

The run-wise lithology of boreholes drilled by MECL and CMPDI (through outsourcing) in the area is given in Annexure-IIA. Graphic logs of boreholes are shown in Plate-III.

1.10 IMPORTANT FEATURES IN THE AREA

The nearly half of the Block area is covered by Reserved Forest and Protected forest. The paddy fields in Patnar Kalan, Chandpur village in the block. However, Chandpur village and majhigawan village are the residential areas/villages falling in the central and northwest parts of the Marwatola Sector I&II Block respectively. A national highway 43 connects the Umaria -Shahdol district HQ passing through the Northwestern boundary part of the block. Whereas the Ghunghuti Township is located on NH-43, just adjacent to the western boundary of the block from where the block is approachability by road in all-weather in the block.

The area is free from any industrial activities and is also devoid of any monument of Historical and Archaeological importance. The southern side of block, Malachua mining project report of SECL are under preparation.

1.11 PRESENT INVESTIGATION

The quantum of schedule of exploration activities by CMPDI in Marwatola Sector I&II Block is given below:

Table No.1.11.A.

The quantum & schedule of exploration activities by MECL and CMPDI in Marwatola Sector I&II Block Sohagpur Coalfield.

S.No.	Activity	Duration		Quantum
		From	To	
1	Reconnaissance/Data Collection :			
a)		April.2004	May.2007	20 BH'S
b)		Dec. 2021	June 2022	15 BH'S
2	Mobilisation :			
a)		April.2004	May.2007	20 BH'S
b)		Dec.2021	June 2022	15 BH'S
3	a) Topographical Survey	Topographical as provided by Survey of India in 2m interval.		
	b) Borehole Survey			
	i)	April.2004	May.2007	20 BHs. (MSM Series)
	ii)	Dec.2021	July.2022	15 BH (CSM I&II Series)
4	Drilling :			
a)		April.2004	May.2007	6866.85m
b)		Dec.2021	June.2022	5882.00m
5	Geological Logging :			
a)		April.2004	May.2007	6866.85m
b)		Dec.2021	June.2022	5882.00m
6	Sampling/Packing			
a)		April.2004	May.2007	19 BHs.
b)		Dec.2021	June.2022	15 Bhs.
7	Geophysical Logging			
		Dec.2021	July.2022	5869.86m

S.No.	Activity	Duration		Quantum
		From	To	
8	Hydrogeological studies			
9	Coal Core Analysis :			
a)		April.2004	May.2007	(19Bh's)
b)		Dec.2021	Aug.2022	187.48m (14Bh's)
10	Physico-mechanical	April.2022	Dec.2022	MSM-20 CSM I&II-12
11	Petrographic studies	Oct.2005 April.2022	Dec.2005	MSM-19 CSM I&II-10
12	Trace Element Test	Oct.2022		(1 BH) CSM I&II-11

Table 1.11.B
Number and Type of Analysis

Seam	Number of intersections	Band by band analysis	Overall proximate analysis on 60% RH & 40°C (>0.90M)		GCV K.cal./kg.	Ultimate analysis	AFT	HGI
			Determined	Calculated				
IV	27	20	10	-	10	-	-	-
IIIA	21	15	03	02	03	1	1	1
III Top	04	02	02	-	02	-	-	-
III Middle	04	02	01	-	01	-	-	-
III Bottom	04	02	03	-	03	-	-	-
IIICOMB	26	22	15	04	15	3	-	-
IIT	15	13	12	-	12	-	-	-
IIB	15	13	08	01	08	-	-	-
III Comb.	14	12	09	01	10	3	1	1
IB	28	26	22	03	22	2	1	1
IA	29	25	10	02	10	3	1	1

1.12 SCOPE AND LIMITATION

The report deals with the potentiality of the seams IV, IIIA, III Top, III Middle, III Bottom, III Comb., II Top, II Bottom, II Comb., IB, and IA. A wide spread occurrence of the thick mantle of soil is observed in the area. The geological information is largely based on sub surface data generated through drilling. The rocks present in the block are not compact in nature. This reflected at places by poor recovery obtained design drilling. As the some area in Marwatola Sector I&II block falls in either protected or Sohagpur Reserved Forest with rugged topography surveying and fixing of boreholes in the area was a very difficult task. The report is based on 35 boreholes which include 20 boreholes of MSM series and 15 boreholes of CSM I&II series. The seams in general are thin in nature therefore in most of the boreholes where the thickness of the seam is less than 0.50m the seams have not been subjected to band-by-band analysis.

Though all care has been taken to delineate precisely to delineate the incrop of the coal seams on the basis of boreholes data there is possibility of minor variation as the exact pattern of local weathering is difficult to ascertain.

Chapter - 2

EXPLORATION

2.1 SCHEME

GSI has involved in establishing existence of coal seams in Sohagpur Coalfield through mapping and regional drilling for a long time. The SKM, SGT, MSM series of boreholes and CSML & CSMS series of boreholes near Marwatola sector I&II Block drilled by GSI, MECL and CMPDIL respectively. The nearby blocks had established the continuity of seams towards Western side Ghungati and Panwari block, in southern side Marwatola Sector V&VI and Eastern side west of shahdol Block. In accordance with the exploration design to examine the quarriable potentiality in particular and the lay and disposition of older seams upto Seam I in general. The Marwatola Sector I&II block cover an area 21.03sq.km and 15 boreholes were drilled by CMPDIL (Throught outsourcing) involved 5882.00m of drilling.

The G-2 stage exploration in the block was initiated by CMPDIL in 15 Nos. of boreholes of CSM I&II series involving 5882.00m of drilling were done during the period of December'2021 to June'2022 and drilling was concluded on 17/06/2022. Earlier MSM series 20 Nos. of boreholes involving 6866.85m of drilling was done by MECL in the block boundary. Thus a total of 35 Nos. of boreholes with a total meterage of 12748.85m had been drilled by MECL and CMPDIL in the area. The coring boreholes of NQ size drilled in the block. Thus the density of the boreholes about 1.67 boreholes per sq.km.

Raniganj seam encounter in few boreholes .In barakar formation total 16 seams from Seam V to Seam I comb were established. Seams III & II have split in to III (Top), III (Middle), III (Bottom), II (Top) & II (Bottom) respectively. These have been referred as III Top, III Middle, III Bottom, III Comb, II Top, II Bottom and Ilcomb respectively in this report. Thus a total of 11 seams have been dealt with in this report. Raniganj seam has inconsistent occurrence throughout the block and

encountered in very few boreholes, also it has not attained workable thickness. Thus, Raniganj seam is not included in reserve estimation.

The core recovery in coal and non-coal is generally above 95% and 88% respectively barring exceptions. The depth range of boreholes varies from a minimum of 58.00m (MSM-49) to a maximum of 708.00m (CSM I&II-15).

2.2 TOPOGRAPHICAL SURVEY

2.2.1 GENERAL

The topographical contour map of 2m has been procured from Survey of India and ground truth has been verified and necessary corrections have been done. The boreholes have been surveyed using DGPS. The closed traverse survey in the area was carried out with reference to a base line Base-1 & Base-2. The coordinates of these two points are given below.

Table 2.2.1A

Coordinates of Base Pillars.

Survey Station	Coordinates						
	UTM			WGS84		Coal Grid	
	Easting(m)	Northing(m)	R.L(m)	Longitude	Latitude	X	Y
Base-1	523520.14	2579104.77	510.02	81° 13' 48.194" E	23° 19' 16.608" N	2717095.1557	1037604.3988
Base-2	527471.31	2580579.86	455.93	81° 16' 7.418" E	23° 20' 4.354" N	2721074.9264	1038997.9675

The area was contoured at 2m interval (Plate-II). The coordinates and reduced level of boreholes are furnished in Annexure-IB.

2.2.2 EQUIPMENT

In order to establish geographical (spherical) co-ordinates of the cardinal points of the block boundary, Differential Global Positioning System (DGPS) consisting of one SmartLink activated receiver is used in field survey.

Brief specifications of DGPS are provided in the table below.

Table Showing Details of Survey Instrument Used in DGPS Survey		
A	DGPS Instrument:	
	Make	Leica
	Model	GS16(Rover) and GS25(Base)
	Signal	GPS (L1, L2, L2C, L5), Glonass (L1, L2, L2C, L32), BeiDou (B1, B2, B32), Galileo (E1, E5a, E5b, Alt-BOC, E62), QZSS (L1, L2C, L5, L62), NavIC L53, SBAS (WAAS, EGNOS, MSAS, GAGAN), L-band
	Channels	555 (more signals, fast acquisition, high sensitivity)
	Measurement Performance & Accuracy:	
	Post Processed - Static (phase) with long observations Static and rapid static (phase)	 Hz 3 mm + 0.1 ppm / V 3.5 mm + 0.4 ppm Hz 3 mm + 0.5 ppm / V 5 mm + 0.5 ppm
	Real Time RTK- (Compliant to ISO17123-8 standard) Single baseline- Network RTK-	 Hz 8 mm + 1 ppm / V 15 mm + 1 ppm Hz 8 mm + 0.5 ppm / V 15 mm + 0.5 ppm
	Power:	
	Internal Battery	Exchangeable Li-Ion battery (2.6 Ah / 7.4 V)
	Communication:	
	USB and Bluetooth	USB and RS232 serial Bluetooth® v2.00 + EDR, class 2
B	Field Controller and Software	
	Leica Captivate software (Inbuilt Leica software for data recording)	Leica CS20 field controller

2.3 GEOLOGICAL MAPPING

Geological mapping has been carried out in the block however, most of the area is covered with weathered mantle, soil and forest cover there are scanty or lack of exposures. The area is mostly covered by Barakar and Raniganj formation.

2.4 REMOTE SENSING

2.4.1 GENERAL

Remote sensing techniques have not been used for generation and interpretation of geological data.

2.5 GEOPHYSICAL LOGGING

Geophysical logging was carried out by M/s APC drilling and construction Pvt Ltd. in 15 nos. of borehole in Marwatola Sector I&II, Sohagpur coalfield during the period from December 2021 to May 2022 for delineation of depth and thickness of coal seams as a supplementary to drilling borehole data.

2.5.1 INSTRUMENTS USED

The geophysical logs were recorded by Digital Geologging Micrologger-2 system, manufactured by M/s Robertson Geologging Ltd., Deganwy, U.K. It is a small and highly portable single unit logging equipment. It acts as an interface between the host PC, winch, cable assembly and the probes also called the sonde. The computer interface of the micro-logger is provided through a Universal Serial Bus (USB). It includes an advanced sonde interface which is able to communicate with all the available sonde without the requirement of an additional module.

For data acquisition, a Windows based Operating System (OS) is provided. The software records different data and then displays the log responses.

2.5.2. DETAILS OF BOREHOLES

All the 15 nos. of borehole have been geophysically logged in the block by APC Drilling & Construction Pvt. Ltd. The logging was carried out by using density and electrical probe. The details of boreholes logged are given in Table 2.7.2.

Table 2.5.2.: Details of Geophysically logged boreholes

Sl. No.	B.H. No.	Type of borehole	Drilling depth (m)	Logging depth (m)	Parameters recorded
1	CSM I&II-01	Coring	356	351.62	NGAM,SP,SPR, Long Normal, Short Normal, LSD, CALP, HRD
2	CSM I&II-02	Coring	292	292	NGAM,SP,SPR, Long Normal, Short Normal, LSD, CALP, HRD
3	CSM I&II-03	Coring	609	605.2	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL, INCL, AZIM, TVD
4	CSM I&II -04	Coring	351	350.34	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL
5	CSM I&II-05	Coring	350	350	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL
6	CSM I&II-06	Coring	367	367	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL
7	CSM I&II-07	Coring	343	343	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL
8	CSM I&II-08	Coring	343	343	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL
9	CSM I&II-09	Coring	417	417	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL
10	CSM I&II-10	Coring	255	255	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD,

Sl. No.	B.H. No.	Type of borehole	Drilling depth (m)	Logging depth (m)	Parameters recorded
					CAL
11	CSM I&II-11	Coring	363	363	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL
12	CSM I&II-12	Coring	438	436	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL
13	CSM I&II-13	Coring	321	320	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL
14	CSM I&II-14	Coring	369	368.7	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL
15	CSM I&II-15	Coring	708	708	SP,SPR,NGAM,Short/ Long Normal,LSD,HRD, CAL, INCL, AZIM, TVD

2.5.3 DATA INTERPRETATION

The geophysical logging of boreholes was carried out to identify the depth and thickness of coal seams encountered in the boreholes. The following parameters have been recorded.

- (i) Self-potential (SP)
- (ii) Single point resistance (SPR)
- (iii) Natural gamma (NGAM)
- (iv) Short normal resistivity (N 16")
- (v) Long normal resistivity (N 64")
- (vi) Density (DEN)
- (vii) Caliper (CALP)

The responses of various logs against coal seams are given in Table-2.5.3.

Table-2.5.3.: Response of various logs against coal seams

Geophysical logs	Response	Not valid when
Resistance/Resistivity	High resistance	(i) Highly resistive bed is adjacent to coal (ii) Heavy cavity in coal bed (iii) If the boreholes is cased
Natural gamma ray	Low gamma Intensity	(i) Very clean sand is adjacent to coal bed (ii) Coal bed is uranium accepted
Density	Low Density	(i)Seam is thinner than tool spacing

2.5.4. DISCUSSION OF RESULTS

The depth and thickness of various coal seams encountered in the borehole are interpreted by using Well-CAD software. The details of borehole wise coal seams are given in Table-2.7.5.

Table-2.5.4: Summarized results of geophysical logs of borehole

(1) CSM I & II-01

CSM I & II-01(P-11)			
Coring; DD-356.00 m;LD-351.62 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done up to 351.62 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
209.54	209.64	0.10	CARB SHALE

CSM I & II-01(P-11)			
Coring; DD-356.00 m;LD-351.62 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done up to 351.62 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
209.64	211.00	1.36	SANDSTONE
211.00	212.48	1.48	COAL
264.94	265.50	0.56	COAL
265.50	265.78	0.28	CARB SHALE
265.78	266.22	0.44	SHALY COAL
279.69	281.30	1.61	COAL
323.17	324.14	0.97	COAL
324.14	324.31	0.17	CARB SHALE
324.31	324.77	0.46	SHALY COAL
324.77	326.51	1.74	SHALE
326.51	328.55	2.04	COAL
341.09	342.90	1.81	COAL
342.90	344.86	1.96	SHALE
344.86	345.63	0.77	COAL
345.63	346.65	1.02	SHALE
346.65	346.91	0.26	CARB SHALE

CSM I & II-01(P-11)			
Coring; DD-356.00 m;LD-351.62 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done up to 351.62 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
346.91	348.08	1.17	SHALE
348.08	348.44	0.36	CARB SHALE
348.44	349.20	0.76	COAL

(2) CSM I & II-02

CSM I & II-02(P-10)			
Coring; DD-292.00 m;LD-292.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
120.21	120.39	0.18	CARB SHALE
136.40	137.07	0.67	COAL
170.55	170.78	0.23	CARB SHALE
170.78	171.22	0.44	SANDSTONE
171.22	171.57	0.35	CARB SHALE
176.78	177.24	0.46	CARB SHALE
177.24	178.14	0.90	COAL
186.74	186.99	0.25	CARB SHALE

CSM I & II-02(P-10)			
Coring; DD-292.00 m;LD-292.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
186.99	187.37	0.38	SHALY COAL
204.81	205.10	0.29	CARB SHALE
205.10	206.04	0.94	COAL
206.04	206.42	0.38	CARB SHALE
206.42	206.95	0.53	COAL
215.10	215.51	0.41	SHALY COAL
238.18	241.63	3.45	COAL
255.26	256.87	1.61	COAL
271.01	272.06	1.05	COAL
278.38	279.00	0.62	COAL
279.00	280.07	1.07	SHALE
280.07	280.89	0.82	COAL

(3) CSM I & II-03

CSM I & II-03 (P-16)			
Coring; DD-609.00 m;LD-605.20 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 605.20 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
75.47	75.74	0.27	CARB SHALE
82.61	82.89	0.28	CARB SHALE
115.13	115.77	0.64	SHALY COAL
115.77	115.96	0.19	CARB SHALE
115.96	116.73	0.77	SHALE
116.73	117.20	0.47	SHALY COAL
117.20	117.66	0.46	SHALE
117.66	118.10	0.44	COAL
118.10	118.32	0.22	SHALY COAL
118.32	118.74	0.42	COAL
146.58	147.71	1.13	COAL
147.71	148.59	0.88	SHALY COAL
148.59	148.81	0.22	CARB SHALE
148.81	149.03	0.22	SHALY COAL
149.03	149.50	0.47	COAL

CSM I & II-03 (P-16)			
Coring; DD-609.00 m;LD-605.20 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 605.20 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
149.50	149.69	0.19	SHALY COAL
149.69	150.07	0.38	CARB SHALE
150.07	150.68	0.61	COAL
150.68	151.04	0.36	CARB SHALE
151.04	151.58	0.54	SHALY COAL
151.58	152.00	0.42	COAL
152.00	152.24	0.24	SHALY COAL
152.24	153.89	1.65	COAL
153.89	154.11	0.22	SHALY COAL
154.11	154.47	0.36	CARB SHALE
154.47	155.03	0.56	COAL
155.03	155.58	0.55	SHALY COAL
155.58	155.88	0.30	COAL
155.88	156.24	0.36	SHALY COAL
156.24	156.73	0.49	COAL
156.73	157.64	0.91	SANDY SHALE

CSM I & II-03 (P-16)			
Coring; DD-609.00 m;LD-605.20 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 605.20 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
157.64	157.94	0.30	CARB SHALE
157.94	158.16	0.22	SHALE
158.16	159.81	1.65	COAL
159.81	160.66	0.85	CARB SHALE
160.66	160.88	0.22	SHALE
160.88	161.46	0.58	SHALY COAL
161.46	161.60	0.14	CARB SHALE
161.60	161.85	0.25	SHALE
161.85	162.53	0.68	COAL
183.97	184.38	0.41	SHALY COAL
184.38	184.66	0.28	CARB SHALE
184.66	184.88	0.22	SHALY COAL
184.88	185.62	0.74	COAL
185.62	185.76	0.14	CARB SHALE
185.76	186.33	0.57	SANDY SHALE
186.33	186.58	0.25	CARB SHALE

CSM I & II-03 (P-16)			
Coring; DD-609.00 m;LD-605.20 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 605.20 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
186.58	187.13	0.55	SANDY SHALE
187.13	187.68	0.55	SHALY COAL
187.68	187.96	0.28	SHALE
187.96	188.56	0.60	SHALY COAL
188.56	189.39	0.83	CARB SHALE
189.39	189.83	0.44	SHALY COAL
189.83	190.33	0.50	COAL
233.46	233.74	0.28	CARB SHALE
264.66	264.93	0.27	CARB SHALE
266.20	266.45	0.25	CARB SHALE
480.33	481.23	0.90	COAL
518.92	519.69	0.77	COAL
519.69	519.91	0.22	CARB SHALE
519.91	520.13	0.22	SHALE
520.13	520.37	0.24	CARB SHALE
545.40	545.65	0.25	CARB SHALE

CSM I & II-03 (P-16)			
Coring; DD-609.00 m;LD-605.20 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 605.20 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
545.65	546.09	0.44	SHALY COAL
584.38	586.17	1.79	COAL
589.33	589.60	0.27	CARB SHALE

(4) CSM I&II-04

CSM I&II-04 (P-6)			
Coring; DD-351.00 m;LD-350.34 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 350.34 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
200.47	201.68	1.21	COAL
239.89	240.55	0.66	SHALY COAL
254.60	255.32	0.72	CARB SHALE
255.32	255.67	0.35	SHALY COAL
255.67	256.17	0.50	SANDY SHALE
256.17	256.61	0.44	SHALY COAL

CSM I&II-04 (P-6)			
Coring; DD-351.00 m;LD-350.34 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 350.34 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
256.61	259.58	2.97	SANDSTONE
259.58	259.88	0.30	CARB SHALE
259.88	261.80	1.92	SANDSTONE
261.80	263.56	1.76	COAL
263.56	263.81	0.25	SHALY COAL
263.81	264.25	0.44	COAL
301.58	302.13	0.55	SHALY COAL
302.13	304.17	2.04	COAL
304.17	304.44	0.27	SHALY COAL
304.44	305.05	0.61	COAL
305.05	305.35	0.30	CARB SHALE
305.35	306.31	0.96	COAL
306.31	306.56	0.25	SHALY COAL
306.56	306.92	0.36	COAL
315.05	316.67	1.62	COAL
332.76	333.69	0.93	COAL

CSM I&II-04 (P-6)			
Coring; DD-351.00 m;LD-350.34 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 350.34 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
341.41	341.93	0.52	COAL
341.93	342.90	0.97	SHALE
342.90	343.72	0.82	COAL

(5) CSM I&II-05

CSM I&II-05 (P-6)			
Coring; DD-351.00 m;LD-350.34 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 350.34 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
200.47	201.68	1.21	COAL
239.89	240.55	0.66	SHALY COAL
254.60	255.32	0.72	CARB SHALE
255.32	255.67	0.35	SHALY COAL
255.67	256.17	0.50	SANDY SHALE
256.17	256.61	0.44	SHALY COAL
256.61	259.58	2.97	SANDSTONE

CSM I&II-05 (P-6)			
Coring; DD-351.00 m;LD-350.34 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 350.34 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
259.58	259.88	0.30	CARB SHALE
259.88	261.80	1.92	SANDSTONE
261.80	263.56	1.76	COAL
263.56	263.81	0.25	SHALY COAL
263.81	264.25	0.44	COAL
301.58	302.13	0.55	SHALY COAL
302.13	304.17	2.04	COAL
304.17	304.44	0.27	SHALY COAL
304.44	305.05	0.61	COAL
305.05	305.35	0.30	CARB SHALE
305.35	306.31	0.96	COAL
306.31	306.56	0.25	SHALY COAL
306.56	306.92	0.36	COAL
315.05	316.67	1.62	COAL
332.76	333.69	0.93	COAL
341.41	341.93	0.52	COAL

CSM I&II-05 (P-6)			
Coring; DD-351.00 m;LD-350.34 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 350.34 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
341.93	342.90	0.97	SHALE
342.90	343.72	0.82	COAL
342.23	342.67	0.44	SHALE
342.67	342.95	0.28	CARB SHALE

(6) CSM I&II-06

CSM I&II-06 (P-8)			
Coring; DD-367.00 m;LD-367.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
208.88	209.41	0.53	COAL
209.41	209.60	0.19	SHALY COAL
257.43	257.95	0.52	SHALY COAL
257.95	258.56	0.61	CARB SHALE
313.67	317.03	3.36	COAL
317.03	317.71	0.68	SHALE
317.71	318.68	0.97	COAL

CSM I&II-06 (P-8)			
Coring; DD-367.00 m;LD-367.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
332.32	334.21	1.89	COAL
348.95	350.05	1.10	COAL
356.78	357.20	0.42	SHALY COAL
359.72	360.55	0.83	COAL

(7) CSM I&II-07

CSM I&II-07 (P-2)			
Coring; DD-343.00 m;LD-343.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
198.55	199.12	0.57	COAL
235.49	235.90	0.41	SHALY COAL
235.90	236.62	0.72	SHALY SANDSTONE
236.62	237.55	0.93	COAL
251.27	252.32	1.05	COAL
252.32	252.87	0.55	SHALY COAL
252.87	253.44	0.57	SHALY SANDSTONE
253.44	253.66	0.22	CARB SHALE

CSM I&II-07 (P-2)			
Coring; DD-343.00 m;LD-343.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
253.66	254.02	0.36	SHALY COAL
254.02	256.38	2.36	SHALE
256.38	257.73	1.35	COAL
257.73	258.01	0.28	SHALY COAL
258.01	258.53	0.52	COAL
289.48	291.73	2.25	COAL
291.73	291.95	0.22	SHALY COAL
291.95	292.31	0.36	COAL
292.31	292.67	0.36	CARB SHALE
292.67	293.55	0.88	COAL
310.32	311.17	0.85	COAL
324.26	325.61	1.35	COAL
331.65	331.87	0.22	CARB SHALE
335.78	336.02	0.24	SHALY COAL
336.02	336.46	0.44	COAL

(8) CSM I&II-08

CSM I&II-08 (P-5)			
Coring; DD-343.00 m;LD-343.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
59.80	60.15	0.35	SHALY COAL
120.67	120.91	0.24	CARB SHALE
189.96	191.04	1.08	COAL
227.08	227.55	0.47	COAL
227.55	228.07	0.52	SANDY SHALE
228.07	228.73	0.66	COAL
241.21	241.65	0.44	SHALY COAL
241.65	241.95	0.30	SHALE
241.95	242.14	0.19	CARB SHALE
252.57	253.47	0.90	COAL
253.47	253.86	0.39	CARB SHALE
253.86	254.52	0.66	COAL
254.52	254.71	0.19	CARB SHALE
254.71	255.20	0.49	COAL
287.50	288.08	0.58	COAL
288.08	288.41	0.33	SANDY SHALE
288.41	288.85	0.44	SHALY COAL

CSM I&II-08 (P-5)			
Coring; DD-343.00 m;LD-343.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
288.85	291.35	2.50	COAL
291.35	293.00	1.65	SHALE
293.00	293.85	0.85	COAL
306.77	307.10	0.33	SHALY COAL
307.10	308.67	1.57	COAL
325.80	326.60	0.80	COAL
332.86	333.11	0.25	CARB SHALE
335.64	336.08	0.44	SHALY COAL
336.08	336.74	0.66	COAL

(9) CSM I&II-09

CSM I&II-09 (P-15)			
Coring; DD-417.00 m;LD-417.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
30.22	30.47	0.25	CARB SHALE
36.57	36.87	0.30	CARB SHALE
158.13	159.95	1.82	COAL

CSM I&II-09 (P-15)			
Coring; DD-417.00 m;LD-417.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
167.53	168.96	1.43	COAL
182.60	183.40	0.80	COAL
190.69	191.02	0.33	CARB SHALE
193.38	194.56	1.18	COAL

(10) CSM I&II-10

CSM I&II-10 (P-12)			
Coring; DD-255.00 m;LD-255.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
96.81	97.41	0.60	COAL
130.97	131.19	0.22	CARB SHALE
131.19	132.43	1.24	SANDSTONE
132.43	132.65	0.22	CARB SHALE
141.14	141.72	0.58	COAL
141.72	141.99	0.27	CARB SHALE
141.99	142.46	0.47	SHALY COAL
142.46	143.61	1.15	COAL

CSM I&II-10 (P-12)			
Coring; DD-255.00 m;LD-255.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
143.61	143.83	0.22	CARB SHALE
161.07	161.62	0.55	COAL
161.62	162.12	0.50	CARB SHALE
162.12	162.42	0.30	SHALY COAL
162.42	163.02	0.60	COAL
195.83	197.34	1.51	COAL
197.34	197.59	0.25	SHALY COAL
197.59	197.97	0.38	COAL
197.97	198.33	0.36	CARB SHALE
198.33	198.96	0.63	COAL
212.40	212.97	0.57	COAL
212.97	213.28	0.31	SHALE
213.28	214.90	1.62	COAL
228.84	229.69	0.85	COAL
236.86	237.14	0.28	CARB SHALE
239.83	240.22	0.39	SHALY COAL
240.22	240.44	0.22	CARB SHALE
240.44	241.07	0.63	COAL

(11) CSM I&II-11

CSM I&II-11 (P-7)			
Coring; DD-363.00 m;LD-363.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
207.37	208.94	1.57	COAL
249.95	250.33	0.38	CARB SHALE
250.33	250.47	0.14	SHALE
250.47	251.13	0.66	COAL
262.68	263.50	0.82	CARB SHALE
275.96	277.83	1.87	COAL
319.06	320.79	1.73	COAL
322.80	324.53	1.73	COAL
334.21	336.03	1.82	COAL
337.95	338.50	0.55	SHALY COAL
342.92	343.17	0.25	CARB SHALE
343.17	344.38	1.21	SHALE
344.38	344.85	0.47	CARB SHALE
344.85	346.03	1.18	COAL
353.43	353.87	0.44	SHALY COAL
353.87	355.11	1.24	SHALE
355.11	356.23	1.12	COAL

(12) CSM I&II-12

CSM I&II-12 (P-1)			
Coring; DD-438.00 m; LD-436.00 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 436.00 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
10.64	10.86	0.22	CARB SHALE
27.17	27.42	0.25	CARB SHALE
213.80	214.60	0.80	COAL
239.28	239.64	0.36	CARB SHALE
239.64	240.22	0.58	SHALE
240.22	240.39	0.17	CARB SHALE
262.90	263.25	0.35	SHALY COAL
263.25	264.05	0.80	COAL
264.05	265.65	1.60	SHALY SANDSTONE
265.65	267.32	1.67	COAL
267.32	267.73	0.41	SHALY COAL
267.73	268.20	0.47	SHALY SANDSTONE
268.20	269.00	0.80	COAL
269.00	270.81	1.81	SHALE
270.81	271.14	0.33	SHALY COAL

CSM I&II-12 (P-1)			
Coring; DD-438.00 m; LD-436.00 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 436.00 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
271.14	271.94	0.80	COAL
271.94	272.33	0.39	CARB SHALE
272.33	272.68	0.35	SHALY COAL
272.68	273.32	0.64	COAL
273.32	273.51	0.19	SHALY COAL
273.51	274.19	0.68	COAL
309.47	309.74	0.27	SHALY COAL
309.74	313.04	3.30	COAL
313.04	313.64	0.60	SANDSTONE
313.64	314.91	1.27	COAL
326.10	327.01	0.91	COAL
327.01	328.00	0.99	SANDY SHALE
328.00	328.69	0.69	COAL
328.69	329.04	0.35	CARB SHALE
329.04	329.73	0.69	COAL
333.28	334.54	1.26	COAL

CSM I&II-12 (P-1)			
Coring; DD-438.00 m; LD-436.00 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 436.00 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
341.53	341.80	0.27	CARB SHALE
344.82	344.96	0.14	CARB SHALE
344.96	345.23	0.27	SANDY SHALE
345.23	345.67	0.44	SHALY COAL
345.67	346.30	0.63	SHALE
346.30	346.52	0.22	CARB SHALE
369.92	370.12	0.20	CARB SHALE
370.12	371.22	1.10	SANDY SHALE
371.22	371.35	0.13	CARB SHALE
371.35	371.54	0.19	SHALE
371.54	371.74	0.20	CARB SHALE

(13) CSM I&II-13

CSM I&II-13 (P-4)			
Coring; DD-321.00 m;LD-320.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
201.95	202.45	0.50	SHALY COAL
212.60	213.06	0.46	SHALY COAL
213.06	213.48	0.42	COAL
213.48	213.83	0.35	SHALY COAL
223.04	223.86	0.82	COAL
223.86	224.14	0.28	CARB SHALE
224.14	224.85	0.71	COAL
224.85	225.04	0.19	CARB SHALE
225.04	225.48	0.44	SHALE
225.48	226.25	0.77	SHALY COAL
286.68	286.96	0.28	CARB SHALE
286.96	288.22	1.26	COAL
303.80	304.49	0.69	COAL
312.30	312.58	0.28	CARB SHALE
312.58	313.57	0.99	SHALE
313.57	313.95	0.38	CARB SHALE

CSM I&II-13 (P-4)			
Coring; DD-321.00 m;LD-320.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
313.95	314.45	0.50	SANDY SHALE
314.45	314.64	0.19	CARB SHALE

(14) CSM I&II-14

CSM I & II-14 (P-9)			
Coring; DD-369.00 m;LD-368.70 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 368.70 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
204.82	205.07	0.25	CARB SHALE
207.25	207.96	0.71	COAL
250.79	251.34	0.55	COAL
260.90	261.78	0.88	COAL
261.78	261.95	0.17	CARB SHALE
320.43	322.35	1.92	COAL
322.35	322.87	0.52	SANDY SHALE
322.87	324.03	1.16	COAL
340.07	340.54	0.47	SHALY COAL

CSM I & II-14 (P-9)			
Coring; DD-369.00 m;LD-368.70 m			
Geophysical parameters recorded: Due to collapsing nature of borehole the GPL done upto 368.70 m.			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
340.54	341.89	1.35	COAL
354.15	354.37	0.22	CARB SHALE
354.37	354.59	0.22	SHALY COAL
354.59	355.78	1.19	COAL
361.33	361.82	0.49	CARB SHALE
361.82	363.28	1.46	SHALY SANDSTONE
363.28	363.77	0.49	SHALY COAL
363.77	363.97	0.20	SHALE
363.97	364.65	0.68	SHALY COAL

(15) CSM I&II-15

CSM I&II-15 (P-14)			
Coring; DD-708.00 m;LD-708.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
71.12	71.43	0.31	CARB SHALE
74.39	74.81	0.42	SHALY COAL

CSM I&II-15 (P-14)			
Coring; DD-708.00 m;LD-708.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
80.15	80.89	0.74	COAL
86.36	86.71	0.35	CARB SHALE
124.54	125.07	0.53	COAL
125.07	125.31	0.24	SHALY COAL
125.31	125.62	0.31	SHALE
125.62	126.03	0.41	COAL
126.03	126.41	0.38	SHALY COAL
126.41	126.85	0.44	COAL
126.85	127.04	0.19	SHALY COAL
155.52	156.79	1.27	COAL
156.79	157.06	0.27	SHALY COAL
157.06	157.36	0.30	CARB SHALE
157.36	157.94	0.58	COAL
157.94	158.57	0.63	CARB SHALE
158.57	159.07	0.50	COAL
159.07	159.34	0.27	SHALE
159.34	159.86	0.52	SHALY COAL
159.86	162.25	2.39	COAL

CSM I&II-15 (P-14)			
Coring; DD-708.00 m;LD-708.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
162.25	162.80	0.55	CARB SHALE
162.80	163.33	0.53	COAL
163.33	163.96	0.63	SHALY COAL
163.96	164.48	0.52	COAL
164.48	164.92	0.44	SHALY COAL
164.92	165.58	0.66	COAL
165.58	166.14	0.56	SHALE
166.14	167.07	0.93	CARB SHALE
167.07	167.73	0.66	COAL
167.73	168.78	1.05	SHALY COAL
168.78	169.13	0.35	CARB SHALE
169.13	169.52	0.39	SHALY COAL
169.52	169.82	0.30	CARB SHALE
169.82	170.45	0.63	SHALY COAL
170.45	170.76	0.31	CARB SHALE
170.76	171.25	0.49	COAL
171.25	171.50	0.25	CARB SHALE
171.50	172.02	0.52	SHALY COAL

CSM I&II-15 (P-14)			
Coring; DD-708.00 m;LD-708.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
192.08	192.82	0.74	SHALY COAL
192.82	193.15	0.33	CARB SHALE
193.15	193.98	0.83	COAL
193.98	194.20	0.22	CARB SHALE
194.20	194.83	0.63	SANDSTONE
194.83	195.10	0.27	CARB SHALE
195.10	195.85	0.75	SANDSTONE
195.85	196.34	0.49	SHALY COAL
196.34	196.48	0.14	SANDSTONE
196.48	197.19	0.71	SHALY COAL
197.19	197.74	0.55	CARB SHALE
197.74	198.16	0.42	SHALY COAL
198.16	198.60	0.44	COAL
198.60	200.42	1.82	SANDY SHALE
200.42	201.27	0.85	SHALY COAL
201.27	201.52	0.25	CARB SHALE
201.52	201.90	0.38	SHALY COAL
201.90	202.12	0.22	CARB SHALE

CSM I&II-15 (P-14)			
Coring; DD-708.00 m;LD-708.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
202.12	202.62	0.50	SHALY COAL
206.38	206.62	0.24	CARB SHALE
232.74	233.26	0.52	CARB SHALE
240.97	241.35	0.38	SHALY COAL
241.35	241.52	0.17	CARB SHALE
269.47	269.83	0.36	SHALY COAL
337.59	337.84	0.25	CARB SHALE
482.63	482.77	0.14	CARB SHALE
594.60	596.19	1.59	COAL
618.19	618.54	0.35	CARB SHALE
649.39	650.05	0.66	COAL
678.48	678.78	0.30	CARB SHALE
682.90	683.34	0.44	SHALY COAL
683.34	683.92	0.58	SHALE
683.92	684.60	0.68	COAL
690.85	691.40	0.55	SHALY COAL
691.40	692.25	0.85	COAL
698.90	699.15	0.25	SHALY COAL

CSM I&II-15 (P-14)			
Coring; DD-708.00 m;LD-708.00 m			
Top Depth	Bottom Depth	Thickness	Lithology
m	m	m	
699.15	700.03	0.88	COAL
700.03	700.22	0.19	SHALY COAL
700.22	700.47	0.25	CARB SHALE
700.47	701.05	0.58	COAL

The geophysical logs along with interpreted results of all the 15 boreholes are given in Plate XI.

2.5.5. CONCLUSION

The report has been prepared on the basis of geophysical logging of boreholes with parameters Natural gamma, resistance, resistivity (including both Short normal & Long normal), SP, Density. The resistivity values over a coal seam vary in very wide range due to quality of coal and moisture contact. Geophysical correlations were also done with the different seams of Marwatola Sector I&II.

2.6 DRILLING

2.6.1 EQUIPMENT

The exploration was initiated by wireline diamond core drills, PRD-04, PRD-06 PRD-09 & PRD-13 with a capacity of 1000m in NQ size. Truck mounted Hydrostatic rigs VEL-200-01 and VEL-200-02 of Velson make, with 2000m capacity, were also deployed to cover the drilling of deep seated coal seams. The drilling was carried out by using rock roller, tungsten- carbide and diamond bits depending upon the type of strata.

2.6.2 DRILLING CONDITIONS

The rocks present in the block are in general not compact. In few borehole reddish sandstone are encountered of raniganj formation. Fine to coarse grained loosely cemented sandstones are found in the weathered zone. Drilling difficulties due to clay formation & water loss etc. were overcome by using bentonite mud.

2.6.3 CORE RECOVERY

In coal the recovery percentage is generally more than 90% and in the non-coal strata it is found mostly more than 80%. The poor recovery in the some boreholes are due to bad strata condition and faults.

2.6.4 BOREHOLE DRILLED

Completed boreholes : 35

Table 2.6.4.A financial year-wise break up of drilling carried out in Marwatola Sector I&II Block.

Agency	Financial year	Drilling	
		No. of Bhs.	Meterage
MECL	2005-06	11	3879.05
	2006-07	9	2987.80
CMPDIL	2020-21	3	1956.00
	2021-22	12	3926.00
Total		35	12748.85

2.7 CORE LOGGING/SAMPLING/PACKING/BOXING

2.7.1 LOGGING

During the detailed exploration in Marwatola Sector I&II Block, a total of 5882.00m of core (Coal 187.48m & Non-coal 5694.52m) have been generated in boreholes CSM-01 to CSM-15. Earlier MECL 6866.85m core was generated and logged. The borehole core was megascopically examined and logged systematically. Physical characteristics (colour, grain size, etc.), mineralogical composition and structural details (core dip, fractures, joints, slickensides, etc.) were recorded in detail. The extrapolation was done against loss of recovery, if any. The processing of the data was done through CEMPGEODOC package to obtain descriptive and graphic litholog which are presented in Annexure-IIA and Plate-IV respectively. The visually recorded roof and floor depths of seams were reconciled on the basis of band by band analysis.

2.7.2 SAMPLING

Completion of each run, i.e 3.00m, of drilling, the core is removed from inner tube of core barrel and systematically laid over the corrugated galvanized iron sheets. The core of each run is marked with wooden pieces mentioning the depth of the core. All the carbonaceous and non carbonaceous horizons viz, coal, shaly coal, carbonaceous shale and obvious dirt bands within seam, if any, as well as non-carbonaceous horizons, viz, sandstones etc., are visually identified and logged. The coal, shaly coal, carbonaceous shale low & High and obvious dirt bands within the seam are systematically sampled.

2.7.3 PACKING AND BOXING

Carbonaceous horizons, viz, coal, shaly coal, carbonaceous shale and obvious dirt bands within seam (less than 1.00m) were picked up for dispatching for chemical analysis. These were properly placed in wooden core boxes with proper labelling of borehole name, sample numbers, depth, top and bottom, roof and floor of coal seams etc.

2.7.4 BOXING AND DISPATCH

After proper packing, each core boxes was serially numbered and suitably identified for each borehole. Care has been taken for properly nailing the boxes with lids, to avoid loss of core during transportation. The packed core boxes were sent to Coal Characterisation Lab CMPDIL HQ, Ranchi, in different lots for chemical analysis.

2.8 COAL ANALYSIS

2.8.1 BAND BY BAND ANALYSIS

On the basis of analytical results, viz, moisture & ash percentage different lithounits are defined as under:

Coal	: Ash + Moisture upto 40%.
Shaly Coal	: Ash + Moisture from >40.1% to 55%.
Carbshale low	: Ash + Moisture from >55.1% to 65%.
Carbshale high/ combustible	
dirt band	: Ash + Moisture >65.1% to 75%.
Obvious dirt bands	: Ash + Moisture > 75%.

2.8.2 OVERALL PROXIMATE ANALYSIS AND GROSS CALORIFIC VALUE (GCV)

After receiving the analytical results, against band by band advice of the coal cores, proximate analysis and GCV determination are advised on 60% RH and 40°C. For high moisture coals a maximum of 50% of boreholes of the Block covering all seams are advised on Bcs, I₃₀ and I₁₀₀, basis which are defined as under:

The analyses were obtained for Bcs, I₃₀ and I₁₀₀ basis which are defined as under

Bcs Sample : only coal and shaly coal and carbonaceous shale low.

I₃₀ Sample : BCS sample + combustible and non-combustible bands upto a thickness of 30cm. This type of sampling has been carried out for assessing potentially under ground reserves.

I₁₀₀ Sample : Bcs sample + high carbonaceous shale ((ash + moisture) upto 75%) bands + grey shale ((ash + moisture) >75%) bands, together upto 1m in thickness, excluding all other obvious bands such as sandy shale, shaly sandstone, sandstone etc. irrespective of their thickness.

2.8.3 ULTIMATE ANALYSIS, ASH ANALYSIS AND ASH FUSION TEMP. RANGE

These special tests are advised for 10% of boreholes (CSM I&II-05 & 15), drilled in the block, for I₃₀ & I₁₀₀ Samples, for which Overall Proximate Analysis are available.

2.8.4 HGI AND TOTAL SULPHUR & SULPHUR DISTRIBUTION

The special tests were advised for BCS samples for which Overall Proximate Analysis were available. A total of 5% of boreholes (CSM I&II-05 & 15), of the block were advised for ultimate analysis and Total Sulphur analysis.

2.8.5 PETROGRAPHY AND PHYSICO-MECHANICAL PROPERTIES

The special test advised for Petrography for 1 borehole (CSM I&II-10) and Physico-Mechanical Properties for 1 borehole (CSM I&II-12) for analysis.

The details of determination of various tests are listed in Table 2.8

Table 2.8:
Details of Coal Analysis of borehole drilled in Marwatola Sector I&II Block
Sohagpur Coalfield.

S. No.	Parameter	Nos. of BH	Agency	Indian Standards
1	Proximate analysis	14	Coal Characterisation Lab CMPDIL Ranchi.	IS 1350 (Part I;1970)
2	GCV	14	Coal Characterisation Lab CMPDIL Ranchi.	IS 1350 (Part II;1970)
3	Ultimate analysis	2	Coal Characterisation Lab CMPDIL Ranchi.	IS 1351 (1959)
4	HGI	2	Coal Characterisation Lab CMPDIL Ranchi.	IS 4433 (1967)
5	Ash Fusion Temp.	2	Coal Characterisation Lab CMPDIL Ranchi.	IS 1355 (1959)
6	Ash Analysis	2	Coal Characterisation Lab CMPDIL Ranchi.	IS 1355 (1959)
7	Petrography	1	Coal Petrography Lab. CMPDIL	
8	Trace Element Analysis	1	Mineral Exploration and Consultancy Limited	
9	Physico-Mechanical Properties	1	Indian institute of Technology, Kharagpur W.B.	

2.9 DATA SYNTHESIS

The interpretation of the geology and structure of Marwatola Sector I&II Block has been done mainly based on sub-surface data generated through boreholes. The study of physical characteristics, delineation and disposition of dirt bands in the seams etc. have been carried out on the basis of megascopic study of the borehole cores while the nature of combustible dirt bands, coal quality etc. have been established through chemical analysis of the cores.

Chapter – 3

GEOLOGY & STRUCTURE

3.1 REGIONAL GEOLOGY

3.1.1 GENERAL

The Sohagpur Coalfield, as described by Fox (1936), extends from near Ghunghuti in the west to the Hasia nala on to the east, for a length of about 103 km. In the north the field extends upto Jaitpur (23° 29': 81° 44'), while the southern margin runs along the latitude near the confluence of Kewai and Tipons river with the Son river and attaining a maximum width of 45 km. The spread of Sohagpur Coalfield is about 3100 sq.km.

The Sohagpur Coalfield can be broadly divided into two parts, i.e. the North and South Sohagpur, along the E - W trending intrabasinal Bahmni Chilpi fault. The North Sohagpur represents the down thrown block of the Sohagpur basin where coal measures i.e. Raniganj and Barakar are surmised to occur at depth, under a thick cover of younger Upper Triassic sediments termed as Supra Barakars.

The coal horizons in South Sohagpur are either exposed or occur under a thin cover of younger sediments; except in the southwest where it is overlain by a thick Lameta Formation and Traps. In South Sohagpur, three sub basins namely Jhagrakhand, Jamuna Kotma & Amlai Burhar, from east to west, have been recognised. The intervening patches between the sub basins were considered devoid of workable coal seams. However, recent exploration in Amadand area by CMPDI has conclusively established workable coal deposit in between Jhagrakhand and Jamuna Kotma sub basin.

Sohagpur Coalfield has been a traditional source of superior quality, low rank, high volatile coals. Such coals have been under active exploitation in the entire tract in South Sohagpur. It was in late'79 and early'80 that existence of high rank caking coals was established in Beheraband area, located in the eastern part of North Sohagpur. Since then, such high rank coals have been located in Urtan,

Keshwai, Pakaria etc. all located along a linear tract to the north of Bahmni Chilpi fault. Sharply contracting occurrence of high rank coals across above intrabasinal dislocation has probably been caused by emplacement of large volume of magma in the area which is well manifested by the presence of large basic bodies in North Sohagpur. This provided the required heat for increase in coal rank and development of caking properties commensurate to the rank. Wherever the heat flow was more, on account of nearness of the intrusive body to coal beds, a depletion of volatile matter and loss of caking properties have been observed.

3.1.2 STRATIGRAPHY

The general stratigraphic succession for the entire coalfield revealed as a result of regional and detailed exploration is as follows:

Table 3.1.2A:
Stratigraphic sequence of Sohagpur Coalfield

Age	Formation	Thickness(m)	Lithology
Recent/Sub-Recent	Alluvium & soil		
Eocene-Upper Cretaceous	Deccan Trap		Effusive and intrusive rocks, mostly basalt & dolerite
Upper Cretaceous	Lameta		Reddish & greyish sandstone & nodular limestones
Late Permian	Raniganj		Heterolith, gray shale, fine to medium grained ripples and cross laminated micaceous sandstone.
Early Late Permian	Barren measure		Greyish white, medium to coarse grained sandstone, green and reddish brown, mottled clay/ shale often with ferruginous concretion.
~~~~~ Unconformity ~~~~~			
Late Early Permian	Barakar	+320m	Coarse to medium grained white to greyish white arkosic to sub arkosic sandstones, with minor shales and coal seams.
~~~~~ Unconformity ~~~~~			
Early Permian	Talchir	+28m	Diamictite, siltstone, sandstone & needle shales.
~~~~~ Unconformity ~~~~~			
Precambrian			Porphyritic granite gneisses with pigmatite veins.

### **3.1.3 DESCRIPTION OF FORMATIONS**

A brief description of various rock formations in Sohagpur Coalfield is furnished below:

#### **Precambrian Archaean**

The Precambrian basement rocks are not exposed in the vicinity of the coalfield. But these rocks crop out as inliers within the Talchir Formation south of the Son river and also in the Hasdo river section east of Manendragarh. The predominant rock type is pink porphyritic gneiss. Archaean comprises of porphyritic granite-gneiss with aplite and pegmatite veins.

#### **Talchir Formation**

Rocks of Talchir Formation lie unconformably over the archaea's. The Talchir are exposed mainly in south and south east of the coalfield. The Talchir beds also continue beneath the Barakar Formation as evidenced by sub-surface data. It is in this coalfield that a thickness of over 430m of Talchir sediments has been recorded in borehole. Rocks of Talchir formation comprise of Khaki green to bluish green needle shale, clay and siltstone with fine to very coarse grained hard sandstone containing pebbles of granite-gneiss and quartzite.

#### **Barakar Formation**

The Barakar Formation shows characteristic vertical variation in the arrangement of lithofacies and this permits sub-dividing into three distinct litho stratigraphic units. The strata grouped within the Lower member conformably overlies the Talchir Formation. The Lower part of this member comprises greyish white, feldspathic multilayered sandstone units with minor amount of shale and is usually devoid of any workable coal seam. The sandstones are compact and moderately sorted with feldspar clasts and disseminations of garnet. This part of the Lower member shows a variation in thickness from 80m to 150m.

The overlying 200m to 300m thick strata define the upper part of the lower member. It includes several workable coal seams which are developed in Malga-Bijuri-Kotma-Dhanpuri area. The sandstone of this part are dominantly

felspathic and are thick bedded. Channeling is quite frequent and this has caused frequent scouring of the coal seams giving rise to multistory sandstone units.

Succeeding this coal bearing unit is a 200m thick sequence of ferruginous sandstones, shale and laminated siltstones which present a sharp lithological contrast with the underlying coal-bearing sequence. Outcrops ascribed to this unit occur down dip of Dhanpura area.

The upper member shows the reappearance of coal seams, which are however of inferior quality. The sandstones of the upper member are fine grained with micaceous siltstone and ferruginous shale bands. The thickness of this member is 300m to 500m.

### **Barren measure**

In the northern part of the Sohagpur Coalfield, a sequence of very coarsegrained, pebbly sandstones and ferruginous sandstones constitute the prominent ridge. Barren measure are restricted to western part of Amlai-Burhar sub-basin of South Sohagpur Coalfield. Barren measure overlie Barakar unconformably. The strata is characterised by pink, buff and red sandstones, with reddish clay and shale.

### **Lameta**

Lametas lie unconformably over Barren measure and comprises of poorly consolidated sandstones having colour reddish and greyish. Towards the top, nodular lime stone appearance. They are mainly observed in Amlai-Burhar sub-basin. Lametas generally occur as capping over Barren measure formation.

### **Raniganj**

Raniganj formation over the Barren measure formation. Variegated shale and appearance of grey shale/ carbonaceous shale mark the beginning of Raniganj sedimentation. It is represented by grey shale, Interlaminated, ripple marks and burrowed fine grained sandstone/siltstone and shale and fine to medium grained cross bedded sandstone and coal.

## **Soil and Alluvium**

The entire Sohagpur Coalfield is mainly capped with dark brown, brown, clayey soil and pinkish brown sandy soil. The thickness of alluvium generally ranges between 3 and 25m

### **3.1.4 INTRUSIVES**

The Barakar measures of Sohagpur Coalfield are traversed by large number of basic dykes and sills. The most prominent sill is noted in the eastern part of the coalfield. Westwards, the sill continues almost uninterruptedly over a strike length of about 70km across the river Son & outcrop attains a width of over 4km at places. The northern limit of this sill is apparently marked by the east-west trending Bhamni-Chilpa fault with down throw towards north. Another prominent dolerite body occurs to the north of the Son River and this extends over a length of over 50km.

Dykes of dolerite are quite common both in the eastern and western parts of the field. They vary in length from a few meters to over 5 km and in width from 2m to about 25m. Most of the dykes trend ENE-WSW to East-West. In a number of cases, the dykes are found to be aligned along prominent fault planes.

## **3.2 REGIONAL STRUCTURE**

### **3.2.1 TECTONIC SET-UP**

The Sohagpur coalfield constitutes the middle segment of the Rewa Gondwana basin. The structural set up of the Sohagpur basin is fairly simple. In conformity with the regional E-W alignment, the rock formation in general strike in an E-W to WNW-ESE direction with low northerly dips ( $1^{\circ}$  to  $4^{\circ}$ ). Small basins, resulting from rolling nature of the beds, are also observed. Local rolls of the strata are also common.

The Sohagpur basin is composed of four well defined sub-basinal structures. These include the (i) Jhagrakhand sub-basin to the east; (ii) Amadand (iii) Jamuna-Kotma sub-basins in the middle and (iv) Burhar-Amlai sub-basin to the west. Basinal

evolution in these sectors was probably controlled by greater subsidence in comparison to the adjoining areas with shallow Barakar cover.

### **3.2.2 BASIN CONFIGURATION**

The prominent structural feature of the Sohagpur basin is the system of ENE-WSW to E-W trending sub-parallel faults. The most important of these faults is the Bamhani-Chilpa fault, which runs along the middle part of the coalfield. The fault hade towards north and at places the drag on the beds is well seen. The throw of the fault is yet to be ascertained precisely but is likely to be over 400m at places. In addition to this major fault there are number of E-W or ENE-WSW trending faults. Two more sets of faults trend NW-SE and NE-SW, but the frequency of these faults is low. It can be summarized that the E-W trending faults were active periodically during sedimentation and the fault movement might have taken place over a protracted period of time. The NE-SW and NW-SE trending faults may be related to the major faulting along the east-west line.

A detailed study of the structural setup of the faults in the eastern part of the basin (Choudhury, 1974, 1977) indicates that most of the dolerite dykes have intruded along the fault planes in the area to the east of the Kulharia nala. In this part cross faults have at places displaced the dolerite dykes.

## **3.3 GENERAL GEOLOGY OF THE BLOCK**

### **3.3.1 LOCATION**

Marwatola Sector I&II Block is located in western part of Amlai-Burhar sub-basin, of Sohagpur coalfield. The sub-basin extends from Bairihia village in the east to perhaps below the intratrappean Lameta Formation and the Deccan Trap in the west. Its southern margin runs approximately south of Dhamni & Patna villages while the northern extent appears truncated against E-W trending Bahmni-Chilpi fault. Raniganj formation occur in northern side of Bahmni –Chilpi fault. The Amlai-Burhar sub-basin covers a total area of (+) 200 sq.km.

### 3.3.2 STRATIGRAPHY

The Amlai-Burhar sub-basin preserves about 300m of Barakar strata of this, the basal 160-200m of Barakar formation is devoid of workable coal seams. The workable coal seams are confined to the succeeding 100-140m Barakar Strata. It contains 9 correlatable coal horizons of which only six have attained workable thickness in one or the other part of the area. The Barakar strata in the deeper part of the sub-basin is covered by younger sediments. Lameta Formation occurs as thin cappings, both over Barakar and Barren measure. It has also been observed to girdle basic bodies.

The Marwatola Sector I&II Block is located in the Western part of the Amlai-Burhar sub-basin. The area is mainly covered by Raniganj in norther part and small patch is cover in western central and remaing part covered by rocks of Barren measure Formation and thick blanket of soil and alluvium. In few boreholes the correlatable coal horizons in Raniganj formation encountered in northern side of Bahmni-Chilpi fault. Rock exposures are largely confined to nala sections. The stratigraphic sequence of the block, established from surface and sub-surface data is given in Table-3.3.2A.

**Table 3.3.2.A**  
**Stratigraphic succession of Marwatola Sector I&II block, Sohagpur Coalfield**

Age	Formation	Lithology
Recent/Sub-recent	Soil & Alluvium (0.20m-6.00m)	Soil & Alluvium.
<----->UNCONFORMITY<----->		
Upper Permian	Raniganj 0.80m(CSM I&II-02) – 351.80(MSM-29)	Gray shale, fine to medium grained ripples and cross laminated micaceous sandstone.
<----->UNCONFORMITY<----->		
Middle Permian	Barren measure 60.33m(MSM-20) – 248.39(CSM I&II-15)	Pink, buff & red sandstone, red shale etc.
<----->Transitional to Uncomformable contact <----->		

Age	Formation	Lithology
Lower Permian	Barakar 39.00m(MSM-49) – 281.10(MSM-23)	Coarse to medium grained sandstone, sub-ordinate shales & coal seams.
Lower Permian to Upper carboniferous	Talchir (Not Fully intersected)	Diamictite sandstone, siltstones & needle shales.
<----->Transitional to Unconformable contact <----->		
Pre-Cambrian	Metamorphics	Porphyritic granite-gneiss with aplite and Pegmatite Veins.

### 3.3.3 DESCRIPTION OF FORMATIONS

The detailed exploration by CMPDIL was confined to talchier, Barakar, Barren Measure and Raniganj Formations only.

**Barakar Formation:** The coal bearing Barakar formation is chiefly composed of grayish white coarse and very coarse-grained sandstone, coal seams, carb shale, shale and sandy shales. Few coal seam occur in Raniganj formation.

The existence of 16 seams viz. Seams V (sporadically occurring), IV, L2 (sporadically occurring), IIIA, IIITop, IIIMiddle, IIBottom, IIIComb, IITop, IIBottom, IIComb, IB, IA, ITop, IBottom and IComb occur in Barakar Formation. The detailed lithology of the formation encountered in boreholes of Marwatola Sector I&II Block is given in Annexure IIA.

**Barren measure:** The overlying Barren measure consist of yellow brown to blue green clay with ferruginous bands at places.

**Raniganj Formation:** Raniganj formation is consist of gray shale, interlaminated, rippled and burrowed fine grained sandstone /siltstone and shale (heterolith) and fine to medium grained cross bedded sandstone and Coal. Coarse grained sandstone unit are also common. Cross bedded Sandstone unit are 2m to 15m thick and occasionally contain gray clay clasts/chips.

## **3.4 GEOLOGICAL STRUCTURE**

### **3.4.1 GENERAL**

As already stated, the interpretation of the geological structure of Marwatola Sector I&II block has been done mainly on the basis of borehole data, since the surface exposure of rocks is patches and the area is mostly covered by soil and vegetation.

Structurally the northern part of the block is less disturbed, whereas, the central part of the block is disturbed by Bahmni-Chilpi fault and southern part traversed by a number of faults.

### **3.4.2 STRIKE AND DIP**

The strata including coal seams exhibit a broad NE-SW trend in the south-western part of the block exhibits a rolling strike. The strata including coal seams show a broad NE-SW trend.

The strata in general show a northwesterly dip. The general dip of the strata varies from 2°-3° but near and in between the faults, the dip of the strata varies from 6°-8°.

### **3.4.3 PATTERN OF FAULTING**

Six faults have been interpreted in the block. These faults are numbered as F₁-F₁ to F₆-F₆ from north to south and fault strike is almost E-W. All the faults F₁-F₁ to F₆-F₆ in the Marwatola Sector-I&II block is continuing in the adjacent Marwatola Sector-III&IV block. This fault F₁-F₁, F₃-F₃ to F₆-F₆ continues in to Ghunghuti Block. The details of faults interpreted in Marwatola Sector I&II Block are listed in the table 3.5.3.A.

Since the area shows variable dip at a few places, the possibility of occurrence of minor faults cannot, however, be ruled out.

**Table 3.5.3A: Details of Faults in Marwatola Sector I&II Block, Sohagpur coalfield.**

Sl.No	Fault No.	Trend	Throw		Evidences
			Direction	Amount(m)	
1	F ₁ -F ₁	W-NNE	Southerly throw	Varies from 10-30m	i) Difference in FRL on either side of the fault.
2	F ₂ -F ₂	E-W	Southerly throw	0 to 2m	i) Difference in FRL on either side of the fault.
3	F ₃ -F ₃	E-W to SE	South-easterly throw	30 to 60m	i) Difference in FRL on either side of the fault.
4	F ₄ -F ₄	NW-SE	Southerly throw	50 to 100m	i) Difference in FRL on either side of the fault.
5	F ₅ -F ₅	NW-SE	Southerly throw	350 to 370m	i) Difference in FRL on either side of the fault.
6	F ₆ -F ₆	Almost E-W	Northerly throw	290 to 660m	i) Difference in FRL on either side of the fault. ii) The Boreholes CSM I&II-3, 9 have been faulted.

## Chapter-4

### DESCRIPTION OF COAL SEAMS

#### 4.1 GENERAL

The existence of 16 seams including splits and local seams well defined coal seams of Barakar Formation namely seam V, IV, L2, IIIA, III Top, III Middle, III Bottom, III Comb., II Top, II Bottom, II Comb., IB, IA, I Top, I Bottom and I Comb. in descending order have been proved in Marwatola Sector I&II block of Sohagpur Coalfield, in which of 11 seams IV, IIIA, III Top, III Middle, III Bottom, III Comb., II Top, II Bottom, II Comb., IB and IA have been found to attain workable seam thickness in the major part of the block. (In few patch of Raniganj seam encounter) The focus of documentation/ interpretation has been confined only within the workable zone of the block.

The coal seams are mainly composed of Coal and Shaly Coal, Carbonaceous Shale Low with thin bands of Carbonaceous Shale High at places. The coal is generally dull in appearance, high in moisture, low in Sulphur (< 1.00%) and low in rank i.e. non-coking type.

For delineation of coal seam, the standard norms applicable to non-coking coal have been followed. In brief they are as under:

Coal	:	Ash% + Moisture% upto 40%.
Shaly Coal	:	Ash% + Moisture% > 40% upto 55%.
Carbonaceous Shale Low:		Ash% + Moisture% > 55% upto 65%.
Carbonaceous Shale High:		Ash% + Moisture% > 65% upto 75%.
Obvious dirt bands	:	Ash% + Moisture% exceeding 75%.
(Grey shale, sandy shale and sandstone)		

Both the Coal, Shaly Coal and Carbonaceous Shale Low are grouped together to form a seam. While delineating the coal seam, the carbonaceous shale high occurring either at the roof or the floor has been excluded from the coal seam. The carbonaceous shale high is (less than 0.30m) treated as combustible dirt band, while the shale is treated as non-combustible dirt band. The in-seam dirt band has been considered as the part of the seam only in those cases where its thickness is less than or equal to the Coal / Shaly coal/ Carbonaceous Shale Low immediately overlying or underlying that dirt band.

The coal bearing Barakar formation are mostly covered by soil, as a result the area is mostly devoid of any coal exposures.

## 4.2 CORRELATION

The coal seams intersected in various boreholes in Marwatola Sector I&II block have been correlated on the basis of their thickness, physical make-up, quality, inter-seam partings and associated lithology's. The graphic litholog of the boreholes are given in Plate IV and the correlation of the coal seams is given in Annexure-III and Plate V.

## 4.3 QUALITY PARAMETER

All the seams are generally either free of dirt bands or a few thin bands are present. Band-by-band analysis of the seam is given in Annexure-IV and details of dirt bands present in the seams are given in Annexure-V. The coals present in the block are high moisture, moderate ash, low rank & non-coking coal.

The results of special tests i.e., Ash Fusion Temperature, Ultimate analysis, Ash analysis, HGI, etc. are presented in Annexure-VII-A. The determined and calculated proximate analysis on 60% RH & 40°C are presented in Annexure-VI. The seam wise average  $M_{100}$  values utilized for calculating analysis on equilibrated basis.

### 4.3.1 $M_{100}$ CALCULATION

$M_{100}$  (moisture related to 100 parts of pure dry coal) value of a particular coal seam can be calculated from the data of boreholes, where determined seam overall proximate analysis on 60% RH & 40°C is available, by using following formula:

$$M_{100} = \frac{M \times 100}{100 - (1.1 A \% + M \%)}$$

Where, M = Moisture%, A = Ash%, determined on 60% RH & 40°C as per IS 1350, Part I, 1984.

#### 4.3.2 WEIGHTED AVERAGE OVERALL CALCULATION

Weighted average of moisture and ash contents of the coal seam from band by band analysis (as available on air dried basis) were calculated as follows: -

$$\text{Average Moisture\%} = \frac{\sum (\text{Weight} \times \text{Moisture \%})}{\sum \text{Weight}} = M$$

$$\text{Average Ash\%} = \frac{\sum (\text{Weight} \times \text{Ash \%})}{\sum \text{Weight}} = A$$

Where moisture%, ash% and weight are of determined band by band proximate analysis as per IS 1350, Part I, 1984.

#### 4.3.3 CONVERSION OF CALCULATED OVERALL TO EQUILIBRIATED OVERALL.

If weighted average moisture and ash contents on air-dried basis, as obtained above, are M and A respectively.

$$\text{Then, Mineral Matter} = 1.1 \times A = MM$$

$$\text{and Pure coal} = 100 - (1.1 \times A + M) = PC$$

On air-dried basis (i.e., 100 - M),

MM

$$\text{Mineral Matter \%} = \frac{\text{-----}}{100 - M} \times 100 = a$$

$$\text{Pure coal \%} = \frac{PC}{100 - M} \times 100 = b$$

Since 100 gram of pure coal is associated with  $M_{100}$  gram of moisture, hence, b gram of pure coal is associated with

$$\text{Moisture} = \frac{M_{100}}{100} \times b = c$$

Thus 100 gram of dry coal on equilibrated basis is associated with c gram of moisture. Hence, after equilibration, this 100 gram of dry coal increases to a weight of (100 + c) gram.

Then on percentage basis, the analyses on equilibrated condition are

$$P_{60} \text{ Moisture} = \frac{c}{100 + c} \times 100$$

$$\text{and } P_{60} \text{ Mineral Matter} = \frac{a}{100 + c} \times 100 = d$$

$$\text{so, } P_{60} \text{ Ash} = \frac{d}{1.1}$$

#### 4.3.4 CALORIFIC VALUE CALCULATION

Mazumder's formula for high moisture coal was used for calculation of Gross Calorific Value as described below:

$$\text{Gross CV in kcal/kg} = \frac{154 \times \{100 - (1.1 \times A + M)\} - 108 \times M}{1.8}$$

Where A = Ash% and M = Moisture % on equilibrated basis, determined as per IS 1350, Part I, 1984.

#### 4.3.5 CALCULATIONS FOR CONVERSION TO DMMF BASIS

##### 4.3.5.1 UNIT VOLATILE MATTER

The following formula for calculating unit VM is used

$$\text{Unit VM} = 100 - \frac{\text{Fixed Carbon} \times 100}{\text{Pure Coal}}$$

where Fixed Carbon was derived by subtracting Moisture, Ash and Volatile Matter, determined in equilibrated basis as per IS 1350, Part I, 1984, from 100.

##### 4.3.5.2 UNIT CALORIFIC VALUE

It appears that CEMPGEODOC package provides Dry Ash Free CV, as such, its output has been modified to gather Unit Calorific Value or DMMF Calorific Value as per following formula:

$$\text{Unit CV} = \frac{\text{CV} \times 100}{\text{Pure Coal}}$$

Where CV is Gross Calorific Value, determined as per IS 1350, Part II, 1970.

##### 4.3.5.3 ULTIMATE ANALYSIS ON DMMF BASIS

Percentages of carbon, hydrogen, etc. as determined by ultimate analysis have been converted to DMMF basis using the following formula, as shown below in case of carbon:

$$\text{Unit Carbon} = \frac{\text{Carbon} \times 100}{\text{Pure Coal}}$$

Where C was carbon percent of coal seam determined by ultimate analysis as per IS 1350, Part IV, 1974. Similar formula is used in case of other constituents except oxygen, which is obtained by subtracting carbon, hydrogen, nitrogen and Sulphur on DMMF basis from 100.

In all the above cases as described in Paragraph 4.3.5.1, 4.3.5.2 and 4.3.5.3, pure coal is derived as follows: -

$$\text{Pure Coal} = 100 - (1.1 \times \text{Ash} + \text{Moisture})$$

Where Ash and Moisture are taken from determined seam overall on equilibrated basis.

Correction for CO₂ as carbonate for above DMMF conversions is required to be done by modifying Pure Coal calculation as per following conditions:

i) When CO₂ ≤ 0.5, there is no need for CO₂ correction.

ii) When CO₂ > 0.5 but ≤ 1.0,

$$\text{Pure Coal} = 100 - (1.1 \times A + M + 0.64 \times \text{CO}_2)$$

iii) When CO₂ > 1.0 but ≤ 1.5,

$$\text{Pure Coal} = 100 - (1.1 \times A + M + 0.7 \times \text{CO}_2)$$

iv) When CO₂ > 1.5,

$$\text{Pure Coal} = 100 - (1.1 \times A + M + 0.8 \times \text{CO}_2)$$

The ultimate analysis data is required to be provided by CIMFR after correction for CO₂. However CIMFR has not provided any CO₂ inspite of advising specifically for CO₂ determination. As a result, it appears, CO₂ correction has not been done.

#### 4.4 SEQUENCE OF COAL SEAMS

The sequence of coal seams, the variation of thickness and their intervening parting, as encountered in the boreholes drilled in Marwatola Sector I&II block, in descending order, is furnished in Table 4.4A.

**Table: 4.4A: Sequence of coal seams Marwatola Sector I&II Block, Sohagpur Coalfield**

Seams / Parting	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		Nos. of Bh
	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	
SOIL	0	0	0.2	6	0.2	6	470.08	542.58			35
			CSM I&II-06	MSM-23	CSM I&II-06	MSM-23		MSM-27			
WM	0.2	6	5.2	46	5.2	46					35
	CSM I&II-06	MSM-23	CSM I&II-01	MSM-55	CSM I&II-01	MSM-55					
Parting	5.2	46	120	201.73	107.56	179.17					3
	CSM I&II-01	MSM-55	CSM I&II-02	MSM-23	CSM I&II-02	MSM-55					
V	120	201.73	120.1	201.83	0.05	0.1	316.56	378.64	316.46	378.54	3
	CSM I&II-02	MSM-23	CSM I&II-02	MSM-23	CSM I&II-13	CSM I&II-02	MSM-23	CSM I&II-02	MSM-23	CSM I&II-02	
Parting	120	201.73	136.2	217.81	15.98	58.49					3
	CSM I&II-02	MSM-23	CSM I&II-02	MSM-23	MSM-23	CSM I&II-13					
IV	99.64	480	100.24	480.65	0.1	1.8	43.28	405.79	42.63	405.19	27
	CSM I&II-10	CSM I&II-03	CSM I&II-10	CSM I&II-03	MSM-51	MSM-24	CSM I&II-03	CSM I&II-10	CSM I&II-03	CSM I&II-10	
Parting	100.24	226.73	135.84	258.5	24.73	52.2					17
	CSM I&II-10	MSM-28	CSM I&II-10	MSM-24	CSM I&II-12	MSM-52					
L2	135.84	258.5	135.94	258.5	0.05	1.35	249.57	369.59	249.13	369.49	19
	CSM	MSM-	CSM	MSM-	CSM	CSM	MSM-	CSM	MSM-	CSM	

Seams / Parting	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		Nos. of Bh
	I&II-10	24	I&II-10	24	I&II-06	I&II-05	54	I&II-10	54	I&II-10	
Parting	135.94	258.5	144.6	595.5	6.51	20.1					13
	CSM I&II-10	MSM-24	CSM-I&II-10	CSM-I&II-15	CSM I&II-02	MSM-28					
IIIA	144.6	595.5	147	597	0.1	2.7	-82.86	360.83	-84.36	358.43	21
	CSM-I&II-10	CSM-I&II-15	CSM-I&II-10	CSM-I&II-15	MSM-18	MSM-28	CSM-I&II-15	CSM-I&II-10	CSM-I&II-15	CSM-I&II-10	
Parting	147	597	164.4	619.1	6.17	25.73	-	-	-	-	21
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-04	CSM I&II-03					
III TOP	251.45	272.3	252.93	273.6	0.3	1.48	227.94	257.96	227.64	256.82	4
	CSM-I&II-07	MSM-55	CSM-I&II-07	MSM-55	MSM-51	CSM-I&II-07	MSM-51	CSM-I&II-12	MSM-51	CSM-I&II-12	
Parting	252.93	273.6	253.93	277	0.62	3.4					4
	CSM-I&II-07	MSM-55	CSM-I&II-07	MSM-55	MSM-51	MSM-55					
III MIDDLE	253.93	277	254.23	278.15	0.3	3.22	227.02	255.1	226.72	251.88	4
	CSM-I&II-07	MSM-55	CSM-I&II-07	MSM-55	CSM-I&II-07	CSM-I&II-12	MSM-51	CSM-I&II-12	MSM-51	CSM-I&II-12	
Parting	254.23	278.15	256.78	280.45	1.08	2.55					4
	CSM-I&II-07	MSM-55	CSM I&II-07	MSM-55	MSM-51	CSM I&II-07					
III BOTTOM	256.78	280.45	258.66	281.9	1.3	3.42	225.64	249.82	224.34	246.4	4
	CSM I&II-07	MSM-55	CSM I&II-07	MSM-55	MSM-51	CSM I&II-12	MSM-51	CSM I&II-12	MSM-51	CSM I&II-12	
III COMB	164.4	619.1	166.3	619.2	0.1	5.06	-106.46	341.03	-106.56	339.13	26
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-15	CSM I&II-05	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-10	
Parting	166.3	619.2	199.3	650.37	18.67	60.96					28
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	MSM-27	MSM-26					
II TOP	237.85	328.65	238.75	330.13	0.4	4.8	188.53	269.52	188.13	268.99	15
	CSM I&II-02	MSM-24	CSM I&II-02	MSM-24	MSM-52	MSM-55	MSM-52	MSM-27	MSM-52	MSM-27	
Parting	238.75	330.13	239.22	332.15	0.47	4.1					15

Seams / Parting	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		Nos. of Bh
	CSM I&II-02	MSM-24	CSM I&II-02	MSM-24	CSM I&II-02	MSM-21					
II BOTTOM	239.22	332.15	241.15	334.4	0.36	2.3	186.88	268.3	185.38	267.94	15
	CSM I&II-02	MSM-24	CSM I&II-02	MSM-24	MSM-27	MSM-21	MSM-52	MSM-27	MSM-52	MSM-27	
II COMB	199.3	650.37	202.3	650.87	0.03	5.6	-137.73	306.13	-138.23	303.13	14
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-13	MSM-17	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-10	
Parting	202.3	650.87	215.9	684.7	7.99	33.83					26
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-04	CSM I&II-15					
I B	20	684.7	20.98	685.3	0.05	2.47	-172.06	467.99	-172.66	467.01	28
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	MSM-27	CSM I&II-10	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	
Parting	20.98	685.3	29.72	691.98	3.9	17.25					28
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	CSM I&II-12	CSM I&II-08					
I A	29.72	691.98	29.83	693.88	0.11	2.11	-179.34	458.27	-181.24	458.16	29
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	MSM-20	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	
Parting	233.1	362.85	243.38	369.4	4.11	11.77					14
	CSM I&II-10	MSM-23	MSM I&II-10	MSM-23	MSM-19	MSM-34					
I TOP	243.38	369.4	243.58	369.7	0.1	0.75	148.2	262.05	147.7	261.85	14
	CSM I&II-10	MSM-23	CSM I&II-10	MSM-23	MSM-19	CSM I&II-02	MSM-34	CSM I&II-10	MSM-34	CSM I&II-10	
Parting	243.58	369.7	246.24	371.7	0.4	6.15					14
	CSM I&II-10	MSM-23	CSM I&II-10	MSM-23	CSM I&II-13	MSM-21					
I BOTTOM	246.24	371.7	247.6	372.25	0.2	1.36	144	259.19	143.8	257.83	14
	CSM I&II-10	MSM-23	CSM I&II-10	MSM-23	CSM I&II-13, MSM-34	CSM I&II-10	MSM-34	CSM I&II-10	MSM-34	CSM I&II-10	
Parting	247.6	372.25	-	-	-	-					14

Seams / Boreholes	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		Nos. of Bh
	CSM I&II-10	MSM- 23									
I COMB	51.8	700.05	52	702.1	0.2	2.05	-187.41	436.19	-189.46	435.99	13
	MSM- 22	CSM I&II-15	MSM- 22	CSM I&II-15	MSM- 17, MSM- 22	CSM I&II-15	CSM I&II-15	MSM- 22	CSM I&II-15	MSM- 22	

While describing the individual seam hereunder, the chemical analyses of seam having 0.50 metre and above thickness in boreholes have only been considered, since seams having thickness of 0.90 metre and above are considered as developed / mineable / workable / potential coal horizons. Accordingly, coal reserves are also estimated in the block.

## 4.5 DESCRIPTION OF INDIVIDUAL COAL SEAMS

### 4.5.1 SEAM V

#### 4.5.1.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.1.1.A: Details of seam references, of the Seam V, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C, VIID, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		-
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		-
13	Cross Section Plans		-
14	OB / Parting Plan		-
15	Seam Folio Plan		-

#### 4.5.1.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

### 4.5.1.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.1.3.A: Data of Borehole Intersection, of the Seam V, Marwatola Sector I&II block, Sohagpur Coalfield**

SI No.	Seam	V
1	No. of full seam intersection in boreholes	3
2	No. of incrop proving boreholes with part thickness	
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	27
6	No. of borehole not deepened to encounter the seam	2
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>33</b>
7	No. of boreholes falling in up dip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.1.3.B: Depth Of Intersection of Roof and Floor of the Seam V, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		V	
ROOF DEPTH	MIN (M)	120.00	CSM I&II-02
	MAX (M)	201.73	MSM-23
FLOOR DEPTH	MIN (M)	120.10	CSM I&II-02
	MAX (M)	201.83	MSM-23
THICK NESS	MIN (M)	0.05	CSM I&II-13
	MAX (M)	0.10	CSM I&II-02, MSM-23

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.1.3.C: Minimum & Maximum RRLs & FRLs of the Seam V, Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	316.56	MSM-23
	MAX (M)	378.64	CSM I&II-02
FRL	MIN (M)	316.46	MSM-23
	MAX (M)	378.54	CSM I&II-02
BH INTERSECTIONS		3	

#### 4.5.1.4 STRATIGRAPHIC POSITION

The Seam V is underlain by Seam IV.

#### 4.5.1.5 PARTING

The details of Parting above and below the Seam V is listed in the Table below.

**Table: 4.5.1.5.A: Parting Details of the Seam V, With Underlying Seams, Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam IV	15.98	58.49	03
	MSM-23	CSM I&II-13	

#### 4.5.1.6 THICKNESS

**Table: 4.5.1.6.A: Thickness data of the Seam V in boreholes,  
Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		V	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.05	CSM I&II-13
	MAX (M)	0.10	CSM I&II-02, MSM-23
GENERAL THICKNESS	MIN (M)	0.05	
	MAX (M)	0.11	
BOREHOLE INTERSECTIONS		3	

#### 4.5.1.7 SPLITTING

The seam does not exhibit any splitting behaviour

#### 4.5.2.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
V	ROOF	3	0	1	1	0	1	0	0
	FLOOR		0	0	0	0	1	0	2

#### 4.5.1.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.1.11 DIRT BANDS/ PROXIMATE / ULTIMATE / SPECIAL TEST:

Since the thickness of the seam is less than 0.90m, the samples have not been analysed.

#### 4.5.1.12 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam V is Not-developed in the block. The seam incrops in the southern part of the block. It is the topmost seam and underlain by Seam IV. The parting between the underlying Seam IV varies from 15.98 to 58.49m.

The seam is unworkable and the thickness of the seam varies from 0.05 to 0.10m. Since the thickness of the seam is less than 0.90m, the samples have not been analysed. Hence the seam has not been considered for resource estimation.

### 4.5.2 SEAM IV

#### 4.5.2.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.2.1.A: Details of seam references, of the Seam IV, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C, VIID, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 1 – V 1
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 1
13	Cross Section Plans		VIII
14	OB / Parting Plan		IX 1
15	Seam Folio Plan		X 1

#### 4.5.2.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

#### 4.5.2.3 DETAILS OF BOREHOLE INTERSECTION

##### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.2.3.A: Data of Borehole Intersection, of the Seam IV, Marwatola Sector I&II block, Sohagpur Coalfield**

SI No.	Seam	IV
1	No. of full seam intersection in boreholes	27
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	3
6	No. of borehole not deepened to encounter the seam	2
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>33</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	-
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

**C. DEPTH OF INTERSECTION (ROOF & FLOOR)****Table: 4.5.2.3.B: Depth Of Intersection of Roof and Floor of the Seam IV, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		IV	
ROOF DEPTH	MIN (M)	99.64	CSM I&II-10
	MAX (M)	480	CSM I&II-03
FLOOR DEPTH	MIN (M)	100.24	CSM I&II-10
	MAX (M)	480.65	CSM I&II-03
THICK NESS	MIN (M)	0.1	MSM-51
	MAX (M)	1.8	MSM-24

**C. MINIMUM & MAXIMUM RRLs & FRLs****Table: 4.5.2.3.C: Minimum & Maximum RRLs & FRLs of the Seam IV, Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	43.28	CSM I&II-03
	MAX (M)	405.79	CSM I&II-10
FRL	MIN (M)	42.63	CSM I&II-03
	MAX (M)	405.19	CSM I&II-10
BH INTERSECTIONS		27	

**4.5.2.4 STRATIGRAPHIC POSITION**

The Seam IV is overlain by Seam V and underlain by Seam L2.

**4.5.2.5 PARTING**

The details of Parting above and below the Seam IV is listed in the Table below.

**Table: 4.5.2.5.A: Parting Details of the Seam IV, With Underlying Seams,  
Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam L2.	24.73	52.20	27
	CSM I&II-12	MSM-52	

#### 4.5.2.6 THICKNESS

**Table: 4.5.2.6.A: Thickness data of the Seam IV in boreholes, Marwatola Sector I&II  
block, Sohagpur Coalfield**

SEAM		IV	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.10	MSM-51
	MAX (M)	1.8	MSM-24
GENERAL THICKNESS	MIN (M)	0.41	
	MAX (M)	1.31	
BOREHOLE INTERSECTIONS		27	

#### 4.5.2.7 SPLITTING

The seam does not exhibit any splitting behaviour

#### 4.5.2.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault(F6-F6) trending east-west of the block. The boreholes MSM-22 and MSM-49 the seam falls in the updip side of the block.

#### 4.5.2.4 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
IV	ROOF	27	0	0	1	0	16	0	10
	FLOOR		0	5	5	0	3	1	13

#### 4.5.2.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.2.11DIRT BANDS

The details of dirt band within seam-IV have been provided in Annexure V. The maximum thickness of the seam including dirt band is 1.80 meters. The seam is devoid of any dirt band.

#### 4.5.2.12 PROXIMATE ANALYSIS

**Table 4.5.2.12.A: Range of Proximate Analysis and Grade of Seam IV,  
Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		IV		IV		IV		
Bhs. considered (Det. & Cal.)		10		10		10		
No. of Determined Boreholes		10		10		4		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)	Min	0.95	MSM-34	0.95	MSM-34	0.95	MSM-34	
	Max	1.80	MSM-24	1.80	MSM-24	1.80	MSM-24	
Range of Analysed Thickness (m)	Min	0.95	MSM-34	0.95	MSM-34	0.95	MSM-34	
	Max	1.80	MSM-24	1.80	MSM-24	1.80	MSM-24	
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	4.7	MSM-34	4.7	MSM-34	6.3	MSM-24
		Max	11.6	MSM-17	11.6	MSM-17	8.9	CSM I&II-11
	Ash%	Min	12.6	MSM-31	12.6	MSM-31	12.4	MSM-31
		Max	25.9	CSM I&II-04	25.9	CSM I&II-04	25.9	CSM I&II-04
	VM%	Min	25.8	CSM I&II-04	25.8	CSM I&II-04	25.8	CSM I&II-04
		Max	31.8	CSM I&II-08	31.8	CSM I&II-08	31.8	CSM I&II-08
	FC%	Min	40.9	CSM I&II-04	40.9	CSM I&II-04	40.9	CSM I&II-04
		Max	52.5	MSM-19	52.5	MSM-19	45.5	CSM I&II-01
Range of UVM%		Min	33.0	MSM-19	33.0	MSM-19	36.3	CSM I&II-04
		Max	42.0	CSM I&II-08	42.0	CSM I&II-08	42.0	CSM I&II-08
Range of CV in k. cal/kg	Gross CVAR	Min	5088	CSM I&II-04	5088	CSM I&II-04	5088	CSM I&II-04
		Max	6368	MSM-31	6368	MSM-31	6402	MSM-31
	Gross CVDF	Min	7278	CSM I&II-08	7278	CSM I&II-08	7278	CSM I&II-08
		Max	8194	MSM-34	8194	MSM-34	8105	CSM I&II-01
Range of UHV in k. cal/kg		Min	4290	CSM I&II-04	4290	CSM I&II-04	4290	CSM I&II-04
		Max	6209	MSM-31	6209	MSM-31	6250	MSM-31
Range of Grade		Min	G-8		G-8		G-8	
		Max	G-4		G-4		G-4	

#### 4.5.2.13 DETERMINED GROSS CV

**Table 4.5.2.13 A: Range of Gross CV of Seam IV, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		IV		IV	
No. of Boreholes (CV Determined)		10		10	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	0.95	MSM-34	0.95	MSM-34
	Max	1.8	MSM-24	1.8	MSM-24
Range of Analysed Thickness (m)	Min	0.95	MSM-34	0.95	MSM-34
	Max	1.80	MSM-24	1.80	MSM-24
Range of Det. CV in k. cal/kg	Gross CVAR	Min	5088	CSM I&II-04	5088
		Max	6368	MSM-31	6368
	Gross CVDF	Min	7278	CSM I&II-08	7278
		Max	8194	MSM-34	8194

#### 4.5.2.14 ULTIMATE ANALYSIS

The Ultimate analysis of seam IV samples have not been analysed.

#### 4.5.2.15 PETROGRAPHIC STUDIES

Results of Petrographic studies are tabulated below.

**Table 4.5.2.15 A: Petrographic Analysis of Seam IV, Marwatola Sector I&II block, Sohagpur Coalfield**

BH. NO.	DEPTH		SEAM THICK NESS	TYPE OF SAMP LE	Maceral Composition %				Maceral Composition (Vmmf%)			Rr%
	FROM	TO										
	(m)	(m)			(m)	Vit	Lipt	Int	VMM	Vit	Lipt	
CSM I&II-10	99.64	100.24	0.60	BCS	52.2	4.3	19.8	23.7	68.3	5.6	26.1	0.58
CSM I&II-10	99.64	100.24	0.60	I30	52.2	4.3	19.8	23.7	68.3	5.6	26.1	0.58
CSM I&II-10	99.64	100.24	0.60	I100	52.2	4.3	19.8	23.7	68.3	5.6	26.1	0.58
MSM-19	175.34	176.74	1.40	BCS	29.1	15.6	35.4	19.9	36.0	19.0	44.2	0.50
MSM-19	175.34	176.74	1.40	I30	29.1	15.6	35.4	19.9	36.0	19.0	44.2	0.50

#### **4.5.2.16 HARDGROVE GRINDABILITY INDEX / ASH FUSION TEMPERATURE RANGE / ASH ANALYSIS**

The Hardgrove grindability index, Ash Fusion Temperature range and Ash analysis of seam IV samples have not been analysed.

#### **4.5.2.17 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS**

The Seam IV is developed in the the block. The seam incrops in the southern part of the block. It is overlain by Seam V and underlain by Seam L2. The parting between Seam IV and the overlying Seam V varies from 15.98 to 58.49m and the parting with the underlying Seam L2 varies from 24.73 to 52.20m.

The seam is less banded and the thickness of the seam varies from 0.10 to 1.80m. The general thickness of the seam varies from 0.41 to 1.31m. The proximate analysis on 60% RH and 40°C of Seam IV on I₃₀ sample shows moisture%, ash% and volatile matter% as 4.7 to 11.6, 12.6 to 25.9 and 25.8 to 31.8 respectively. The grade of the seam varies from G8 to G4.

### 4.5.3 SEAM L2

#### 4.5.3.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.2.1.A: Details of seam references, of the Seam L2, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C, VIID, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		-
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		-
13	Cross Section Plans		-
14	OB / Parting Plan		-
15	Seam Folio Plan		-

#### 4.5.3.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

### 4.5.3.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.3.3.A: Data of Borehole Intersection, of the Seam L2, Marwatola Sector I&II block, Sohagpur Coalfield**

S.No.	Seam	L2
1	No. of full seam intersection in boreholes	19
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	11
6	No. of borehole not deepened to encounter the seam	2
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>33</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	-
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.3.3.B: Depth Of Intersection of Roof and Floor of the Seam L2, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		L2	
ROOF DEPTH	MIN (M)	135.84	CSM I&II-10
	MAX (M)	258.5	MSM-24
FLOOR DEPTH	MIN (M)	135.94	CSM I&II-10
	MAX (M)	259	MSM-24
THICK NESS	MIN (M)	0.05	CSM I&II-02, CSM I&II-06
	MAX (M)	1.35	CSM I&II-05

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.3.3.C: Minimum & Maximum RRLs & FRLs of the Seam L2, Marwatola Sector I&II block, Sohagpur Coalfield.**

<b>RRL</b>	<b>MIN (M)</b>	249.57	MSM-54
	<b>MAX (M)</b>	369.59	CSM I&II-10
<b>FRL</b>	<b>MIN (M)</b>	249.13	MSM-54
	<b>MAX (M)</b>	369.49	CSM I&II-10
<b>BH INTERSECTIONS</b>		<b>19</b>	

#### 4.5.3.4 STRATIGRAPHIC POSITION

The Seam L2 is overlain by Seam IV and underlain by Seam IIIA.

#### 4.5.3.5 PARTING

The details of Parting above and below the Seam L2 is listed in the Table below.

**Table: 4.5.3.5.A: Parting Details of the Seam L2, With Underlying Seams, Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam IIIA	6.51	20.10	19
	CSM I&II-02	MSM-28	

#### 4.5.3.6 THICKNESS

**Table: 4.5.3.6.A: Thickness data of the Seam IV in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		L2	
DETAILS	MIN /MAX (M)	THICK NESS	BORE HOLE NO.
THICKNESS	MIN (M)	0.05	CSM I&II-02, CSM I&II-06
	MAX (M)	1.35	CSM I&II-05
GENERAL THICKNESS	MIN (M)	0.16	
	MAX (M)	0.78	
BOREHOLE INTERSECTIONS		19	

#### 4.5.3.7 SPLITTING

The seam does not exhibit any splitting behaviour

#### 4.5.3.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
L2	ROOF	19	0	1	1	0	6	0	11
	FLOOR		0	0	0	0	3	1	15

#### 4.5.3.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

**4.5.3.11DIRT BANDS/ PROXIMATE / ULTIMATE / SPECIAL TEST:**

Since the thickness of the seam is not workable and the seam thickness is less than 0.90m except one borehole number CSM I&II-05 having thickness of 1.35m, the samples have not been analysed.

**4.5.3.12 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS**

The Seam L2 is not-developed in the block. The seam incrops in the southern part of the block. It is overlain by Seam IV and underlain by Seam IIIA. The parting between Seam L2 and the overlying Seam IV varies from 24.73 to 52.20m and the parting with the underlying Seam IIIA varies from 6.51 to 20.10m.

The seam thickness varies from 0.05 to 1.35m. The general thickness of the seam varies from 0.16 to 0.78m. The seam thickness is less than 0.90m except one borehole CSM-I&II-05 having thickness of 1.35m, the samples have not been analysed. Hence the seam has not been considered for resource estimation

## 4.5.4 SEAM IIIA

### 4.5.4.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.4.1.A: Details of seam references, of the Seam IIIA, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A, VII B, VII C, VIID, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 2
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 2
13	Cross Section Plans		VIII 1
14	OB / Parting Plan		IX 1
15	Seam Folio Plan		X 2

### 4.5.4.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

#### 4.5.4.3 DETAILS OF BOREHOLE INTERSECTION

##### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.4.3.A: Data of Borehole Intersection, of the Seam IIIA, Marwatola Sector I&II block, Sohagpur Coalfield**

SI No.	Seam	IIIA
1	No. of full seam intersection in boreholes	21
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	9
6	No. of borehole not deepened to encounter the seam	2
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>33</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

##### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.4.3.B: Depth Of Intersection of Roof and Floor of the Seam IIIA, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		IIIA	
ROOF DEPTH	MIN (M)	144.6	CSM I&II-10
	MAX (M)	595.5	CSM I&II-15
FLOOR DEPTH	MIN (M)	147	CSM I&II-10
	MAX (M)	597	CSM I&II-15
THICK NESS	MIN (M)	0.1	MSM-18
	MAX (M)	2.7	MSM-28

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.4.3.C: Minimum & Maximum RRLs & FRLs of the Seam IIIA, Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	-82.86	CSM I&II-15
	MAX (M)	360.83	CSM I&II-10
FRL	MIN (M)	-84.36	CSM I&II-15
	MAX (M)	358.43	CSM I&II-10
BH INTERSECTIONS		21	

#### 4.5.4.4 STRATIGRAPHIC POSITION

The Seam IIIA is overlain by Seam L2 and underlain by Seam III Top.

#### 4.5.4.5 PARTING

The details of Parting above and below the Seam IIIA is listed in the Table below.

**Table: 4.5.4.5.A: Parting Details of the Seam IIIA, With Underlying Seams, Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam III Top	6.17	25.73	21
	CSM I&II-04	CSM I&II-03	

#### 4.5.4.6 THICKNESS

**Table: 4.5.3.6.A: Thickness data of the Seam IIIA in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		IIIA	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.1	MSM-18
	MAX (M)	2.7	MSM-28
GENERAL THICKNESS	MIN (M)	0.22	
	MAX (M)	1.56	
BOREHOLE INTERSECTIONS		21	

#### 4.5.4.7 SPLITTING

The seam does not exhibit any splitting behavior

#### 4.5.4.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault (F6-F6) trending east-west of the block. The boreholes MSM-22 and MSM-49 the seam falls in the updip side of the block.

#### 4.5.4.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
IIIA	ROOF	21	0	0	1	0	7	0	13
	FLOOR		0	4	2	0	4	0	11

#### 4.5.3.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.4.11 DIRT BANDS

The details of dirt band within seam-IIIA have been provided in Annexure V. The maximum thickness of the seam including dirt band is 2.40 meters. The abstract of dirt bands within the seam having thickness more than 0.90m, is given in the following table:

**Table 4.5.4.11 A: Abstract of Dirt Bands in Seam IIIA having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name				IIIA	
Type of Sample				I30	
Total Seam Thickness		Min.	(m)	0.96	MSM-17
		Max.	(m)	2.70	MSM-28
Analysed Seam Thickness		Min.	(m)	0.91	MSM-17
		Max.	(m)	2.40	CSM I&II-10
Effective Dirt Bands	Included	No.	Min.	0.00	CSM I&II-10
			Max.	1.00	MSM-17
		Thick (m)	Min.	0.00	CSM I&II-10
			Max.	0.20	MSM-17
	Excluded	No.	Min.	0.00	CSM I&II-10
			Max.	1.00	CSM I&II-13
		Thick (m)	Min.	0.00	CSM I&II-10
			Max.	0.07	CSM I&II-13
No. of Bhs				5	

The abstract of combustible and non-combustible dirt band constituents within the seam is tabulated below.

**Table 4.5.4.11 B: Abstract Of Combustible and Non-combustible Dirt Band Constituents  
in Seam IIIA having thickness more than 0.90m, Marwatola Sector I&II block,  
Sohagpur Coalfield**

Seam			IIIA		
Total Seam Thickness	Min.	(m)	0.96		
	Max.	(m)	2.70		
Analysed Seam Thickness	Min.	(m)	0.91		
	Max.	(m)	2.40		
Combustible Dirt Bands (Carbonaceous Shale)	No.	Min	0.00	CSM I&II-10	
		Max	1.00	MSM-17	
	Thick (m)	Min	0.00	CSM I&II-10	
		Max	0.20	MSM-17	
	%	Min	0.00	CSM I&II-10	
		Max	21.00	MSM-17	
	Non-Combustible Dirt Bands ( Shale & Sandstone )	No.	Min	0.00	CSM I&II-10
			Max	1.00	CSM I&II-13
Thick (m)		Min	0.00	CSM I&II-10	
		Max	0.07	CSM I&II-13	
%		Min	0.00	CSM I&II-10	
		Max	6.00	CSM I&II-13	
Total bands (Carb. Shale, Shale and Sandstone)	No.	Min	0.00	CSM I&II-10	
		Max	1.00	CSM I&II-13	
	Thick (m)	Min	0.00	CSM I&II-10	
		Max	0.20	MSM-17	
	%	Min	0.00	CSM I&II-10	
		Max	21.00	MSM-17	
Nos of Bh			5		

## 4.5.4.12 PROXIMATE ANALYSIS

**Table 4.5.4.12.A: Range of Proximate Analysis and Grade of Seam IIIA having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		IIIA		IIIA		IIIA		
Bhs. considered (Det. & Cal.)		4		5		5		
No. of Determined Boreholes		3		3		3		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)	Min	1.22	CSM I&II-13	0.96	MSM-17	0.96	MSM-17	
	Max	2.70	MSM-28	2.70	MSM-28	2.70	MSM-28	
Range of Analysed Thickness (m)	Min	1.15	CSM I&II-13	0.91	MSM-17	0.91	MSM-17	
	Max	2.40	CSM I&II-10	2.40	CSM I&II-10	2.40	CSM I&II-10	
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	2.3	CSM I&II-15	2.3	CSM I&II-15	2.3	CSM I&II-15
		Max	4.8	MSM-28	4.8	MSM-28	4.8	MSM-28
	Ash%	Min	20.1	CSM I&II-15	20.1	CSM I&II-15	20.1	CSM I&II-15
		Max	45.4	CSM I&II-13	55.1	MSM-17	55.1	MSM-17
	VM%	Min	19.5	CSM I&II-13	19.5	CSM I&II-13	19.5	CSM I&II-13
		Max	28.1	CSM I&II-15	28.1	CSM I&II-15	28.1	CSM I&II-15
	FC%	Min	31.2	CSM I&II-13	31.2	CSM I&II-13	31.2	CSM I&II-13
		Max	49.5	CSM I&II-15	49.5	CSM I&II-15	49.5	CSM I&II-15
Range of UVM%		Min	32.4	CSM I&II-13	32.4	CSM I&II-13	32.4	CSM I&II-13
		Max	38.0	CSM I&II-10	38.0	CSM I&II-10	38.0	CSM I&II-10
Range of CV in k. cal/kg	Gross CVAR	Min	3814	CSM I&II-13	2934	MSM-17	2934	MSM-17
		Max	6439	CSM I&II-15	6439	CSM I&II-15	6439	CSM I&II-15
	Gross CVDF	Min	8061	MSM-28	8061	MSM-28	8061	MSM-28
		Max	8518	CSM I&II-15	8518	CSM I&II-15	8518	CSM I&II-15
Range of UHV in k. cal/kg		Min	2096	CSM I&II-13	882	MSM-17	882	MSM-17
		Max	5808	CSM I&II-15	5808	CSM I&II-15	5808	CSM I&II-15
Range of Grade		Min	G-12		G-15		G-15	
		Max	G-3		G-3		G-3	

## 4.5.4.13 DETERMINED GROSS CV

**Table 4.5.2.13 A: Range of Gross CV of Seam IIIA having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		IIIA		IIIA	
No. of Boreholes (CV Determined)		3		3	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	1.22	CSM I&II-13	1.22	CSM I&II-13
	Max	2.4	CSM I&II-10	2.4	CSM I&II-10
Range of Analysed Thickness (m)	Min	1.15	CSM I&II-13	1.15	CSM I&II-13
	Max	2.40	CSM I&II-10	2.40	CSM I&II-10
Range of Det. CV in k. cal/kg	Gross CVAR	Min	<b>3814</b>	CSM I&II-13	<b>3814</b>
		Max	<b>6439</b>	CSM I&II-15	<b>6439</b>
	Gross CVDF	Min	8263	CSM I&II-13	8263
		Max	8518	CSM I&II-15	8518

## 4.5.4.14 ULTIMATE ANALYSIS

The ultimate analysis has been done in only one borehole no CSM I&II-15.

**Table 4.5.2.14 A: Range of Ultimate Analysis Seam IIIA, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			III A	
Boreholes considered			1	
Type of Samples			BCS/I30/I100	
Range of Seam Thickness (m)		Min	1.50	CSM I&II-15
		Max	1.50	CSM I&II-15
Range of Ultimate analyses	C%	Min	66.12	CSM I&II-15
		Max	66.12	CSM I&II-15
	H%	Min	3.92	CSM I&II-15
		Max	3.92	CSM I&II-15
	N%	Min	1.25	CSM I&II-15
		Max	1.25	CSM I&II-15
	S%	Min	0.88	CSM I&II-15
		Max	0.88	CSM I&II-15
	O% by diff.	Min	4.40	CSM I&II-15
		Max	4.40	CSM I&II-15

#### 4.5.4.15 PETROGRAPHIC STUDIES

The Petrographic study of seam IIIA samples have been analysed in only one borehole no CSM I&II-10.

**Table 4.5.2.15 A: Petrographical Analysis of Seam IIIA, Marwatola Sector I&II block, Sohagpur Coalfield**

BH. NO.	DEPTH		SEAM THICK NESS	TYPE OF SAMPLE	Maceral Composition %				Maceral Composition (Vmmf%)			Rr%
	FROM	TO										
	(m)	(m)			Vit	Lipt	Int	VMM	Vit	Lipt	Int	
CSM I&II-10	144.60	147.00	2.40	BCS/13 0/1100	27.3	12.7	25.2	34.8	41.8	19.4	38.8	0.53

#### 4.5.4.16 HARDGROVE GRINDABILITY INDEX

The Hardgrove grindability index of seam IV samples have been analysed in only one borehole no CSM I&II-15.

**Table 4.5.2.16 A: Range of HGI of Seam IIIA, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name	Type of Samples	Range of HGI		Boreholes considered
		Min	Max	
IIIA	BCS / I30/ I100	58	58	1
		CSM I&II-15	CSM I&II-15	

#### 4.5.4.17 ASH FUSION TEMPERATURE RANGE

The Ash Fusion temperature range of seam IIIA samples have been analysed in only one borehole no CSM I&II-15.

**Table 4.5.2.17 A: Ash Fusion Temperature Range under Mildly Reducing Atmosphere for Seam IIIA, Marwatola Sector I&II block, Sohagpur Coalfield**

Range of Ash fusion Temperature						under Mildly Reducing
Atmosphere on I ₃₀ samples						
SEAM IIIA						
IDT (°C)		HT (°C)		FT (°C)		No. of boreholes
Min	Max	Min	Max	Min	Max	
1190	1190	1260	1260	1274	1274	
CSM I&II-15						1

#### 4.5.4.18 ASH ANALYSIS

The Ash analysis of seam IIIA samples have been analysed in only one borehole no CSM I&II-15.

**Table 4.5.2.18 A: Ash analysis Range for Seam IIIA, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			IIIA	
Boreholes considered			1	
Type of Samples			BCS / I30 /I100	
Range of Ash Analyses	SiO ₂ %	Min	57.35	CSM I&II-15
		Max	57.35	CSM I&II-15
	Al ₂ O ₃ %	Min	22.93	CSM I&II-15
		Max	22.93	CSM I&II-15
	Fe ₂ O ₃ %	Min	7.31	CSM I&II-15
		Max	7.31	CSM I&II-15
	TiO ₂ %	Min	2.35	CSM I&II-15
		Max	2.35	CSM I&II-15
	MgO%	Min	2.07	CSM I&II-15
		Max	2.07	CSM I&II-15
	P ₂ O ₅ %	Min	0.13	CSM I&II-15
		Max	0.13	CSM I&II-15
	MnO%	Min	0.27	CSM I&II-15
		Max	0.27	CSM I&II-15
	Na ₂ O	Min	0.68	CSM I&II-15
		Max	0.68	CSM I&II-15
	K ₂ O	Min	2.841	CSM I&II-15
		Max	2.84	CSM I&II-15

#### 4.5.4.19 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam IIIA is developed in the block. The seam incrops in the southern part of the block. It is overlain by Seam L2 and underlain by Seam III Top/Comb. The parting between Seam IIIA and the overlying Seam L2 varies from 6.51 to 20.10m and the parting with the underlying Seam III Top/Comb. varies from 6.17 to 25.73m.

The seam is less banded and the thickness of the seam varies from 0.10 to 2.7m. The general thickness of the seam varies from 0.22 to 1.56m. The proximate analysis on 60% RH and 40°C of Seam IIIA on I₃₀ sample shows moisture%, ash% and volatile matter% as 2.3 to 4.8, 20.1 to 55.1 and 19.5 to 28.1 respectively. The grade of the seam varies from G15 to G3.

### 4.5.5 SEAM III Top

#### 4.5.5.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.5.1.A: Details of seam references, of the Seam III Top, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A, VII B, VII C, VIID, VIIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 3 – V 4
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 3
13	Cross Section Plans		VIII 1
14	OB / Parting Plan		IX 3
15	Seam Folio Plan		X 3

#### 4.5.5.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

#### 4.5.5.3 DETAILS OF BOREHOLE INTERSECTION

##### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.5.3.A: Data of Borehole Intersection, of the Seam III Top, Marwatola Sector I&II block, Sohagpur Coalfield**

S.No.	Seam	III Top
1	No. of full seam intersection in boreholes	04
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	-
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	-
6	No. of borehole not deepened to encounter the seam	-
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>04</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	29
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

## B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.5.3.B: Depth Of Intersection of Roof and Floor of the Seam III Top, Marwatola  
Sector I&II block, Sohagpur Coalfield**

SEAM		III Top	
ROOF DEPTH	MIN (M)	251.45	CSM I&II-07
	MAX (M)	272.3	MSM-55
FLOOR DEPTH	MIN (M)	252.93	CSM I&II-07
	MAX (M)	273.6	MSM-55
THICK NESS	MIN (M)	0.3	MSM-51
	MAX (M)	1.48	CSM I&II-07

## C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.3.3.C: Minimum & Maximum RRLs & FRLs of the Seam III Top, Marwatola  
Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	227.94	MSM-51
	MAX (M)	257.96	CSM I&II-12
FRL	MIN (M)	227.64	MSM-51
	MAX (M)	256.82	CSM I&II-12
BH INTERSECTIONS		04	

### 4.5.5.4 STRATIGRAPHIC POSITION

The Seam III Top is overlain by Seam IIIA and underlain by Seam III Middle.

### 4.5.5.5 PARTING

The details of Parting above and below the Seam III Top is listed in the Table below.

**Table: 4.5.5.5.A: Parting Details of the Seam III Top, With Underlying Seams, Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam III Middle	0.62 MSM-51	3.40 MSM-55	04

#### 4.5.5.6 THICKNESS

**Table: 4.5.5.6.A: Thickness data of the Seam III Top in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		III Top	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.3	MSM-51
	MAX (M)	1.48	CSM I&II-07
GENERAL THICKNESS	MIN (M)	0.54	
	MAX (M)	1.58	
BOREHOLE INTERSECTIONS		04	

#### 4.5.5.7 SPLITTING

The seam III exhibits splitting behavior and the seam is splitted in to top, middle and bottom splits. Even though the Seam III Top is a split of Seam III, Seam III Top itself does not exhibit any split.

#### 4.5.5.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault (F6-F6) trending east-west in the block. In the boreholes MSM-22 and MSM-49 the seam falls in the updipside of the block.

#### 4.5.5.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
III Top	ROOF	4	0	0	0	0	1	0	3
	FLOOR		0	1	1	0	1	0	1

#### 4.5.5.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.5.11 DIRT BANDS

The details of dirt band within seam III Top have been provided in Annexure V. The maximum thickness of the seam including dirt band is 1.38 meters. The abstract of dirt bands within the seam having thickness more than 0.90m, is given in the following table:

**Table 4.5.5.11 A: Abstract of Dirt Bands in Seam III Top having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name				III Top	
Type of Sample				I30	
Total Seam Thickness		Min.	(m)	1.30	MSM-55
		Max.	(m)	1.48	CSM I&II-07
Analysed Seam Thickness		Min.	(m)	1.30	MSM-55
		Max.	(m)	1.38	CSM I&II-07
Effective Dirt Bands	Included	No.	Min.	0.00	CSM I&II-07
			Max.	0.00	CSM I&II-07
		Thick (m)	Min.	0.00	CSM I&II-07
			Max.	0.00	CSM I&II-07
	Excluded	No.	Min.	0.00	MSM-55
			Max.	1.00	CSM I&II-07
		Thick (m)	Min.	0.00	MSM-55
			Max.	0.10	CSM I&II-07
No. of Bhs				2	

The abstract of combustible and non-combustible dirt band constituents within the seam is tabulated below.

**Table 4.5.5.11 B: Abstract Of Combustible and Non-combustible Dirt Band Constituents in Seam III Top having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam			III Top	
Total Seam Thickness	Min.	(m)	1.30	
	Max.	(m)	1.48	
Analysed Seam Thickness	Min.	(m)	1.30	
	Max.	(m)	1.38	
Combustible Dirt Bands (Carbonaceous Shale)	No.	Min	0.00	CSM I&II-07
		Max	0.00	CSM I&II-07
	Thick (m)	Min	0.00	CSM I&II-07
		Max	0.00	CSM I&II-07
	%	Min	0.00	CSM I&II-07
		Max	0.00	CSM I&II-07
Non-Combustible Dirt Bands ( Shale & Sandstone )	No.	Min	0.00	MSM-55
		Max	1.00	CSM I&II-07
	Thick (m)	Min	0.00	MSM-55
		Max	0.10	CSM I&II-07
	%	Min	0.00	MSM-55
		Max	7.00	CSM I&II-07
Total bands (Carb. Shale, Shale and Sandstone)	No.	Min	0.00	MSM-55
		Max	1.00	CSM I&II-07
	Thick (m)	Min	0.00	MSM-55
		Max	0.10	CSM I&II-07
	%	Min	0.00	MSM-55
		Max	7.00	CSM I&II-07
Nos of Bh			2	

#### 4.5.5.12 PROXIMATE ANALYSIS

**Table 4.5.5.12.A: Range of Proximate Analysis and Grade of Seam III Top having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		III Top		III Top		III Top		
Bhs. considered (Det. & Cal.)		2		2		2		
No. of Determined Boreholes		2		2		1		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)	Min	1.3	MSM-55	1.30	MSM-55	1.30	MSM-55	
	Max	1.48	CSM I&II-07	1.48	CSM I&II-07	1.48	CSM I&II-07	
Range of Analysed Thickness (m)	Min	1.3	MSM-55	1.30	MSM-55	1.30	MSM-55	
	Max	1.38	CSM I&II-07	1.38	CSM I&II-07	1.38	CSM I&II-07	
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	3.7	MSM-55	3.7	MSM-55	5.6	CSM I&II-07
		Max	5.6	CSM I&II-07	5.6	CSM I&II-07	7.3	MSM-55
	Ash%	Min	15.4	MSM-55	15.4	MSM-55	14.8	MSM-55
		Max	33.7	CSM I&II-07	33.7	CSM I&II-07	33.7	CSM I&II-07
	VM%	Min	23.4	CSM I&II-07	23.4	CSM I&II-07	23.4	CSM I&II-07
		Max	30.8	MSM-55	30.8	MSM-55	23.4	CSM I&II-07
	FC%	Min	37.3	CSM I&II-07	37.3	CSM I&II-07	37.3	CSM I&II-07
		Max	50.2	MSM-55	50.2	MSM-55	37.3	CSM I&II-07
Range of UVM%		Min	34.9	CSM I&II-07	34.9	CSM I&II-07	34.9	CSM I&II-07
		Max	36.9	MSM-55	36.9	MSM-55	34.9	CSM I&II-07
Range of CV in k. cal/kg	Gross CVAR	Min	4715	CSM I&II-07	4715	CSM I&II-07	4715	CSM I&II-07
		Max	6571	MSM-55	6571	MSM-55	6103	MSM-55
	Gross CVDF	Min	8224	CSM I&II-07	8224	CSM I&II-07	7986	MSM-55
		Max	8280	MSM-55	8280	MSM-55	8224	CSM I&II-07
Range of UHV in k. cal/kg		Min	3476	CSM I&II-07	3476	CSM I&II-07	3476	CSM I&II-07
		Max	6264	MSM-55	6264	MSM-55	5850	MSM-55
Range of Grade		Min	G-9		G-9		G-9	
		Max	G-3		G-3		G-4	

#### 4.5.5.13 DETERMINED GROSS CV

**Table 4.5.5.13 A: Range of Gross CV of Seam III Top having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		III Top		III Top	
No. of Boreholes (CV Determined)		2		2	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	1.3	MSM-55	1.3	MSM-55
	Max	1.48	CSM I&II-07	1.48	CSM I&II-07
Range of Analysed Thickness (m)	Min	1.30	MSM-55	1.30	MSM-55
	Max	1.38	CSM I&II-07	1.38	CSM I&II-07
Range of Det. CV in k. cal/kg	Gross CVAR	Min	4715	CSM I&II-07	4715
		Max	6571	MSM-55	6571
	Gross CVDF	Min	8224	CSM I&II-07	8224
		Max	8280	MSM-55	8280

#### 4.5.5.14 ULTIMATE ANALYSIS / HARDGROVE GRINDABILITY INDEX / ASH FUSION TEMPERATURE RANGE / ASH ANALYSIS / PETROGRAPHIC STUDIES

The Ultimate analysis, HGI, Ash Fusion temperature range, Ash analysis and Petrographic studies for seam III TOP samples have not been analysed.

#### 4.5.5.15 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam III Top is developed in the block. The seam incrops in the southern part of the block. It is overlain by Seam IIIA and underlain by Seam III Middle. The parting between Seam III Top and the overlying Seam IIIA varies from 6.17 to 25.73m and the parting with the underlying Seam III Middle varies from 0.62 to 3.40m.

The seam is less banded and the thickness of the seam varies from 0.30 to 1.48m. The general thickness of the seam varies from 0.54 to 1.58m. The proximate analysis on 60% RH and 40°C of Seam IX on I₃₀ sample shows moisture%, ash% and volatile matter% as 3.7 to 5.6, 15.4 to 33.7 and 23.4 to 30.8 respectively. The grade of the seam varies from G9 to G3.

#### 4.5.6 SEAM III Middle

##### 4.5.6.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.6.1.A: Details of seam references, of the Seam III Middle, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII Bottom,VII C,VII D	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 3 – V 4
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 3
13	Cross Section Plans		VIII 1
14	OB / Parting Plan		IX 3
15	Seam Folio Plan		X 3

##### 4.5.6.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block.

### 4.5.6.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.6.3.A: Data of Borehole Intersection, of the Seam III Middle, Marwatola Sector I&II block, Sohagpur Coalfield**

S.No	Seam	III Middle
1	No. of full seam intersection in boreholes	04
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	-
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	-
6	No. of borehole not deepened to encounter the seam	-
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>04</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	29
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.6.3.B: Depth Of Intersection of Roof and Floor of the Seam III Middle, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		III Middle	
ROOF DEPTH	MIN (M)	253.93	CSM I&II-07
	MAX (M)	277.00	MSM-55
FLOOR DEPTH	MIN (M)	254.23	CSM I&II-07
	MAX (M)	278.15	MSM-55
THICK NESS	MIN (M)	0.30	CSM I&II-07, MSM-51
	MAX (M)	3.22	CSM I&II-12

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.6.3.C: Minimum & Maximum RRLs & FRLs of the Seam III Middle, Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	227.02	MSM-51
	MAX (M)	255.1	CSM I&II-12
FRL	MIN (M)	226.72	MSM-51
	MAX (M)	251.88	CSM I&II-12
BH INTERSECTIONS		04	

#### 4.5.6.4 STRATIGRAPHIC POSITION

The Seam III Middle is overlain by Seam III Top and underlain by Seam III Bottom.

#### 4.5.6.5 PARTING

The details of Parting above and below the Seam III Middle is listed in the Table below.

**Table: 4.5.6.5.A: Parting Details of the Seam III Middle, With Underlying Seams, Marwatola Sector -I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam III Bottom	1.08	2.55	04
	MSM-51	CSM I&II-07	

#### 4.5.6.6 THICKNESS

**Table: 4.5.6.6.A: Thickness data of the Seam III Middle in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		III Middle	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.30	CSM I&II-07, MSM-51
	MAX (M)	3.22	CSM I&II-12
GENERAL THICKNESS	MIN (M)	0.30	
	MAX (M)	1.15	
BOREHOLE INTERSECTIONS		04	

#### 4.5.6.7 SPLITTING

The seam III exhibits splitting behavior and the seam is splitted in to top, middle and bottom splits. Even though the Seam III Middle is a split of Seam III, Seam III Middle itself does not exhibit any split.

#### 4.5.6.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault (F6-F6) trending east-west in the block. In the boreholes MSM-22 and MSM-49 the seam falls in the updipside of the block.

#### 4.5.6.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
III Middle	ROOF	4	0	1	0	0	1	0	2
	FLOOR		0	2	1	0	0	1	0

#### 4.5.6.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.6.11 DIRT BANDS

The details of dirt band within seam-III Middle have been provided in Annexure V. The maximum thickness of the seam including dirt band is 1.15 meters. The seam is devoid of any dirt band.

#### 4.5.6.12 PROXIMATE ANALYSIS

**Table 4.5.6.12.A: Range of Proximate Analysis and Grade of Seam III Middle, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		III Middle		III Middle		III Middle		
Bhs. considered (Det. & Cal.)		1		1		1		
No. of Determined Boreholes		1		1		1		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)		Min	1.15	MSM-55	1.15	MSM-55	1.15	MSM-55
		Max	1.15	MSM-55	1.15	MSM-55	1.15	MSM-55
Range of Analysed Thickness (m)		Min	1.15	MSM-55	1.15	MSM-55	1.15	MSM-55
		Max	1.15	MSM-55	1.15	MSM-55	1.15	MSM-55
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	3.4	MSM-55	3.4	MSM-55	3.4	MSM-55
		Max	3.4	MSM-55	3.4	MSM-55	3.4	MSM-55
	Ash%	Min	24.0	MSM-55	24.0	MSM-55	24.0	MSM-55
		Max	24.0	MSM-55	24.0	MSM-55	24.0	MSM-55
	VM%	Min	28.7	MSM-55	28.7	MSM-55	28.7	MSM-55
		Max	28.7	MSM-55	28.7	MSM-55	28.7	MSM-55
	FC%	Min	43.9	MSM-55	43.9	MSM-55	43.9	MSM-55
		Max	43.9	MSM-55	43.9	MSM-55	43.9	MSM-55
Range of UVM%		Min	37.5	MSM-55	37.5	MSM-55	37.5	MSM-55
		Max	37.5	MSM-55	37.5	MSM-55	37.5	MSM-55
Range of CV in k. cal/kg	Gross CVAR	Min	5805	MSM-55	5805	MSM-55	5805	MSM-55
		Max	5805	MSM-55	5805	MSM-55	5805	MSM-55
	Gross CVDF	Min	8269	MSM-55	8269	MSM-55	8269	MSM-55
		Max	8269	MSM-55	8269	MSM-55	8269	MSM-55
Range of UHV in k. cal/kg		Min	5118	MSM-55	5118	MSM-55	5118	MSM-55
		Max	5118	MSM-55	5118	MSM-55	5118	MSM-55
Range of Grade		Min	G-5		G-5		G-5	
		Max	G-5		G-5		G-5	

#### 4.5.6.13 DETERMINED GROSS CV

**Table 4.5.6.13 A: Range of Gross CV of Seam III Middle, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		III Middle		III Middle	
No. of Boreholes (CV Determined)		1		1	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	1.15	MSM-55	1.15	MSM-55
	Max	1.15	MSM-55	1.15	MSM-55
Range of Analysed Thickness (m)	Min	1.15	MSM-55	1.15	MSM-55
	Max	1.15	MSM-55	1.15	MSM-55
Range of Det. CV in k. cal/kg	Gross CVAR	Min	5805	MSM-55	<b>5805</b>
		Max	5805	MSM-55	<b>5805</b>
	Gross CVDF	Min	8269	MSM-55	8269
		Max	8269	MSM-55	8269

#### 4.5.6.14 ULTIMATE ANALYSIS / HARDGROVE GRINDABILITY INDEX / ASH FUSION TEMPERATURE RANGE / ASH ANALYSIS / PETROGRAPHIC STUDIES

The Ultimate analysis, HGI, Ash Fusion temperature range, Ash analysis and Petrographic studies for seam III MIDDLE samples have not been analysed.

#### 4.5.6.15 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam III Middle is not-developed mostly in the block. The seam incrops in the southern part of the block. It is overlain by Seam III Top and underlain by Seam III Bottom. The parting between Seam III Middle and the overlying Seam III Top varies from 0.62 to 3.40m and the parting with the underlying Seam III Bottom varies from 1.08 to 2.55m.

The seam is devoid of any dirt and the thickness of the seam varies from 0.30 to 3.22m. The general thickness of the seam varies from 0.20 to 1.15m. The proximate analysis on 60% RH and 40°C of Seam III middle on I30 sample based on one borehole analysis data shows moisture%, ash% and volatile matter% as 3.4, 24.0 and 28.7 respectively. The grade of the seam is G5 based on the single borehole analysis data.

## 4.5.7 SEAM III Bottom

### 4.5.7.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.7.1.A: Details of seam references, of the Seam III Bottom, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A, VII B, VII C, VIID, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 3 – V 4
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 5
13	Cross Section Plans		VIII 1
14	OB / Parting Plan		IX 5
15	Seam Folio Plan		X 5

### 4.5.7.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

### 4.5.7.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.7.3.A: Data of Borehole Intersection, of the Seam III Bottom, Marwatola Sector I&II block, Sohagpur Coalfield**

S.No.	Seam	III Bottom
1	No. of full seam intersection in boreholes	04
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	-
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	-
6	No. of borehole not deepened to encounter the seam	-
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>04</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	<b>29</b>
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.7.3.B: Depth Of Intersection of Roof and Floor of the Seam III Bottom, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		III Bottom	
ROOF DEPTH	MIN (M)	256.78	CSM I&II-07
	MAX (M)	280.45	MSM-55
FLOOR DEPTH	MIN (M)	258.66	CSM I&II-07
	MAX (M)	281.90	MSM-55
THICK NESS	MIN (M)	1.30	MSM-51
	MAX (M)	3.42	CSM I&II-12

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.7.3.C: Minimum & Maximum RRLs & FRLs of the Seam III Bottom, Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	225.64	MSM-51
	MAX (M)	249.82	CSM I&II-12
FRL	MIN (M)	224.34	MSM-51
	MAX (M)	246.40	CSM I&II-12
BH INTERSECTIONS		04	

#### 4.5.7.4 STRATIGRAPHIC POSITION

The Seam III Bottom is overlain by Seam III Middle and underlain by Seam IITop.

#### 4.5.7.5 PARTING

The details of Parting below the Seam III bottom/combined is listed in the Table below.

**Table: 4.5.7.5.A: Parting Details of the Seam III Bottom, With Underlying Seams, Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam II Top /COMB.	18.67	60.96	28
	MSM-27	MSM-26	

#### 4.5.7.6 THICKNESS

**Table: 4.5.7.6.A: Thickness data of the Seam III Bottom in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		III Bottom	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	1.30	MSM-51
	MAX (M)	3.42	CSM I&II-12
GENERAL THICKNESS	MIN (M)	1.04	
	MAX (M)	2.98	
BOREHOLE INTERSECTIONS		04	

#### 4.5.7.7 SPLITTING

Even though the Seam III Bottom is a split of Seam III, Seam III Bottom itself does not exhibit any split.

#### 4.5.7.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault (F6-F6) trending east-west in the block. In the boreholes MSM-22 and MSM-49 the seam falls in the updip side of the block.

#### 4.5.7.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
III Bottom	ROOF	4	0	1	0	0	0	1	2
	FLOOR		0	1	0	0	0	1	2

#### 4.5.7.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.7.11 DIRT BANDS

The details of dirt band within seam III Bottom have been provided in Annexure V. The maximum thickness of the seam including dirt band is 1.75 meters. The abstract of dirt bands within the seam having thickness more than 0.90m, is given in the following table:

**Table 4.5.7.11 A: Abstract of Dirt Bands in Seam III Bottom having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			III Bottom		
Type of Sample			I30		
Total Seam Thickness		Min.	(m)	1.30	MSM-51
		Max.	(m)	1.88	CSM I&II-07
Analysed Seam Thickness		Min.	(m)	1.30	MSM-51
		Max.	(m)	1.75	CSM I&II-07
Effective Dirt Bands	Included	No.	Min.	0.00	CSM I&II-07
			Max.	0.00	CSM I&II-07
		Thick (m)	Min.	0.00	CSM I&II-07
			Max.	0.00	CSM I&II-07
	Excluded	No.	Min.	0.00	MSM-51
			Max.	1.00	CSM I&II-07
		Thick (m)	Min.	0.00	MSM-51
			Max.	0.130	CSM I&II-07
No. of Bhs			3		

The abstract of combustible and non-combustible dirt band constituents within the seam is tabulated below.

**Table 4.5.7.11 B: Abstract Of Combustible and Non-combustible Dirt Band Constituents in Seam III Bottom having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam			III Bottom	
Total Seam Thickness	Min.	(m)	1.30	
	Max.	(m)	1.88	
Analysed Seam Thickness	Min.	(m)	1.30	
	Max.	(m)	1.75	
Combustible Dirt Bands (Carbonaceous Shale)	No.	Min	0.00	CSM I&II-07
		Max	0.00	CSM I&II-07
	Thick (m)	Min	0.00	CSM I&II-07
		Max	0.00	CSM I&II-07
	%	Min	0.00	CSM I&II-07
		Max	0.00	CSM I&II-07
Non-Combustible Dirt Bands ( Shale & Sandstone )	No.	Min	0.00	MSM-51
		Max	1.00	CSM I&II-07
	Thick (m)	Min	0.00	MSM-51
		Max	0.13	CSM I&II-07
	%	Min	0.00	MSM-51
		Max	7.00	CSM I&II-07
Total bands (Carb. Shale, Shale and Sandstone)	No.	Min	0.00	MSM-51
		Max	1.00	CSM I&II-07
	Thick (m)	Min	0.00	MSM-51
		Max	0.13	CSM I&II-07
	%	Min	0.00	MSM-51
		Max	7.00	CSM I&II-07
Nos of Bh			3	

## 4.5.7.12 PROXIMATE ANALYSIS

**Table 4.5.7.12.A: Range of Proximate Analysis and Grade of Seam III Bottom having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		III Bottom		III Bottom		III Bottom		
Bhs. considered (Det. & Cal.)		3		3		3		
No. of Determined Boreholes		3		3		1		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)		Min	1.30	MSM-51	1.30	MSM-51	1.30	MSM-51
		Max	1.88	CSM I&II-07	1.88	CSM I&II-07	1.88	CSM I&II-07
Range of Analysed Thickness (m)		Min	1.30	MSM-51	1.30	MSM-51	1.30	MSM-51
		Max	1.75	CSM I&II-07	1.75	CSM I&II-07	1.75	CSM I&II-07
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	3.2	MSM-55	3.2	MSM-55	5.3	MSM-55
		Max	6.7	CSM I&II-07	6.7	CSM I&II-07	6.7	CSM I&II-07
	Ash%	Min	15.2	CSM I&II-07	15.2	CSM I&II-07	15.2	CSM I&II-07
		Max	32.8	MSM-55	32.8	MSM-55	31.7	MSM-55
	VM%	Min	23.8	MSM-51	23.8	MSM-51	30.7	CSM I&II-07
		Max	30.7	CSM I&II-07	30.7	CSM I&II-07	30.7	CSM I&II-07
	FC%	Min	39.7	MSM-55	39.7	MSM-55	47.4	CSM I&II-07
		Max	47.4	CSM I&II-07	47.4	CSM I&II-07	47.4	CSM I&II-07
Range of UVM%		Min	33.1	MSM-51	33.1	MSM-51	38.1	CSM I&II-07
		Max	38.1	CSM I&II-07	38.1	CSM I&II-07	38.1	CSM I&II-07
Range of CV in k. cal/kg	Gross CVAR	Min	5005	MSM-55	5005	MSM-55	4803	MSM-55
		Max	6192	CSM I&II-07	6192	CSM I&II-07	6192	CSM I&II-07
	Gross CVDF	Min	8013	MSM-51	8013	MSM-51	8028	MSM-55
		Max	8243	MSM-55	8243	MSM-55	8086	CSM I&II-07
Range of UHV in k. cal/kg		Min	3932	MSM-55	3932	MSM-55	3793	MSM-55
		Max	5877	CSM I&II-07	5877	CSM I&II-07	5877	CSM I&II-07
Range of Grade		Min	G-8		G-8		G-8	
		Max	G-4		G-4		G-4	

#### 4.5.7.13 DETERMINED GROSS CV

**Table 4.5.7.13 A: Range of Gross CV of Seam III Bottom having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		III Bottom		III Bottom	
No. of Boreholes (CV Determined)		3		3	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	1.3	MSM-51	1.3	MSM-51
	Max	1.88	CSM I&II-07	1.88	CSM I&II-07
Range of Analysed Thickness (m)	Min	1.30	MSM-51	1.30	MSM-51
	Max	1.75	CSM I&II-07	1.75	CSM I&II-07
Range of Det. CV in k. cal/kg	Gross CVAR	Min	<b>5005</b>	MSM-55	<b>5005</b>
		Max	<b>6192</b>	CSM I&II-07	<b>6192</b>
	Gross CVDF	Min	8013	MSM-51	8013
		Max	8243	MSM-55	8243

#### 4.5.7.14 ULTIMATE ANALYSIS / HARDGROVE GRINDABILITY INDEX / ASH FUSION TEMPERATURE RANGE / ASH ANALYSIS / PETROGRAPHIC STUDIES

The Ultimate Analysis, Hardgrove Grindability Index, Ash Fusion Temperature Range, Ash Analysis and Petrographic studies for seam III BOTTOM samples have not been analysed.

#### 4.5.7.15 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam III Bottom is developed in the block. The seam incrops in the southern part of the block. It is overlain by Seam III Middle and underlain by Seam II Top/Comb. The parting between Seam III Bottom and the overlying Seam III Middle varies from 1.08 to 2.55m and the parting with the underlying Seam II Top/Comb. varies from 18.67 to 60.96m.

The seam is less banded and the thickness of the seam varies from 1.30 to 3.42m. The general thickness of the seam varies from 1.04 to 2.98m. The proximate analysis on 60% RH and 40°C of Seam III Bottom on I₃₀ sample shows moisture%, ash% and volatile matter% as 3.2 to 6.7, 15.2 to 32.8 and 23.8 to 30.7 respectively. The grade of the seam varies from G8 to G4.

## 4.5.8 SEAM III Comb.

### 4.5.8.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.8.1.A: Details of seam references, of the Seam III Comb., Marwatola Sector  
I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C, VII D, VII E	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 3 – V 4
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 5
13	Cross Section Plans		VIII 1
14	OB / Parting Plan		IX 5
15	Seam Folio Plan		X 5

### 4.5.8.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

### 4.5.8.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.8.3.A: Data of Borehole Intersection, of the Seam III Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

SI No.	Seam	III Comb.
1	No. of full seam intersection in boreholes	26
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	-
6	No. of borehole not deepened to encounter the seam	2
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>29</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	4
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.8.3.B: Depth Of Intersection of Roof and Floor of the Seam III Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		III Comb.	
ROOF DEPTH	MIN (M)	164.40	CSM I&II-10
	MAX (M)	619.10	CSM I&II-15
FLOOR DEPTH	MIN (M)	166.30	CSM I&II-10
	MAX (M)	619.20	CSM I&II-15
THICK NESS	MIN (M)	0.10	CSM I&II-15
	MAX (M)	5.06	CSM I&II-05

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.8.3.C: Minimum & Maximum RRLs & FRLs of the Seam III Comb.,  
Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	-106.46	CSM I&II-15
	MAX (M)	341.03	CSM I&II-10
FRL	MIN (M)	-106.56	CSM I&II-15
	MAX (M)	339.13	CSM I&II-10
BH INTERSECTIONS		26	

#### 4.5.8.4 STRATIGRAPHIC POSITION

The Seam III Comb. is overlain by Seam IIIA and underlain by Seam IITop/Comb.

#### 4.5.8.5 PARTING

The details of Parting below the Seam III Comb./Bottom is listed in the Table below.

**Table: 4.5.8.5.A: Parting Details of the Seam III Comb., With Underlying Seams,  
Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam II Top/Comb.	18.67	60.96	28
	MSM-27	MSM-26	

#### 4.5.8.6 THICKNESS

**Table: 4.5.8.6.A: Thickness data of the Seam III Comb. in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		III Comb.	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.10	CSM I&II-15
	MAX (M)	5.06	CSM I&II-05
GENERAL THICKNESS	MIN (M)	0.60	
	MAX (M)	3.22	
BOREHOLE INTERSECTIONS		26	

#### 4.5.8.7 SPLITTING

The seam III Comb. is splitted in to three sections into Top, Middle and Bottom.

#### 4.5.8.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault (F6-F6) trending east-west in the block. In the boreholes MSM-22 and MSM-49 the seam falls in the updipside of the block.

#### 4.5.8.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
III Comb.	ROOF	26	0	1	1	0	13	0	11
	FLOOR		0	1	3	0	5	3	14

#### 4.5.8.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.8.11 DIRT BANDS

The details of dirt band within seam III Comb. have been provided in Annexure V. The maximum thickness of the seam including dirt band is 4.81 meters. The abstract of dirt bands within the seam having thickness more than 0.90m, is given in the following table:

**Table 4.5.8.11 A: Abstract of Dirt Bands in Seam III Comb. having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			III Comb.		
Type of Sample			I30		
Total Seam Thickness		Min.	(m)	1.10	CSM I&II-14
		Max.	(m)	5.06	CSM I&II-05
Analysed Seam Thickness		Min.	(m)	1.02	CSM I&II-14
		Max.	(m)	4.81	CSM I&II-05
Effective Dirt Bands	Included	No.	Min.	0.00	CSM I&II-01
			Max.	0.00	CSM I&II-01
		Thick (m)	Min.	0.00	CSM I&II-01
			Max.	0.00	CSM I&II-01
	Excluded	No.	Min.	0.00	CSM I&II-01
			Max.	3.00	CSM I&II-13
		Thick (m)	Min.	0.00	CSM I&II-01
			Max.	0.41	CSM I&II-13
No. of Bhs			19		

The abstract of combustible and non-combustible dirt band constituents within the seam is tabulated below.

**Table 4.5.8.11 B: Abstract Of Combustible and Non-combustible Dirt Band Constituents in Seam III Comb. having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam			III Comb.	
Total Seam Thickness	Min.	(m)	1.10	
	Max.	(m)	5.06	
Analysed Seam Thickness	Min.	(m)	1.02	
	Max.	(m)	4.81	
Combustible Dirt Bands (Carbonaceous Shale)	No.	Min	0.00	CSM I&II-01
		Max	0.00	CSM I&II-01
	Thick (m)	Min	0.00	CSM I&II-01
		Max	0.00	CSM I&II-01
	%	Min	0.00	CSM I&II-01
		Max	0.00	CSM I&II-01
Non-Combustible Dirt Bands ( Shale & Sandstone )	No.	Min	0.00	CSM I&II-01
		Max	3.00	CSM I&II-13
	Thick (m)	Min	0.00	CSM I&II-01
		Max	0.41	CSM I&II-13
	%	Min	0.00	CSM I&II-01
		Max	18.00	MSM-17
Total bands (Carb. Shale, Shale and Sandstone)	No.	Min	0.00	CSM I&II-01
		Max	3.00	CSM I&II-13
	Thick (m)	Min	0.00	CSM I&II-01
		Max	0.41	CSM I&II-13
	%	Min	0.00	CSM I&II-01
		Max	18.00	MSM-17
Nos of Bh			19	

## 4.5.8.12 PROXIMATE ANALYSIS

**Table 4.5.8.12.A: Range of Proximate Analysis and Grade of Seam III Comb. having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		III Comb.		III Comb.		III Comb.		
Bhs. considered (Det. & Cal.)		19		19		19		
No. of Determined Boreholes		15		15		8		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)		Min	1.10	CSM I&II-14	1.10	CSM I&II-14	1.10	CSM I&II-14
		Max	5.06	CSM I&II-05	5.06	CSM I&II-05	5.06	CSM I&II-05
Range of Analysed Thickness (m)		Min	0.90	MSM-26	1.02	CSM I&II-14	1.02	CSM I&II-14
		Max	4.61	CSM I&II-05	4.81	CSM I&II-05	4.81	CSM I&II-05
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	2.0	CSM I&II-01	2.0	CSM I&II-01	2.0	CSM I&II-01
		Max	10.0	MSM-17	10.0	MSM-17	8.8	CSM I&II-11
	Ash%	Min	13.4	MSM-28	13.4	MSM-28	13.4	MSM-28
		Max	49.6	MSM-19	49.6	MSM-19	49.6	MSM-19
	VM%	Min	21.2	CSM I&II-01	21.2	CSM I&II-01	21.2	CSM I&II-01
		Max	31.1	CSM I&II-11	31.1	CSM I&II-11	31.1	CSM I&II-11
	FC%	Min	36.6	CSM I&II-10	35.2	CSM I&II-13	35.2	CSM I&II-13
		Max	48.2	CSM I&II-01	48.2	CSM I&II-01	48.2	CSM I&II-01
Range of UVM%		Min	27.6	CSM I&II-01	27.6	CSM I&II-01	27.6	CSM I&II-01
		Max	39.9	CSM I&II-11	39.9	CSM I&II-11	39.9	CSM I&II-11
Range of CV in k. cal/kg	Gross CVAR	Min	3394	MSM-19	3394	MSM-19	3394	MSM-19
		Max	6366	MSM-28	6366	MSM-28	6366	MSM-28
	Gross CVDF	Min	7755	MSM-17	7755	MSM-17	7960	CSM I&II-11
		Max	8487	CSM I&II-01	8487	CSM I&II-01	8487	CSM I&II-01
Range of UHV in k. cal/kg		Min	1586	MSM-19	1586	MSM-19	1586	MSM-19
		Max	6167	MSM-28	6167	MSM-28	6167	MSM-28
Range of Grade		Min	G-14		G-14		G-14	
		Max	G-4		G-4		G-4	

## 4.5.8.13 DETERMINED GROSS CV

Table 4.5.8.13 A: Range of Gross CV of Seam III Comb. having thickness more than 0.90m, Marwatola Sector I&amp;II block, Sohagpur Coalfield

Seam Name		III Comb.		III Comb.	
No. of Boreholes (CV Determined)		15		15	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	1.1	CSM I&II-14	1.1	CSM I&II-14
	Max	5.06	CSM I&II-05	5.06	CSM I&II-05
Range of Analysed Thickness (m)	Min	0.90	MSM-26	1.02	CSM I&II-14
	Max	4.61	CSM I&II-05	4.81	CSM I&II-05
Range of Det. CV in k. cal/kg	Gross CVAR	Min	4570	CSM I&II-10	4388
		Max	6049	MSM-34	6049
	Gross CVDF	Min	7755	MSM-17	7755
		Max	8487	CSM I&II-01	8487

## 4.5.8.14 ULTIMATE ANALYSIS

Table 4.5.2.14 A: Range of Ultimate Analysis Seam III Comb., Marwatola Sector I&amp;II block, Sohagpur Coalfield

Seam Name			III Comb.	
Boreholes considered			2	
Type of Samples			BCS/I30I100	
Range of Seam Thickness (m)		Min	2.20	MSM-17
		Max	3.50	MSM-52
Range of Ultimate analyses	C%	Min	55.51	MSM-52
		Max	62.35	MSM-17
	H%	Min	2.97	MSM-52
		Max	3.18	MSM-17
	N%	Min	0.29	MSM-52
		Max	0.82	MSM-17
	S%	Min	0.50	MSM-52
		Max	0.66	MSM-17
	O% by diff.	Min	6.83	MSM-52
		Max	7.59	MSM-17

#### 4.5.8.15 PETROGRAPHIC STUDIES

Results of Petrographic studies are tabulated below.

**Table 4.5.8.15 A: Petrographic Analysis of Seam III Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

BH. NO.	DEPTH		SEA M THICKNESS	TYPE OF SAMPLE	Maceral Composition %				Maceral Composition (Vmmf%)			Rr%
	FROM	TO			Vit	Lipt	Int	VMM	Vit	Lipt	Int	
	(m)	(m)										
CSM I&II - 10	164.40	166.30	1.90	BCS/I30/ I100	21.3	8.5	31.0	39.2	35.0	14.0	51.0	0.54
MSM-19	229.00	231.00	1.10	BCS/130/ I100	25.6	14.8	31.2	28.4	36.0	21.0	43.6	0.51

#### 4.5.8.16 HARDGROVE GRINDABILITY INDEX / ASH FUSION TEMPERATURE RANGE / ASH ANALYSIS

The Hardgrove Grindability Index, Ash Fusion Temperature range and Ash Analysis for seam III Comb. samples have not been analysed.

#### 4.5.8.17 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam **III Comb.** is developed in the block. The seam incrops in the southern part of the block. It is overlain by Seam IV and underlain by Seam II Top/Comb. The parting between Seam III Comb. and the overlying Seam IV varies from 6.17 to 25.73m and the parting with the underlying Seam II Top/Comb. varies from 18.67 to 60.96m.

The seam is less banded and the thickness of the seam varies from 0.10 to 5.06m. The general thickness of the seam varies from 0.60 to 3.22m. The proximate analysis on 60% RH and 40°C of Seam III Comb. on I₃₀ sample shows moisture%, ash% and volatile matter% as 2.0 to 10.0, 13.4 to 49.6 and 21.2 to 31.1 respectively. The grade of the seam varies from G14 to G4.

## 4.5.9 SEAM II Top

### 4.5.9.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.9.1.A: Details of seam references, of the Seam II Top, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C, VII D, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 5 – V 6
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 6
13	Cross Section Plans		VIII 1
14	OB / Parting Plan		IX 6
15	Seam Folio Plan		X 6

### 4.5.9.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

#### 4.5.9.3 DETAILS OF BOREHOLE INTERSECTION

##### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.9.3.A: Data of Borehole Intersection, of the Seam II Top, Marwatola Sector I&II block, Sohagpur Coalfield**

SI No.	Seam	II Top
1	No. of full seam intersection in boreholes	15
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	-
6	No. of borehole not deepened to encounter the seam	-
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>16</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	<b>17</b>
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

##### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.9.3.B: Depth Of Intersection of Roof and Floor of the Seam II Top, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		II Top	
ROOF DEPTH	MIN (M)	237.85	CSM I&II-02
	MAX (M)	328.65	MSM-24
FLOOR DEPTH	MIN (M)	238.75	CSM I&II-02
	MAX (M)	330.13	MSM-24
THICK NESS	MIN (M)	0.40	MSM-52
	MAX (M)	4.80	MSM-55

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.9.3.C: Minimum & Maximum RRLs & FRLs of the Seam II Top, Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	188.53	MSM-52
	MAX (M)	269.52	MSM-27
FRL	MIN (M)	188.13	MSM-52
	MAX (M)	268.99	MSM-27
BH INTERSECTIONS		15	

#### 4.5.9.4 STRATIGRAPHIC POSITION

The Seam II Top is overlain by Seam III Comb. and underlain by Seam II Bottom.

#### 4.5.9.5 PARTING

The details of Parting below the Seam II Top is listed in the Table below.

**Table: 4.5.9.5.A: Parting Details of the Seam II Top, With Underlying Seams, Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam II Bottom	0.47	4.1	15
	CSM I&II-02	MSM-21	

#### 4.5.9.6 THICKNESS

**Table: 4.5.9.6.A: Thickness data of the Seam II Top in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		II Top	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.40	MSM-52
	MAX (M)	4.80	MSM-55
GENERAL THICKNESS	MIN (M)	0.89	
	MAX (M)	3.61	
BOREHOLE INTERSECTIONS		15	

#### 4.5.9.7 SPLITTING

The seam II has been splitted in to II Top and Bottom Sections. However, the present seam II TOP does not exhibit any splitting behavior.

#### 4.5.9.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault (F6-F6) trending east-west in the block. In the boreholes MSM-22 and MSM-49 the seam falls in the updipside of the block.

#### 4.5.9.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
II Top	ROOF	15	0	2	1	0	5	0	7
	FLOOR		0	3	6	0	0	0	6

#### 4.5.9.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.9.11 DIRT BANDS

The details of dirt band within seam-II Top have been provided in Annexure V. The maximum thickness of the seam including dirt band is 4.80 meters. The abstract of dirt bands within the seam having thickness more than 0.90m, is given in the following table:

**Table 4.5.9.11 A: Abstract of Dirt Bands in Seam II Top having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			II Top		
Type of Sample			I30		
Total Seam Thickness		Min.	(m)	0.90	CSM I&II-02
		Max.	(m)	4.80	MSM-55
Analysed Seam Thickness		Min.	(m)	0.90	CSM I&II-02
		Max.	(m)	4.80	MSM-55
Effective Dirt Bands	Included	No.	Min.	0.00	CSM I&II-01
			Max.	0.00	CSM I&II-01
		Thick (m)	Min.	0.00	CSM I&II-01
			Max.	0.00	CSM I&II-01
	Excluded	No.	Min.	0.00	CSM I&II-02
			Max.	3.00	CSM I&II-08
		Thick (m)	Min.	0.00	CSM I&II-02
			Max.	0.49	CSM I&II-08
No. of Bhs			12		

The abstract of combustible and non-combustible dirt band constituents within the seam is tabulated below.

**Table 4.5.9.11 B: Abstract Of Combustible and Non-combustible Dirt Band Constituents  
in Seam II Top having thickness more than 0.90m, Marwatola Sector I&II block,  
Sohagpur Coalfield**

Seam			II Top	
Total Seam Thickness	Min.	(m)	0.90	
	Max.	(m)	4.80	
Analysed Seam Thickness	Min.	(m)	0.90	
	Max.	(m)	4.80	
Combustible Dirt Bands (Carbonaceous Shale)	No.	Min	0.00	CSM I&II-01
		Max	0.00	CSM I&II-01
	Thick (m)	Min	0.00	CSM I&II-01
		Max	0.00	CSM I&II-01
	%	Min	0.00	CSM I&II-01
		Max	0.00	CSM I&II-01
Non-Combustible Dirt Bands ( Shale & Sandstone )	No.	Min	0.00	CSM I&II-02
		Max	3.00	CSM I&II-08
	Thick (m)	Min	0.00	CSM I&II-02
		Max	0.49	CSM I&II-08
	%	Min	0.00	CSM I&II-02
		Max	12.00	CSM I&II-08
Total bands (Carb. Shale, Shale and Sandstone)	No.	Min	0.00	CSM I&II-02
		Max	3.00	CSM I&II-08
	Thick (m)	Min	0.00	CSM I&II-02
		Max	0.49	CSM I&II-08
	%	Min	0.00	CSM I&II-02
		Max	12.00	CSM I&II-08
Nos of Bh			12	

## 4.5.9.12 PROXIMATE ANALYSIS

**Table 4.5.9.12.A: Range of Proximate Analysis and Grade of Seam II Top having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		II Top		II Top		II Top		
Bhs. considered (Det. & Cal.)		12		12		12		
No. of Determined Boreholes		12		12		7		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)	Min	0.90	CSM I&II-02	0.90	CSM I&II-02	0.90	CSM I&II-02	
	Max	4.80	MSM-55	4.80	MSM-55	4.80	MSM-55	
Range of Analysed Thickness (m)	Min	0.90	CSM I&II-02	0.90	CSM I&II-02	0.90	CSM I&II-02	
	Max	4.80	MSM-55	4.80	MSM-55	4.80	MSM-55	
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	2.8	CSM I&II-01	2.8	CSM I&II-01	2.8	CSM I&II-01
		Max	7.6	CSM I&II-11	7.6	CSM I&II-11	7.6	CSM I&II-11
	Ash%	Min	11.6	MSM-24	11.6	MSM-24	11.6	MSM-24
		Max	30.2	CSM I&II-01	30.2	CSM I&II-01	30.2	CSM I&II-01
	VM%	Min	16.4	CSM I&II-01	16.4	CSM I&II-01	16.4	CSM I&II-01
		Max	30.9	MSM-28	30.9	MSM-28	29.4	CSM I&II-11
	FC%	Min	42.5	CSM I&II-05	42.5	CSM I&II-05	42.5	CSM I&II-05
		Max	55.5	MSM-24	55.5	MSM-24	51.2	CSM I&II-11
Range of UVM%		Min	20.9	CSM I&II-01	20.9	CSM I&II-01	20.9	CSM I&II-01
		Max	37.4	MSM-28	37.4	MSM-28	35.6	CSM I&II-11
Range of CV in k. cal/kg	Gross CVAR	Min	5192	CSM I&II-05	5192	CSM I&II-05	5192	CSM I&II-05
		Max	6637	MSM-24	6637	MSM-24	6637	MSM-24
	Gross CVDF	Min	8085	CSM I&II-08	8085	CSM I&II-08	8085	CSM I&II-08
		Max	8428	CSM I&II-01	8428	CSM I&II-01	8428	CSM I&II-01
Range of UHV in k. cal/kg		Min	4304	CSM I&II-05	4304	CSM I&II-05	4304	CSM I&II-05
		Max	6512	MSM-24	6512	MSM-24	6512	MSM-24
Range of Grade		Min	G-8		G-8		G-8	
		Max	G-3		G-3		G-3	

#### 4.5.9.13 DETERMINED GROSS CV

**Table 4.5.9.13 A: Range of Gross CV of Seam II Top having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		II Top		II Top	
No. of Boreholes (CV Determined)		12		12	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	0.9	CSM I&II-02	0.9	CSM I&II-02
	Max	4.8	MSM-55	4.8	MSM-55
Range of Analysed Thickness (m)	Min	0.90	CSM I&II-02	0.90	CSM I&II-02
	Max	4.80	MSM-55	4.80	MSM-55
Range of Det. CV in k. cal/kg	Gross CVAR	Min	5192	CSM I&II-05	5192
		Max	6637	MSM-24	6637
	Gross CVDF	Min	8085	CSM I&II-08	8085
		Max	8428	CSM I&II-01	8428

#### 4.5.9.14 ULTIMATE ANALYSIS / HARDGROVE GRINDABILITY INDEX / ASH FUSION TEMPERATURE RANGE / ASH ANALYSIS / PETROGRAPHIC STUDIES

The Ultimate analysis, HGI, Ash Fusion temperature range, Ash analysis and Petrographic studies for seam III Top samples have not been analysed

#### 4.5.9.15 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam II Top is developed in the block. The seam incrops in the southern part of the block. It is overlain by Seam III Bottom/Comb. and underlain by Seam II Bottom. The parting between Seam II Top and the overlying Seam III Bottom/Comb. varies from 18.67 to 60.96m and the parting with the underlying Seam II Bottom varies from 0.47 to 4.10m.

The seam is less banded and the thickness of the seam varies from 0.40 to 4.80m. The general thickness of the seam varies from 0.89 to 3.61m. The proximate analysis on 60% RH and 40°C of Seam IX on I₃₀ sample shows moisture%, ash% and volatile matter% as 2.8 to 7.6, 11.6 to 30.2 and 16.4 to 30.9 respectively. The grade of the seam varies from G8 to G3.

## 4.5.10 SEAM II Bottom

### 4.5.10.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.10.1.A: Details of seam references, of the Seam II Bottom, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C,VIID, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 5 – V 6
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 7
13	Cross Section Plans		VIII 1
14	OB / Parting Plan		IX 7
15	Seam Folio Plan		X 7

### 4.5.10.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

### 4.5.10.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.10.3.A: Data of Borehole Intersection, of the Seam II Bottom, Marwatola Sector I&II block, Sohagpur Coalfield**

S.No.	Seam	II Bottom
1	No. of full seam intersection in boreholes	15
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	-
6	No. of borehole not deepened to encounter the seam	-
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>16</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	17
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.10.3.B: Depth Of Intersection of Roof and Floor of the Seam II Bottom, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		II Bottom	
ROOF DEPTH	MIN (M)	239.22	CSM I&II-02
	MAX (M)	332.15	MSM-24
FLOOR DEPTH	MIN (M)	241.15	CSM I&II-02
	MAX (M)	334.40	MSM-24
THICK NESS	MIN (M)	0.36	MSM-27
	MAX (M)	2.30	MSM-21

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.10.3.C: Minimum & Maximum RRLs & FRLs of the Seam II Bottom, Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	186.88	MSM-52
	MAX (M)	268.3	MSM-27
FRL	MIN (M)	185.38	MSM-52
	MAX (M)	267.94	MSM-27
BH INTERSECTIONS		15	

#### 4.5.10.4 STRATIGRAPHIC POSITION

The Seam II Bottom is overlain by Seam II Top and underlain by Seam IB.

#### 4.5.10.5 PARTING

The details of Parting below the Seam II Bottom/Comb. is listed in the Table below.

**Table: 4.5.10.5.A: Parting Details of the Seam II Bottom/Comb, With Underlying Seams, Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam IB	7.99	33.83	26
	CSM I&II-04	CSM I&II-15	

#### 4.5.10.6 THICKNESS

**Table: 4.5.10.6.A: Thickness data of the Seam II Bottom in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		II Bottom	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.36	MSM-27
	MAX (M)	2.30	MSM-21
GENERAL THICKNESS	MIN (M)	0.59	
	MAX (M)	1.93	
BOREHOLE INTERSECTIONS		15	

#### 4.5.10.7 SPLITTING

The seam II has been splitted in to II Top and Bottom Sections. However, the present seam II Bottom does not exhibit any splitting behavior.

#### 4.5.10.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault (F6-F6) trending east-west in the block. In the boreholes MSM-22 and MSM-49 the seam falls in the updipside of the block.

#### 4.5.10.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
II Bottom	ROOF	15	0	3	4	0	0	1	7
	FLOOR		0	0	1	0	3	2	9

#### 4.5.10.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.10.11 DIRT BANDS

The details of dirt band within seam-II Bottom have been provided in Annexure V. The maximum thickness of the seam including dirt band is 2.30 meters. The seam is devoid of any dirt band.

#### 4.5.10.12 PROXIMATE ANALYSIS

**Table 4.5.10.12.A: Range of Proximate Analysis and Grade of Seam II Bottom having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		II Bottom		II Bottom		II Bottom		
Bhs. considered (Det. & Cal.)		9		9		9		
No. of Determined Boreholes		8		8		5		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)		Min	0.94	CSM I&II-14	0.94	CSM I&II-14	0.94	CSM I&II-14
		Max	2.3	MSM-21	2.30	MSM-21	2.30	MSM-21
Range of Analysed Thickness (m)		Min	0.94	CSM I&II-14	0.94	CSM I&II-14	0.94	CSM I&II-14
		Max	2.3	MSM-21	2.30	MSM-21	2.30	MSM-21
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	2.5	CSM I&II-01	2.5	CSM I&II-01	2.5	CSM I&II-01
		Max	6.7	CSM I&II-11	6.7	CSM I&II-11	6.7	CSM I&II-11
	Ash%	Min	13.4	MSM-24	13.4	MSM-24	13.6	MSM-24
		Max	35.1	MSM-52	35.1	MSM-52	35.1	MSM-52
	VM%	Min	14.2	CSM I&II-01	14.2	CSM I&II-01	14.2	CSM I&II-01
		Max	31.2	CSM I&II-11	31.2	CSM I&II-11	31.2	CSM I&II-11
	FC%	Min	38.1	CSM I&II-14	38.1	CSM I&II-14	38.1	CSM I&II-14
		Max	51.5	MSM-24	51.5	MSM-24	51.4	CSM I&II-01
Range of UVM%		Min	17.7	CSM I&II-01	17.7	CSM I&II-01	17.7	CSM I&II-01
		Max	39.8	CSM I&II-11	39.8	CSM I&II-11	39.8	CSM I&II-11
Range of CV in k. cal/kg	Gross CVAR	Min	4643	MSM-52	4643	MSM-52	4643	MSM-52
		Max	6402	MSM-28	6402	MSM-28	6420	MSM-24
	Gross CVDF	Min	7967	CSM I&II-14	7967	CSM I&II-14	7967	CSM I&II-14
		Max	8660	CSM I&II-01	8660	CSM I&II-01	8660	CSM I&II-01
Range of UHV in k. cal/kg		Min	3476	MSM-52	3476	MSM-52	3476	MSM-52
		Max	6167	MSM-24	6167	MSM-24	6209	MSM-24
Range of Grade		Min	G-9		G-9		G-9	
		Max	G-3		G-3		G-3	

#### 4.5.10.13 DETERMINED GROSS CV

**Table 4.5.10.13 A: Range of Gross CV of Seam II Bottom having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		II Bottom		II Bottom	
No. of Boreholes (CV Determined)		8		8	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	0.94	CSM I&II-14	0.94	CSM I&II-14
	Max	2.3	MSM-21	2.3	MSM-21
Range of Analysed Thickness (m)	Min	0.94	CSM I&II-14	0.94	CSM I&II-14
	Max	2.30	MSM-21	2.30	MSM-21
Range of Det. CV in k. cal/kg	Gross CVAR	Min	4831	CSM I&II-14	4831
		Max	6402	MSM-28	6402
	Gross CVDF	Min	7967	CSM I&II-14	7967
		Max	8660	CSM I&II-01	8660

#### 4.5.10.14 ULTIMATE ANALYSIS / HARDGROVE GRINDABILITY INDEX / ASH FUSION TEMPERATURE RANGE / ASH ANALYSIS / PETROGRAPHIC STUDIES

The Ultimate analysis, HGI, Ash Fusion temperature range, Ash analysis and Petrographic studies for seam II Bottom samples have not been analysed

#### 4.5.10.15 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam II Bottom is developed in the block. The seam incrops in the southern part of the block. It is overlain by Seam II Top and underlain by Seam IB. The parting between Seam II Bottom and the overlying Seam II Top varies from 0.47 to 4.10m and the parting with the underlying Seam IB varies from 7.99 to 33.83m.

The seam is devoid of any dirt band and the thickness of the seam varies from 0.36 to 2.30m. The general thickness of the seam varies from 0.59 to 1.93m. The proximate analysis on 60% RH and 40°C of Seam II Bottom on I₃₀ sample shows moisture%, ash% and volatile matter% as 2.5 to 6.7, 13.4 to 35.1 and 14.2 to 31.2 respectively. The grade of the seam varies from G9 to G3.

#### 4.5.11 SEAM II Comb.

##### 4.5.11.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.11.1.A: Details of seam references, of the Seam II Comb., Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A, VII B, VII C, VIID, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 5 – V 6
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 7
13	Cross Section Plans		VIII 1
14	OB / Parting Plan		IX 7
15	Seam Folio Plan		X 7

##### 4.5.11.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

### 4.5.11.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.11.3.A: Data of Borehole Intersection, of the Seam II Comb., Marwatola Sector -I&II block, Sohagpur Coalfield**

S.No.	Seam	II Comb.
1	No. of full seam intersection in boreholes	14
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	1
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	1
6	No. of borehole not deepened to encounter the seam	2
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>19</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	14
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.11.3.B: Depth Of Intersection of Roof and Floor of the Seam II Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		II Comb.	
ROOF DEPTH	MIN (M)	199.30	CSM I&II-10
	MAX (M)	650.37	CSM I&II-15
FLOOR DEPTH	MIN (M)	202.30	CSM I&II-10
	MAX (M)	650.87	CSM I&II-15
THICK NESS	MIN (M)	0.03	CSM I&II-13
	MAX (M)	5.60	MSM-17

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.11.3.C: Minimum & Maximum RRLs & FRLs of the Seam II Comb., Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	137.73	CSM I&II-15
	MAX (M)	306.13	CSM I&II-10
FRL	MIN (M)	138.23	CSM I&II-15
	MAX (M)	303.13	CSM I&II-10
BH INTERSECTIONS		14	

#### 4.5.11.4 STRATIGRAPHIC POSITION

The Seam II Comb. is overlain by Seam III Bottom/Comb. and underlain by Seam IB.

#### 4.5.11.5 PARTING

The details of Parting above and below the Seam II Comb. is listed in the Table below.

**Table: 4.5.11.5.A: Parting Details of the Seam II Comb., With Underlying Seams, Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam IB	7.99	33.83	14
	CSM I&II-04	CSM I&II-15	

#### 4.5.11.6 THICKNESS

**Table: 4.5.11.6.A: Thickness data of the Seam II Comb. in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		II Comb.	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.03	CSM I&II-13
	MAX (M)	5.60	MSM-17
GENERAL THICKNESS	MIN (M)	0.92	
	MAX (M)	4.84	
BOREHOLE INTERSECTIONS		14	

#### 4.5.11.7 SPLITTING

The seam II Comb. has been splitted in to II Top and Bottom section.

#### 4.5.11.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault (F6-F6) trending east-west in the block. In the boreholes MSM-22 and MSM-49 the seam falls in the updipside of the block.

#### 4.5.11.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
II Comb.	ROOF	15	0	3	4	0	0	1	7
	FLOOR		0	0	1	0	3	2	9

#### 4.5.11.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.11.11 DIRT BANDS

The details of dirt band within seam-II Comb. have been provided in Annexure V. The maximum thickness of the seam including dirt band is 5.08 meters. The abstract of dirt bands within the seam having thickness more than 0.90m, is given in the following table:

**Table 4.5.11.11 A: Abstract of Dirt Bands in Seam II Comb. having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			II Comb.		
Type of Sample			I30		
Total Seam Thickness		Min.	(m)	1.39	MSM-23
		Max.	(m)	5.60	MSM-17
Analysed Seam Thickness		Min.	(m)	1.39	MSM-23
		Max.	(m)	5.08	MSM-19
Effective Dirt Bands	Included	No.	Min.	0.00	CSM I&II-03
			Max.	1.00	CSM I&II-10
		Thick (m)	Min.	0.00	CSM I&II-03
			Max.	0.22	CSM I&II-10
	Excluded	No.	Min.	0.00	CSM I&II-03
			Max.	2.00	CSM I&II-04
		Thick (m)	Min.	0.00	CSM I&II-03
			Max.	0.46	MSM-54
No. of Bhs			10		

The abstract of combustible and non-combustible dirt band constituents within the seam is tabulated below.

**Table 4.5.11.11 B: Abstract Of Combustible and Non-combustible Dirt Band Constituents in Seam II Comb. having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam			II Comb.	
Total Seam Thickness	Min.	(m)	1.39	
	Max.	(m)	5.60	
Analysed Seam Thickness	Min.	(m)	1.39	
	Max.	(m)	5.08	
Combustible Dirt Bands (Carbonaceous Shale)	No.	Min	0.00	CSM I&II-03
		Max	1.00	CSM I&II-10
	Thick (m)	Min	0.00	CSM I&II-03
		Max	0.22	CSM I&II-10
	%	Min	0.00	CSM I&II-03
		Max	7.00	CSM I&II-10

Seam			II Comb.	
Non-Combustible Dirt Bands ( Shale & Sandstone )	No.	Min	0.00	CSM I&II-03
		Max	2.00	CSM I&II-04
	Thick (m)	Min	0.00	CSM I&II-03
		Max	0.46	MSM-54
	%	Min	0.00	CSM I&II-03
		Max	11.00	MSM-54
Total bands (Carb. Shale, Shale and Sandstone)	No.	Min	0.00	CSM I&II-03
		Max	2.00	CSM I&II-04
	Thick (m)	Min	0.00	CSM I&II-03
		Max	0.46	MSM-54
	%	Min	0.00	CSM I&II-03
		Max	11.00	MSM-54
Nos of Bh			10	

#### 4.5.11.12 PROXIMATE ANALYSIS

**Table 4.5.11.12.A: Range of Proximate Analysis and Grade of Seam II Comb. having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		II Comb.		II Comb.		II Comb.		
Bhs. considered (Det. & Cal.)		10		10		10		
No. of Determined Boreholes		9		9		4		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)		Min	1.39	MSM-23	1.39	MSM-23	1.39	MSM-23
		Max	5.60	MSM-17	5.60	MSM-17	5.60	MSM-17
Range of Analysed Thickness (m)		Min	1.39	MSM-23	1.39	MSM-23	1.39	MSM-23
		Max	5.08	MSM-19	5.08	MSM-19	5.08	MSM-19
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	1.9	CSM I&II-03	1.9	CSM I&II-03	1.9	CSM I&II-03
		Max	7.4	MSM-18	7.4	MSM-18	6.3	MSM-23
	Ash%	Min	12.0	MSM-23	12.0	MSM-23	12.2	MSM-23
		Max	31.1	MSM-26	33.8	CSM I&II-10	33.8	CSM I&II-10
	VM%	Min	22.9	CSM I&II-10	22.1	CSM I&II-10	22.1	CSM I&II-10
		Max	28.5	MSM-23	28.5	MSM-23	26.3	CSM I&II-03
	FC%	Min	38.6	MSM-26	38.6	MSM-26	40.9	CSM I&II-10
		Max	57.5	CSM I&II-03	57.5	CSM I&II-03	57.5	CSM I&II-03
Range of UVM%		Min	30.2	CSM I&II-03	30.2	CSM I&II-03	30.2	CSM I&II-03
		Max	36.9	MSM-26	36.9	MSM-26	33.7	CSM I&II-07
Range of CV in k. cal/kg	Gross CVAR	Min	4961	MSM-26	4961	MSM-26	4989	MSM-26
		Max	7025	CSM I&II-03	7025	CSM I&II-03	7025	CSM I&II-03
	Gross CVDF	Min	8035	MSM-18	8035	MSM-18	8057	CSM I&II-04
		Max	8540	CSM I&II-03	8540	CSM I&II-03	8540	CSM I&II-03
Range of UHV in k. cal/kg		Min	3973	MSM-26	3791	CSM I&II-10	3791	CSM I&II-10
		Max	6650	CSM I&II-03	6650	CSM I&II-03	6650	CSM I&II-03
Range of Grade		Min	G-8		G-8		G-8	
		Max	G-1		G-1		G-1	

#### 4.5.11.13 DETERMINED GROSS CV

**Table 4.5.11.13 A: Range of Gross CV of Seam II Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		II Comb.		II Comb.	
No. of Boreholes (CV Determined)		9		9	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	1.39	MSM-23	1.39	MSM-23
	Max	5.41	CSM I&II-04	5.41	CSM I&II-04
Range of Analysed Thickness (m)	Min	1.39	MSM-23	1.39	MSM-23
	Max	5.08	MSM-19	5.08	MSM-19
Range of Det. CV in k. cal/kg	Gross CVAR	Min	4961	MSM-26	4961
		Max	7025	CSM I&II-03	7025
	Gross CVDF	Min	8035	MSM-18	8035
		Max	8540	CSM I&II-03	8540

#### 4.5.11.14 ULTIMATE ANALYSIS

**Table 4.5.11.14 A: Range of Ultimate Analysis Seam II Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		II Comb.	
Boreholes considered		3	
Type of Samples		BCS/I30/I100	
Range of Seam Thickness (m)	Min	0.50	CSM I&II-15
	Max	5.60	MSM-17
Range of Ultimate analyses	C%	Min	44.08
		Max	60.22
	H%	Min	2.84
		Max	3.34
	N%	Min	0.80
		Max	0.93
	S%	Min	0.36
		Max	0.44
	O% by diff.	Min	2.50
		Max	6.20

#### 4.5.11.15 PETROGRAPHIC STUDIES

Results of Petrographic studies are tabulated below.

**Table 4.5.11.15 A: Petrographic Analysis of Seam II Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

BH. NO.	DEPTH		SEAM THICK .	TYPE OF SAMPLE	Maceral Composition %				Maceral Composition (Vmmf%)			Rr%
	FROM	TO			Vit	Lipt	Int	VMM	Vit	Lipt	Int	
	(m)	(m)										
CSM I&II-10	199.30	202.30	3.00	BCS	34.7	10.2	28.5	26.6	47.3	13.9	38.8	0.53
CSM I&II-10	199.30	202.30	3.00	I30 / I100	39.4	7.5	20.6	32.5	58.3	11.1	30.6	0.59
MSM-19	284.00	289.08	5.08	BCS/130/	23.5	13.0	43.4	20.0	29.0	16.0	54.3	0.51

#### 4.5.11.16 HARDGROVE GRINDABILITY INDEX

**Table 4.5.11.16 A: Range of HGI of Seam II Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name	Type of Samples	Range of HGI		Boreholes considered
		Min	Max	
II Comb.	BCS	71	71	1
		CSM I&II-15	CSM I&II-15	

#### 4.5.11.17 ASH FUSION TEMPERATURE RANGE

**Table 4.5.8.17 A: Ash Fusion Temperature Range under Mildly Reducing Atmosphere for Seam II Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

Range of Ash fusion Temperature under Mildly Reducing Atmosphere on I ₃₀ samples						
II Comb.						
IDT (°C)		HT (°C)		FT (°C)		No. of boreholes
Min	Max	Min	Max	Min	Max	
1216	1216	>1400	>1400	>1400	>1400	1
CSM I&II-15						

#### 4.5.11.18 ASH ANALYSIS

**Table 4.5.11.18 A: Ash analysis Range for Seam II Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			II COMB	
Boreholes considered			1	
Type of Samples			BCS/ I30/ I100	
<b>Range of Ash Analyses</b>	<b>SiO₂%</b>	<b>Min</b>	63.14	CSM I&II-15
		<b>Max</b>	63.14	CSM I&II-15
	<b>Al₂O₃%</b>	<b>Min</b>	21.35	CSM I&II-15
		<b>Max</b>	21.35	CSM I&II-15
	<b>Fe₂O₃%</b>	<b>Min</b>	3.93	CSM I&II-15
		<b>Max</b>	3.93	CSM I&II-15
	<b>TiO₂%</b>	<b>Min</b>	2.55	CSM I&II-15
		<b>Max</b>	2.55	CSM I&II-15
	<b>MgO%</b>	<b>Min</b>	1.33	CSM I&II-15
		<b>Max</b>	1.33	CSM I&II-15
	<b>P₂O₅%</b>	<b>Min</b>	0.22	CSM I&II-15
		<b>Max</b>	0.22	CSM I&II-15
	<b>SO₃</b>	<b>Min</b>	0.07	CSM I&II-15
		<b>Max</b>	0.07	CSM I&II-15
	<b>MnO%</b>	<b>Min</b>	0.06	CSM I&II-15
		<b>Max</b>	0.06	CSM I&II-15
	<b>K₂O</b>	<b>Min</b>	2.76	CSM I&II-15
		<b>Max</b>	2.76	CSM I&II-15

#### 4.5.11.20 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam II Comb. is developed in the block. The seam incrops in the southern part of the block. It is overlain by Seam III Bottom/Top and underlain by Seam IB. The parting between Seam II Comb. and the overlying Seam III Bottom/Top varies from 18.67 to 60.96m and the parting with the underlying Seam IB varies from 7.99 to 33.83m.

The seam is less banded and the thickness of the seam varies from 0.03 to 5.60m. The general thickness of the seam varies from 0.92 to 4.84m. The proximate analysis on 60% RH and 40°C of Seam II Comb. on I₃₀ sample shows moisture%, ash% and volatile matter% as 1.9 to 7.4, 12.0 to 33.8 and 22.1 to 28.5 respectively. The grade of the seam varies from G8 to G1.

## 4.5.12 SEAM IB

### 4.5.12.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.11.1.A: Details of seam references, of the Seam IB, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C,VIID, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 7
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 8
13	Cross Section Plans		VIII 1
14	OB / Parting Plan		IX 8
15	Seam Folio Plan		X 8

### 4.5.12.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

### 4.5.12.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.12.3.A: Data of Borehole Intersection, of the Seam IB, Marwatola Sector I&II block, Sohagpur Coalfield**

S.No.	Seam	IB
1	No. of full seam intersection in boreholes	28
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	1
6	No. of borehole not deepened to encounter the seam	3
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>33</b>
7	No. of boreholes falling in updip area of seam incrop	2
8	No. of boreholes falling in combined / split zone	-
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.12.3.B: Depth Of Intersection of Roof and Floor of the Seam IB, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		IB	
ROOF DEPTH	MIN (M)	20.00	MSM-22
	MAX (M)	684.70	CSM I&II-15
FLOOR DEPTH	MIN (M)	20.98	MSM-22
	MAX (M)	685.30	CSM I&II-15
THICK NESS	MIN (M)	0.05	MSM-27
	MAX (M)	2.47	CSM I&II-10

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.12.3.C: Minimum & Maximum RRLs & FRLs of the Seam II Comb., Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	-172.06	CSM I&II-15
	MAX (M)	467.99	MSM-22
FRL	MIN (M)	-172.66	CSM I&II-15
	MAX (M)	467.01	MSM-22
BH INTERSECTIONS		28	

#### 4.5.12.4 STRATIGRAPHIC POSITION

The Seam IB is overlain by Seam II Bottom/Comb. and underlain by Seam IA.

#### 4.5.12.5 PARTING

The details of Parting below the Seam IB is listed in the Table below.

**Table: 4.5.12.5.A: Parting Details of the Seam IB, With Underlying Seams, Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam IA	3.90	17.25	28
	CSM I&II-12	CSM I&II-08	

#### 4.5.12.6 THICKNESS

**Table: 4.5.12.6.A: Thickness data of the Seam IB in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		IB	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.05	MSM-27
	MAX (M)	2.47	CSM I&II-10
GENERAL THICKNESS	MIN (M)	0.99	
	MAX (M)	1.95	
BOREHOLE INTERSECTIONS		28	

#### 4.5.12.7 SPLITTING

The seam does not exhibit any splitting behaviour

#### 4.5.12.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault (F6-F6) trending east-west in the block. In the borehole MSM-49 the seam falls in the updip side of the block. The length of the incrop is around 2.9kms.

#### 4.5.12.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
IB	ROOF	28	1	5	4	0	11	0	7
	FLOOR		0	8	4	0	4	3	9

#### 4.5.12.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.12.11 DIRT BANDS

The details of dirt band within seam-IB have been provided in Annexure V. The maximum thickness of the seam including dirt band is 2.10 meters. The abstract of dirt bands within the seam having thickness more than 0.90m, is given in the following table:

**Table 4.5.12.11 A: Abstract of Dirt Bands in Seam IB having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name				IB	
Type of Sample				I30	
Total Seam Thickness		Min.	(m)	0.98	MSM-22
		Max.	(m)	2.47	CSM I&II-10
Analysed Seam Thickness		Min.	(m)	0.98	MSM-22
		Max.	(m)	2.10	CSM I&II-10
Effective Dirt Bands	Included	No.	Min.	0.00	CSM I&II-01
			Max.	0.00	CSM I&II-01
		Thick (m)	Min.	0.00	CSM I&II-01
			Max.	0.00	CSM I&II-01
	Excluded	No.	Min.	0.00	CSM I&II-01
			Max.	1.00	CSM I&II-04
		Thick (m)	Min.	0.00	CSM I&II-01
			Max.	0.37	CSM I&II-10
No. of Bhs				25	

The abstract of combustible and non-combustible dirt band constituents within the seam is tabulated below.

**Table 4.5.12.11 B: Abstract Of Combustible and Non-combustible Dirt Band Constituents in Seam IB having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam			IB	
Total Seam Thickness	Min.	(m)	0.98	
	Max.	(m)	2.47	
Analysed Seam Thickness	Min.	(m)	0.98	
	Max.	(m)	2.10	
Combustible Dirt Bands (Carbonaceous Shale)	No.	Min	0.00	CSM I&II-01
		Max	0.00	CSM I&II-01
	Thick (m)	Min	0.00	CSM I&II-01
		Max	0.00	CSM I&II-01
	%	Min	0.00	CSM I&II-01
		Max	0.00	CSM I&II-01
Non-Combustible Dirt	No.	Min	0.00	CSM I&II-01

Seam			IB	
Bands ( Shale & Sandstone )		Max	1.00	CSM I&II-04
	Thick (m)	Min	0.00	CSM I&II-01
		Max	0.37	CSM I&II-10
	%	Min	0.00	CSM I&II-01
		Max	15.00	CSM I&II-10
Total bands (Carb. Shale, Shale and Sandstone)	No.	Min	0.00	CSM I&II-01
		Max	1.00	CSM I&II-04
	Thick (m)	Min	0.00	CSM I&II-01
		Max	0.37	CSM I&II-10
	%	Min	0.00	CSM I&II-01
		Max	15.00	CSM I&II-10
Nos of Bh			25	

#### 4.5.12.12 PROXIMATE ANALYSIS

**Table 4.5.12.12.A: Range of Proximate Analysis and Grade of Seam IB having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		IB		IB		IB		
Bhs. considered (Det. & Cal.)		25		25		24		
No. of Determined Boreholes		22		22		10		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)		Min	0.98	MSM-22	0.98	MSM-22	0.98	MSM-22
		Max	2.47	CSM I&II-10	2.47	CSM I&II-10	2.47	CSM I&II-10
Range of Analysed Thickness (m)		Min	0.98	MSM-22	0.98	MSM-22	0.98	MSM-22
		Max	2.10	CSM I&II-10	2.10	CSM I&II-10	2.10	CSM I&II-10
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	1.5	MSM-22	1.5	MSM-22	3.0	CSM I&II-01
		Max	7.9	MSM-17	7.9	MSM-17	7.7	CSM I&II-11
	Ash%	Min	9.6	MSM-31	9.6	MSM-31	9.6	MSM-31
		Max	32.8	CSM I&II-01	32.8	CSM I&II-01	32.8	CSM I&II-01
	VM%	Min	13.9	MSM-22	13.9	MSM-22	14.5	CSM I&II-01
		Max	31.2	CSM I&II-11	31.2	CSM I&II-11	31.2	CSM I&II-11
	FC%	Min	40.4	CSM I&II-05	38.2	CSM I&II-14	38.2	CSM I&II-14
		Max	69.5	MSM-22	69.5	MSM-22	49.8	CSM I&II-01
Range of UVM%		Min	15.1	MSM-22	15.1	MSM-22	18.4	CSM I&II-01
		Max	37.7	CSM I&II-11	37.7	CSM I&II-11	37.7	CSM I&II-11
Range of	Gross	Min	4918	CSM I&II-05	4918	CSM I&II-05	4918	CSM I&II-05

Seam Name			IB		IB		IB	
Bhs. considered (Det. & Cal.)			25		25		24	
No. of Determined Boreholes			22		22		10	
Type of Samples			BCS		I30		I100	
CV in k. cal/kg	CVAR	Max	6910	MSM-22	6910	MSM-22	6782	MSM-31
	Gross	Min	7869	MSM-18	7869	MSM-18	8094	CSM I&II-04
	CVDF	Max	8485	CSM I&II-01	8485	CSM I&II-01	8485	CSM I&II-01
Range of UHV in k. cal/kg		Min	3890	CSM I&II-05	3890	CSM I&II-05	3890	CSM I&II-05
		Max	6719	MSM-31	6719	MSM-31	6747	MSM-31
Range of Grade		Min	G-8		G-8		G-8	
		Max	G-2		G-2		G-2	

#### 4.5.12.13 DETERMINED GROSS CV

**Table 4.5.12.13 A: Range of Gross CV of Seam IB, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			IB		IB	
No. of Boreholes (CV Determined)			22		22	
Type of Samples			BCS		I30	
Range of Seam Thickness (m)		Min	0.98	MSM-22	0.98	MSM-22
		Max	2.47	CSM I&II-10	2.47	CSM I&II-10
Range of Analysed Thickness (m)		Min	0.98	MSM-22	0.98	MSM-22
		Max	2.10	CSM I&II-10	2.10	CSM I&II-10
Range of Det. CV in k. cal/kg	Gross CVAR	Min	4918	CSM I&II-05	4918	CSM I&II-05
		Max	6910	MSM-22	6910	MSM-22
	Gross CVDF	Min	7869	MSM-18	7869	MSM-18
		Max	8485	CSM I&II-01	8485	CSM I&II-01

#### 4.5.12.14 ULTIMATE ANALYSIS

**Table 4.5.12.14 A: Range of Ultimate Analysis Seam IB, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			IB	
Boreholes considered			2	
Type of Samples			BCS / I30	
Range of Seam	Thickness (m)	Min	060	CSM I&II-15
		Max	1.83	MSM-17
Range of Ultimate analyses	C%	Min	61.96	MSM-17
		Max	67.67	CSM I&II-15
	H%	Min	3.17	MSM-17
		Max	3.81	CSM I&II-15
	N%	Min	0.78	MSM-17
		Max	1.36	CSM I&II-15
	S%	Min	0.44	MSM-17
		Max	1.28	CSM I&II-15
	O% by diff.	Min	2.90	CSM I&II-15
		Max	7.55	MSM-17

#### 4.5.12.15 PETROGRAPHIC STUDIES

Results of Petrographic studies are tabulated below.

**Table 4.5.11.15 A: Petrographical Analysis of Seam IB, Marwatola Sector I&II block, Sohagpur Coalfield**

BH. NO.	DEPTH		SEAM THICK NESS	TYPE OF SAMPLE	Maceral Composition %				Maceral Composition (Vmmf%)			Rr%
	FROM	TO			Vit	Lipt	Int	VMM	Vit	Lipt	Int	
	(m)	(m)	(m)									
CSM I&II -10	215.90	218.37	2.47	BCS / I30	32.8	10.9	37.4	18.9	40.4	13.4	46.1	0.59
MSM-19	300.73	301.98	1.25	130/1 100	23.6	16.3	44.5	15.6	28.0	19.0	52.7	0.45

#### 4.5.12.16 HARDGROVE GRINDABILITY INDEX

**Table 4.5.11.16 A: Range of HGI of Seam IB, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name	Type of Samples	Range of HGI		Boreholes considered
		Min	Max	
IB	BCS/ I30/ I100	63	63	1
		CSM I&II-15	CSM I&II-15	

#### 4.5.12.17 ASH FUSION TEMPERATURE RANGE

**Table 4.5.12.17 A: Ash Fusion Temperature Range under Mildly Reducing Atmosphere for Seam IB, Marwatola Sector I&II block, Sohagpur Coalfield**

Range of Ash fusion Temperature						under
Mildly Reducing Atmosphere on I ₃₀ samples						
IB						
IDT (°C)		HT (°C)		FT (°C)		No. of boreholes
Min	Max	Min	Max	Min	Max	
1127	1127	1224	1224	1246	1246	1
CSM I&II-15						

#### 4.5.12.18 ASH ANALYSIS

**Table 4.5.11.18 A: Ash analysis Range for Seam IB, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			IB	
Boreholes considered			1	
Type of Samples			BCS / I30 /I100	
Range of Ash Analyses	SiO ₂ %	Min	53.43	CSM I&II-15
		Max	53.43	CSM I&II-15
	Al ₂ O ₃ %	Min	24.53	CSM I&II-15
		Max	24.53	CSM I&II-15
	Fe ₂ O ₃ %	Min	10.33	CSM I&II-15

Seam Name			IB	
Boreholes considered			1	
Type of Samples			BCS / I30 /I100	
		Max	10.33	CSM I&II-15
	TiO ₂ %	Min	1.55	CSM I&II-15
		Max	1.55	CSM I&II-15
	MgO%	Min	1.14	CSM I&II-15
		Max	1.14	CSM I&II-15
	P ₂ O ₅ %	Min	0.13	CSM I&II-15
		Max	0.13	CSM I&II-15
	SO ₃	Min	1.45	CSM I&II-15
		Max	1.45	CSM I&II-15
	MnO%	Min	0.17	CSM I&II-15
		Max	0.17	CSM I&II-15
	NA ₂ O	Min	0.12	CSM I&II-15
		Max	0.12	CSM I&II-15
	K ₂ O	Min	2.87	CSM I&II-15
		Max	2.87	CSM I&II-15

#### 4.5.12.19 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam IB is developed in the block. The seam incrops in the southern part of the block. It is overlain by Seam II Bottom/Comb. and underlain by Seam IA. The parting between Seam IB and the overlying Seam II Bottom/Comb. varies from 7.99 to 33.83m and the parting with the underlying Seam IA varies from 3.9 to 17.25m.

The seam is less banded and the thickness of the seam varies from 0.05 to 2.47m. The general thickness of the seam varies from 0.99 to 1.95m. The proximate analysis on 60% RH and 40°C of Seam IB on I₃₀ sample shows moisture%, ash% and volatile matter% as 1.5 to 7.9, 9.6 to 32.8 and 13.9 to 31.2 respectively. The grade of the seam varies from G8 to G2.

### 4.5.13 SEAM IA

#### 4.5.13.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.13.1.A: Details of seam references, of the Seam IA, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C,VIID, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		V 8
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		VII 9
13	Cross Section Plans		VIII 1
14	OB / Parting Plan		-
15	Seam Folio Plan		X 9

#### 4.5.13.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

### 4.5.13.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.13.3.A: Data of Borehole Intersection, of the Seam IA, Marwatola Sector I&II block, Sohagpur Coalfield**

S.No.	Seam	IA
1	No. of full seam intersection in boreholes	29
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	1
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	-
6	No. of borehole not deepened to encounter the seam	3
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>34</b>
7	No. of boreholes falling in updip area of seam incrop	1
8	No. of boreholes falling in combined / split zone	-
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.13.3.B: Depth Of Intersection of Roof and Floor of the Seam IA, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		IA	
ROOF DEPTH	MIN (M)	29.72	MSM-22
	MAX (M)	691.98	CSM I&II-15
FLOOR DEPTH	MIN (M)	29.83	MSM-22
	MAX (M)	693.88	CSM I&II-15
THICK NESS	MIN (M)	0.11	MSM-22
	MAX (M)	2.11	MSM-20

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.13.3.C: Minimum & Maximum RRLs & FRLs of the Seam IA, Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	-179.34	CSM I&II-15
	MAX (M)	458.27	MSM-22
FRL	MIN (M)	-181.24	CSM I&II-15
	MAX (M)	458.16	MSM-22
BH INTERSECTIONS		29	

#### 4.5.13.4 STRATIGRAPHIC POSITION

The Seam IA is overlain by Seam IB and underlain by Seam I Top.

#### 4.5.13.5 PARTING

The details of Parting above and below the Seam II Comb. is listed in the Table below.

**Table: 4.5.13.5.A: Parting Details of the Seam IA, With Underlying Seams, Marwatola Sector -I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam IA	4.11	11.77	29
	MSM-19	MSM-34	

#### 4.5.13.6 THICKNESS

**Table: 4.5.13.6.A: Thickness data of the Seam IA in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		IA	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.11	MSM-22
	MAX (M)	2.11	MSM-20
GENERAL THICKNESS	MIN (M)	0.53	
	MAX (M)	1.49	
BOREHOLE INTERSECTIONS		29	

#### 4.5.13.7 SPLITTING

The seam does not exhibit any splitting behaviour

#### 4.5.13.8 OUTCROP/INCROP

The seam incrops in the southern part of the block due to fault (F6-F6) trending east-west in the block. In the borehole MSM-49 the seam falls in the updip side of the block. The length of the incrop is around 3.1 kms.

#### 4.5.13.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
IA	ROOF	29	0	1	0	0	10	1	17
	FLOOR		0	5	4	0	6	3	11

#### 4.5.13.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.13.11 DIRT BANDS

The details of dirt band within seam-IA have been provided in Annexure V. The maximum thickness of the seam including dirt band is 2.00 meters. The seam is devoid of any dirt band.

#### 4.5.13.12 PROXIMATE ANALYSIS

**Table 4.5.13.12.A: Range of Proximate Analysis and Grade of Seam IA having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		IA		IA		IA		
Bhs. considered (Det. & Cal.)		12		12		12		
No. of Determined Boreholes		10		10		6		
Type of Samples		BCS		I30		I100		
Range of Seam Thickness (m)	Min	0.90	CSM I&II-02	0.90	CSM I&II-02	0.90	CSM I&II-02	
	Max	2.00	MSM-31	2.00	MSM-31	2.00	MSM-31	
Range of Analysed Thickness (m)	Min	0.90	CSM I&II-02	0.90	CSM I&II-02	0.90	CSM I&II-02	
	Max	2.00	MSM-31	2.00	MSM-31	2.00	MSM-31	
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	2.7	CSM I&II-01	2.7	CSM I&II-01	2.7	CSM I&II-01
		Max	7.6	MSM-17	7.6	MSM-17	6.8	CSM I&II-05
	Ash%	Min	11.3	MSM-17	11.3	MSM-17	11.3	MSM-17
		Max	42.2	CSM I&II-01	42.2	CSM I&II-01	42.2	CSM I&II-01
	VM%	Min	9.8	CSM I&II-01	9.8	CSM I&II-01	9.8	CSM I&II-01
		Max	29.6	MSM-19	29.6	MSM-19	28.2	CSM I&II-02
	FC%	Min	45.2	CSM I&II-11	45.2	CSM I&II-11	45.2	CSM I&II-11
		Max	54.0	MSM-17	54.0	MSM-17	53.8	CSM I&II-02
Range of UVM%		Min	11.0	CSM I&II-01	11.0	CSM I&II-01	11.0	CSM I&II-01
		Max	36.4	MSM-19	36.4	MSM-19	34.2	CSM I&II-14
Range of CV in k. cal/kg	Gross CVAR	Min	4324	CSM I&II-01	4324	CSM I&II-01	4324	CSM I&II-01
		Max	6737	CSM I&II-02	6737	CSM I&II-02	6737	CSM I&II-02
	Gross CVDF	Min	7989	MSM-17	7989	MSM-17	8039	CSM I&II-11
		Max	8498	CSM I&II-01	8498	CSM I&II-01	8498	CSM I&II-01
Range of UHV in k. cal/kg		Min	2703	CSM I&II-01	2703	CSM I&II-01	2703	CSM I&II-01
		Max	6416	CSM I&II-02	6416	CSM I&II-02	6512	MSM-17
Range of Grade		Min	G-10		G-10		G-10	
		Max	G-2		G-2		G-2	

## 4.5.13.13 DETERMINED GROSS CV

**Table 4.5.13.13 A: Range of Gross CV of Seam IA having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		IA		IA	
No. of Boreholes (CV Determined)		10		10	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	0.9	CSM I&II-02	0.9	CSM I&II-02
	Max	2	MSM-31	2	MSM-31
Range of Analysed Thickness (m)	Min	0.90	CSM I&II-02	0.90	CSM I&II-02
	Max	2.00	MSM-31	2.00	MSM-31
Range of Det. CV in k. cal/kg	Gross CVAR	Min	4324	CSM I&II-01	4324
		Max	6737	CSM I&II-02	6737
	Gross CVDF	Min	7989	MSM-17	7989
		Max	8498	CSM I&II-01	8498

## 4.5.13.14 ULTIMATE ANALYSIS

**Table 4.5.13.14 A: Range of Ultimate Analysis Seam IA, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			IA	
Boreholes considered			3	
Type of Samples			BCS /I30	
Range of Seam Thickness (m)		Min	1.12	MSM-17
		Max	2.00	MSM-31
Range of Ultimate analyses	C%	Min	57.32	CSM I&II-15
		Max	67.84	MSM-17
	H%	Min	3.33	CSM I&II-15
		Max	4.16	MSM-17
	N%	Min	0.48	MSM-31
		Max	1.23	CSM I&II-15
	S%	Min	0.50	MSM-17
		Max	0.58	MSM-31
	O%	Min	0.60	CSM I&II-15
		Max	6.66	MSM-17

#### 4.5.13.15 PETROGRAPHIC STUDIES

Results of Petrographic studies are tabulated below.

**Table 4.5.13.15 A: Petrographic Analysis of Seam IA, Marwatola Sector I&II block, Sohagpur Coalfield**

BH. NO.	DEPTH		SEAM THICKNESS	TYPE OF SAMPLE	Maceral Composition %				Maceral Composition (Vmmf%)			Rr%
	FROM	TO										
	(m)	(m)			Vit	Lipt	Int	VM M	Vit	Lipt	Int	
MSM-19	316.97	318.30	1.33	BCS/ I30	30.10	10.9	45.5	13.5	35.0	13.0	52.6	0.48

#### 4.5.13.16 HARDGROVE GRINDABILITY INDEX

**Table 4.5.13.16 A: Range of HGI of Seam IA, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name	Type of Samples	Range of HGI		Boreholes considered
		Min	Max	
IA	BCS	71	71	1
		CSM I&II-15	CSM I&II-15	

#### 4.5.13.17 ASH FUSION TEMPERATURE RANGE

**Table 4.5.13.17 A: Ash Fusion Temperature Range under Mildly Reducing Atmosphere for Seam IA, Marwatola Sector I&II block, Sohagpur Coalfield**

Range of Ash fusion Temperature						under
Mildly Reducing Atmosphere on I ₃₀ samples						
IA						
IDT (°C)		HT (°C)		FT (°C)		No. of boreholes
Min	Max	Min	Max	Min	Max	
>1325	>1325	>1400	>1400	>1400	>1400	1
CSM I&II-15						

#### 4.5.13.18 ASH ANALYSIS

**Table 4.5.13.18 A: Ash analysis Range for Seam IA, Marwatola Sector -I&II block, Sohagpur Coalfield**

Seam Name			IA	
Boreholes considered			1	
Type of Samples			BCS/ I30	
<b>Range of Ash Analyses</b>	<b>SiO₂%</b>	<b>Min</b>	61.83	CSM I&II-15
		<b>Max</b>	62.56	MSM-31
	<b>Al₂O₃%</b>	<b>Min</b>	21.07	CSM I&II-15
		<b>Max</b>	25.59	MSM-31
	<b>Fe₂O₃%</b>	<b>Min</b>	5.25	MSM-31
		<b>Max</b>	9.58	CSM I&II-15
	<b>TiO₂%</b>	<b>Min</b>	1.48	CSM I&II-15
		<b>Max</b>	2.20	MSM-31
	<b>MgO%</b>	<b>Min</b>	1.09	CSM I&II-15
		<b>Max</b>	1.23	MSM-31
	<b>P₂O₅%</b>	<b>Min</b>	0.05	CSM I&II-15
		<b>Max</b>	0.12	MSM-31
	<b>SO₃%</b>	<b>Min</b>	0.42	MSM-31
		<b>Max</b>	0.72	CSM I&II-15
	<b>MnO%</b>	<b>Min</b>	0.12	CSM I&II-15
		<b>Max</b>	0.12	CSM I&II-15
		<b>Min</b>	0.07	CSM I&II-15
		<b>Max</b>	0.12	MSM-31
	<b>K₂O %</b>	<b>Min</b>	0.96	MSM-31
		<b>Max</b>	1.09	CSM I&II-15

#### 4.5.13.20 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam IA is developed in the block. The seam incrops in the southern part of the block. It is overlain by Seam IB and underlain by Seam I Top. The parting between Seam IA and the overlying Seam IB varies from 3.9 to 17.25m and the parting with the underlying Seam I Top varies from 4.11 to 11.77m.

The seam is less banded and the thickness of the seam varies from 0.11 to 2.11m. The general thickness of the seam varies from 0.99 to 1.95m. The proximate analysis on 60% RH and 40°C of Seam IA on I₃₀ sample shows moisture%, ash% and volatile matter% as 2.7 to 7.6, 11.3 to 42.2 and 9.8 to 29.6 respectively. The grade of the seam varies from G10 to G2.

#### 4.5.14 SEAM I Top

##### 4.5.14.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.14.1.A: Details of seam references, of the Seam I Top, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C,VIID, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		-
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		-
13	Cross Section Plans		-
14	OB / Parting Plan		-
15	Seam Folio Plan		-

##### 4.5.14.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

#### 4.5.14.3 DETAILS OF BOREHOLE INTERSECTION

##### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.14.3.A: Data of Borehole Intersection, of the Seam I Top, Marwatola Sector I&II block, Sohagpur Coalfield**

S.No.	Seam	I Top
1	No. of full seam intersection in boreholes	14
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	1
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	1
6	No. of borehole not deepened to encounter the seam	
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>17</b>
7	No. of boreholes falling in updip area of seam incrop	1
8	No. of boreholes falling in combined / split zone	17
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

##### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.14.3.B: Depth Of Intersection of Roof and Floor of the Seam I Top, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		I Top	
ROOF DEPTH	MIN (M)	243.38	CSM I&II-10
	MAX (M)	369.40	MSM-23
FLOOR DEPTH	MIN (M)	243.58	CSM I&II-10
	MAX (M)	369.70	MSM-23
THICK NESS	MIN (M)	0.10	MSM-19
	MAX (M)	0.75	CSM I&II-02

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.14.3.C: Minimum & Maximum RRLs & FRLs of the Seam I Top, Marwatola Sector I&II block, Sohagpur Coalfield.**

RRL	MIN (M)	148.2	MSM-34
	MAX (M)	262.05	CSM I&II-10
FRL	MIN (M)	147.7	MSM-34
	MAX (M)	261.85	CSM I&II-10
BH INTERSECTIONS		14	

#### 4.5.14.4 STRATIGRAPHIC POSITION

The Seam I Top is overlain by Seam IA and underlain by Seam I Bottom.

#### 4.5.14.5 PARTING

The details of Parting below the Seam I Top is listed in the Table below.

**Table: 4.5.14.5.A: Parting Details of the Seam I Top, With Underlying Seams, Marwatola Sector I&II block, Sohagpur Coalfield**

Seams / Parting	THICKNESS		Nos. of Bh
	MIN (M)	MAX (M)	
Parting with underlying Seam I Bottom	0.40	6.15	14
	CSM I&II-13	MSM-21	

#### 4.5.14.6 THICKNESS

**Table: 4.5.14.6.A: Thickness data of the Seam I Top in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		I Top	
DETAILS	MIN /MAX (M)	THICK NESS	BOREHOLE NO.
THICKNESS	MIN (M)	0.10	MSM-19
	MAX (M)	0.75	CSM I&II-02
GENERAL THICKNESS	MIN (M)	0.21	
	MAX (M)	0.59	
BOREHOLE INTERSECTIONS		14	

#### 4.5.14.7 SPLITTING

The seam I has been splitted in to I Top and Bottom sections. However, the preset seam I Top does not exhibit any splitting behavior.

#### 4.5.14.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
I Top	ROOF	14	0	0	1	0	5	1	7
	FLOOR		1	2	4	0	1	3	3

#### 4.5.14.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.14.11 DIRT BANDS

Since the thickness of the seam is not workable and the seam thickness of the seam is less than 0.90m, the samples have not been analysed.

#### 4.5.14.12 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam I Top is Not-developed in the block. It is overlain by Seam IA and underlain by Seam I Bottom/Comb. The parting between Seam I Top and the overlying Seam IA varies from 4.11 to 11.77m and the parting with the underlying Seam I Bottom varies from 0.4 to 6.15m.

The seam is unworkable and the thickness of the seam varies from 0.10 to 0.75m. The general thickness of the seam varies from 0.21 to 0.59m. Since the thickness of the seam is less than 0.90m, the samples have not been analysed. Hence the seam has not been considered for resource estimation.

#### 4.5.15 SEAM IBOT

##### 4.5.15.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.15.1.A: Details of seam references, of the Seam I Bottom, Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C, VII D, VIIE	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		-
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		-
13	Cross Section Plans		-
14	OB / Parting Plan		-
15	Seam Folio Plan		-

##### 4.5.15.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

### 4.5.15.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.15.3.A: Data of Borehole Intersection, of the Seam I Bottom, Marwatola Sector I&II block, Sohagpur Coalfield**

S.No.	Seam	I Bottom
1	No. of full seam intersection in boreholes	14
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	1
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	1
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	1
6	No. of borehole not deepened to encounter the seam	
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>35</b>
7	No. of boreholes falling in updip area of seam incrop	1
8	No. of boreholes falling in combined / split zone	17
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.15.3.B: Depth Of Intersection of Roof and Floor of the Seam I Bottom, Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		I Bottom	
ROOF DEPTH	MIN (M)	246.24	CSM I&II-10
	MAX (M)	371.70	MSM-23
FLOOR DEPTH	MIN (M)	247.60	CSM I&II-10
	MAX (M)	372.25	MSM-23
THICK NESS	MIN (M)	0.20	CSM I&II-13, MSM-34
	MAX (M)	1.36	CSM I&II-10

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.15.3.C: Minimum & Maximum RRLs & FRLs of the Seam I Bottom, Marwatola Sector I&II block, Sohagpur Coalfield.**

<b>RRL</b>	<b>MIN (M)</b>	144.00	MSM-34
	<b>MAX (M)</b>	259.19	CSM I&II-10
<b>FRL</b>	<b>MIN (M)</b>	143.80	MSM-34
	<b>MAX (M)</b>	257.83	CSM I&II-10
<b>BH INTERSECTIONS</b>		<b>14</b>	

#### 4.5.15.4 STRATIGRAPHIC POSITION

The Seam I Bottom is overlain by Seam I Top and underlain by Seam Local.

#### 4.5.15.5 PARTING

The Seam I Bottom is lower most seam in the Marwatola sector I&II block.

#### 4.5.15.6 THICKNESS

**Table: 4.5.15.6.A: Thickness data of the Seam I Bottom in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

<b>SEAM</b>		<b>I Bottom</b>	
<b>DETAILS</b>	<b>MIN /MAX (M)</b>	<b>THICK NESS</b>	<b>BOREHOLE NO.</b>
<b>THICKNESS</b>	<b>MIN (M)</b>	0.20	CSM I&II-13, MSM-34
	<b>MAX (M)</b>	1.36	CSM I&II-10
<b>GENERAL THICKNESS</b>	<b>MIN (M)</b>	<b>0.26</b>	
	<b>MAX (M)</b>	<b>1.02</b>	
<b>BOREHOLE INTERSECTIONS</b>		<b>14</b>	

#### 4.5.15.7 SPLITTING

The seam I has been splitted in to I Top and Bottom sections. However, the preset seam I Bottom does not exhibit any splitting behavior.

#### 4.5.15.9 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
I Bottom	ROOF	14	0	3	6	0	2	1	2
	FLOOR		0	0	5	0	4	2	3

#### 4.5.15.10 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.15.11 DIRT BANDS

The details of dirt band within seam I Bottom have been provided in Annexure V. The maximum thickness of the seam including dirt band is 1.36 meters. The abstract of dirt bands within the seam having thickness more than 0.90m, is given in the following table:

**Table 4.5.15.11 A: Abstract of Dirt Bands in Seam I Bottom having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			I Bottom		
Type of Sample			I30		
Total Seam Thickness		Min.	(m)	1.36	CSM I&II-10
		Max.	(m)	1.36	CSM I&II-10
Analysed Seam Thickness		Min.	(m)	1.36	CSM I&II-10
		Max.	(m)	1.36	CSM I&II-10
Effective Dirt Bands	Included	No.	Min.	1.00	CSM I&II-10
			Max.	1.00	CSM I&II-10
		Thick (m)	Min.	0.30	CSM I&II-10
			Max.	0.30	CSM I&II-10
	Excluded	No.	Min.	0.00	CSM I&II-10
			Max.	0.00	CSM I&II-10
		Thick (m)	Min.	0.00	CSM I&II-10
			Max.	0.00	CSM I&II-10
No. of Bhs			1		

The abstract of combustible and non-combustible dirt band constituents within the seam is tabulated below.

**Table 4.5.15.11 B: Abstract Of Combustible and Non-combustible Dirt Band Constituents in Seam I Bottom having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam			IBOT	
Total Seam Thickness	Min.	(m)	1.36	
	Max.	(m)	1.36	
Analysed Seam Thickness	Min.	(m)	1.36	
	Max.	(m)	1.36	
Combustible Dirt Bands (Carbonaceous Shale)	No.	Min	1	CSM I&II-10
		Max	1	CSM I&II-10
	Thick (m)	Min	0.3	CSM I&II-10
		Max	0.3	CSM I&II-10
	%	Min	22	CSM I&II-10
		Max	22	CSM I&II-10
Non-Combustible Dirt Bands ( Shale & Sandstone )	No.	Min	0	CSM I&II-10
		Max	0	CSM I&II-10
	Thick (m)	Min	0	CSM I&II-10
		Max	0	CSM I&II-10
	%	Min	0	CSM I&II-10
		Max	0	CSM I&II-10
Total bands (Carb. Shale, Shale and Sandstone)	No.	Min	1	CSM I&II-10
		Max	1	CSM I&II-10
	Thick (m)	Min	0.3	CSM I&II-10
		Max	0.3	CSM I&II-10
	%	Min	22	CSM I&II-10
		Max	22	CSM I&II-10
Nos of Bh			1	

## 4.5.15.12 PROXIMATE ANALYSIS

**Table 4.5.15.12.A: Range of Proximate Analysis and Grade of Seam I Bottom having thickness more than 0.90m, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name			I Bottom		I Bottom		I Bottom	
Bhs. considered (Det. & Cal.)			1		1		1	
No. of Determined Boreholes			1		1		1	
Type of Samples			BCS		I30		I100	
Range of Seam Thickness (m)		Min	1.06	CSM I&II-10	1.06	CSM I&II-10	1.06	CSM I&II-10
		Max	1.06	CSM I&II-10	1.06	CSM I&II-10	1.06	CSM I&II-10
Range of Analysed Thickness (m)		Min	1.06	CSM I&II-10	1.06	CSM I&II-10	1.06	CSM I&II-10
		Max	1.06	CSM I&II-10	1.06	CSM I&II-10	1.06	CSM I&II-10
Range of Proximate Analysis on 60% RH & 40°C	M%	Min	3.4	CSM I&II-10	3.4	CSM I&II-10	3.4	CSM I&II-10
		Max	3.4	CSM I&II-10	3.4	CSM I&II-10	3.4	CSM I&II-10
	Ash%	Min	35.8	CSM I&II-10	35.8	CSM I&II-10	35.8	CSM I&II-10
		Max	35.8	CSM I&II-10	35.8	CSM I&II-10	35.8	CSM I&II-10
	VM%	Min	22.4	CSM I&II-10	22.4	CSM I&II-10	22.4	CSM I&II-10
		Max	22.4	CSM I&II-10	22.4	CSM I&II-10	22.4	CSM I&II-10
	FC%	Min	38.4	CSM I&II-10	38.4	CSM I&II-10	38.4	CSM I&II-10
		Max	38.4	CSM I&II-10	38.4	CSM I&II-10	38.4	CSM I&II-10
Range of UVM%		Min	32.9	CSM I&II-10	32.9	CSM I&II-10	32.9	CSM I&II-10
		Max	32.9	CSM I&II-10	32.9	CSM I&II-10	32.9	CSM I&II-10
Range of CV in k. cal/kg	Gross CVAR	Min	4892	CSM I&II-10	4892	CSM I&II-10	4892	CSM I&II-10
		Max	4892	CSM I&II-10	4892	CSM I&II-10	4892	CSM I&II-10
	Gross CVDF	Min	8552	CSM I&II-10	8552	CSM I&II-10	8552	CSM I&II-10
		Max	8552	CSM I&II-10	8552	CSM I&II-10	8552	CSM I&II-10
Range of UHV in k. cal/kg		Min	3488	CSM I&II-10	3488	CSM I&II-10	3488	CSM I&II-10
		Max	3488	CSM I&II-10	3488	CSM I&II-10	3488	CSM I&II-10
Range of Grade		Min	G-8		G-8		G-8	
		Max	G-8		G-8		G-8	

#### 4.5.15.13 DETERMINED GROSS CV

**Table 4.5.15.13 A: Range of Gross CV of Seam I Bottom, Marwatola Sector I&II block, Sohagpur Coalfield**

Seam Name		I Bottom		I Bottom	
No. of Boreholes (CV Determined)		1		1	
Type of Samples		BCS		I30	
Range of Seam Thickness (m)	Min	1.06	CSM I&II-10	1.36	CSM I&II-10
	Max	1.06	CSM I&II-10	1.36	CSM I&II-10
Range of Analysed Thickness (m)	Min	1.06	CSM I&II-10	1.36	CSM I&II-10
	Max	1.06	CSM I&II-10	1.36	CSM I&II-10
Range of Det. CV in k. cal/kg	Gross CVAR	Min	4892	CSM I&II-10	4166
		Max	4892	CSM I&II-10	4166
	Gross CVDF	Min	8552	CSM I&II-10	8544
		Max	8552	CSM I&II-10	8544

#### 4.5.15.14 ULTIMATE ANALYSIS / HARDGROVE GRINDABILITY INDEX / ASH FUSION TEMPERATURE RANGE / ASH ANALYSIS / PETROGRAPHIC STUDIES

The Ultimate analysis, HGI, Ash Fusion temperature range, Ash analysis and Petrographic studies for seam I Bottom samples have not been analysed.

#### 4.5.15.16 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam I Bottom is Not-developed in the block. It is overlain by Seam I Top. The parting between Seam I Bottom and the overlying Seam ITop varies from 0.40 to 6.15m.

The seam is less banded and the thickness of the seam varies from 0.20 to 1.36m. The general thickness of the seam varies from 0.26 to 1.02m. The proximate analysis (ONLY ONE BOREHOLE) on 60% RH and 40°C of Seam I Bottom on I₃₀ sample shows moisture%, ash% and volatile matter% as 3.0 to 3.0, 43.9 to 43.9 and 19.9 to 19.9 respectively. Hence the seam has not been considered for resource estimation.

#### 4.5.16 SEAM ICOMB

##### 4.5.16.1 REFERENCE

Details of seam references are tabulated below.

**Table: 4.5.16.1.A: Details of seam references, of the Seam I Comb., Marwatola Sector I&II Block, Sohagpur Coalfield**

S.No.	Particular	Annexure No.	Plate No.
1	Descriptive Lithologs	II A	
2	Correlation Chart	III	
4	Details of Dirt Bands	V	
5	Overall Quality	VI	
6	Special Test	VII A,VII B,VII C, VII D, VII E	
7	Reserves	VIII	
8	OB, Parting etc	-	
9	Graphic Lithologs		IV 1 – IV 4
10	Seam Structures		-
11	Correlation Diagram		VI 1 – VI 2
12	Floor Contour Plan		-
13	Cross Section Plans		-
14	OB / Parting Plan		-
15	Seam Folio Plan		-

##### 4.5.16.2 STATUS OF MINING

The block is virgin and no mining activity is noticed in the block

### 4.5.16.3 DETAILS OF BOREHOLE INTERSECTION

#### A. INTERSECTIONS OF BOREHOLES

**Table: 4.5.16.3.A: Data of Borehole Intersection, of the Seam I Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

S.No.	Seam	I Comb.
1	No. of full seam intersection in boreholes	13
2	No. of incrop proving boreholes with part thickness	-
3	No. of boreholes with part thickness due to faulting / drilling difficulties / roof or floor deterioration	-
4	No. of boreholes where seam is totally omitted due to fault / drilling difficulties	-
5	No. of boreholes where seam is not encountered due to non-development (thinning out) of coal seams	-
6	No. of borehole not deepened to encounter the seam	4
	<b>Total nos. of boreholes falling within coal seam area</b>	<b>17</b>
7	No. of boreholes falling in updip area of seam incrop	1
8	No. of boreholes falling in combined / split zone	17
	<b>Nos. of boreholes drilled in the block</b>	<b>35</b>

#### B. DEPTH OF INTERSECTION (ROOF & FLOOR)

**Table: 4.5.16.3.B: Depth Of Intersection of Roof and Floor of the Seam I Comb., Marwatola Sector I&II block, Sohagpur Coalfield**

SEAM		I Comb.	
ROOF DEPTH	MIN (M)	51.80	MSM-22
	MAX (M)	700.05	CSM I&II-15
FLOOR DEPTH	MIN (M)	52.00	MSM-22
	MAX (M)	702.10	CSM I&II-15
THICK NESS	MIN (M)	0.20	MSM-17, MSM-22
	MAX (M)	2.05	CSM I&II-15

### C. MINIMUM & MAXIMUM RRLs & FRLs

**Table: 4.5.16.3.C: Minimum & Maximum RRLs & FRLs of the Seam I Comb., Marwatola Sector I&II block, Sohagpur Coalfield.**

<b>RRL</b>	<b>MIN (M)</b>	-187.41	CSM I&II-15
	<b>MAX (M)</b>	436.19	MSM-22
<b>FRL</b>	<b>MIN (M)</b>	-189.46	CSM I&II-15
	<b>MAX (M)</b>	435.99	MSM-22
<b>BH INTERSECTIONS</b>		<b>13</b>	

#### 4.5.16.4 STRATIGRAPHIC POSITION

The Seam I Comb. is overlain by Seam IA and is the bottom-most seam in the block.

#### 4.5.16.5 THICKNESS

**Table: 4.5.16.6.A: Thickness data of the Seam I Comb. in boreholes, Marwatola Sector I&II block, Sohagpur Coalfield**

<b>SEAM</b>		<b>I Comb.</b>	
<b>DETAILS</b>	<b>MIN /MAX (M)</b>	<b>THICK NESS</b>	<b>BOREHOLE NO.</b>
<b>THICKNESS</b>	<b>MIN (M)</b>	0.20	MSM-17, MSM-22
	<b>MAX (M)</b>	2.05	CSM I&II-15
<b>GENERAL THICKNESS</b>	<b>MIN (M)</b>	<b>0.35</b>	
	<b>MAX (M)</b>	<b>1.65</b>	
<b>BOREHOLE INTERSECTIONS</b>		<b>13</b>	

#### 4.5.16.6 SPLITTING

The seam I has been split in to I Top and Bottom sections.

#### 4.5.16.8 ROOF & FLOOR

SEAM	Stratum	Total no. of BHs	Immediate Roof and Floor strata as encountered in boreholes						
			Carb Shale High	Shale	Sandy Shale	Shaly Sandstone	Sandstone	Intercalation	Others
<b>I Comb.</b>	ROOF	13	0	1	1	0	7	0	4
	FLOOR		0	3	2	0	2	0	6

#### 4.5.16.9 PYROLITISATION

The area is devoid of any igneous activity as such the seam is free from pyrolitisation.

#### 4.5.16.10 DIRT BANDS

Since the thickness of the seam is not workable and the seam thickness is less than 0.90m except few borehole number CSM I&II-08, 15, MSM-17, 26, 27, 27, 31 and 54 the samples have not been analyzed

#### 4.5.16.11 SUMMARY OF SEAM PARAMETERS AND SPECIAL OBSERVATIONS

The Seam I Comb is not-developed in the block. The parting between Seam I Comb and the overlying Seam IA varies from 4.11 to 11.77m.

The seam thickness varies from 0.20 to 2.05m. The general thickness of the seam varies from 0.35 to 1.65m. Since the thickness of the seam mostly in the block is less than 0.90m, the samples have not been analyzed. Hence the seam has not been considered for resource estimation

## Chapter-5

# RESOURCES

### 5.1 GENERAL

In Marwatola sector I & II Block, 16 correlatable seams including splits (V, IV, L2, IIIA, III Top, III Middle, III Bottom, III Comb., II Top, II Bottom, II Comb., IB, IA, I Top, I Bottom and I Comb. have been proved to exist. Among them regional seams IV, IIIA, III Top, III Middle, III Bottom, III Comb., II top, II Bottom, II Comb., IB and IA have been assessed for resource in the block. Seam III occurs in 3 splits, viz, Top, Middle & Bottom sections and are referred in this report as Seams III Top, III Middle and III Bottom. Seam II occurs in 2 splits, viz, Top & Bottom sections and are referred in this report as Seams II Top and II Bottom.

The reserves assessment has been done considering that the seams will be worked by underground method. The physical and chemical parameters of these seams including estimation of reserves described in this report are as per norms applicable for underground mining and deal with only the workable thickness of the seams.

### 5.2 BASIC ASSUMPTIONS AND NORMS

#### 5.2.1 DEFINITIONS

For calculation of resources assumption of following parameters has been made.

- 1 Resources estimation of regional Seams IV, IIIA, III Top, III Middle, III Bottom, III Comb, II top, II Bottom, II Comb, IB and IA have been computed.
- 2 Resources estimation of the seams have been done in accordance with Underground project norms and placed in the indicated category.
- 3 The assessment of resources of 11 regional seams have been made in indicated category, as the borehole density is lower up to 1.54 bhs/sq,km. The indicated resources have been computed thickness-wise and grade-

wise, by using iso-chore and iso-grade lines drawn through MINEX 6.1.3 software, for each seam separately.

- 4** The minimum workable thickness considered is 0.90m and resources have been estimated up to this workable limit.
- 5** Iso-chore and iso-grade lines have been drawn for workable thickness of seams ( $I_{30}$  basis), including all types of dirt bands up to 0.30m of their thickness and excluding obvious dirt bands irrespective of their thickness, for estimation of resources and grade determination.
- 6** Resources estimation for the areas within the heave zone of faults, have not been done.
- 7** Resources have been estimated separately for each seam, depth-wise.
- 8** Iso-grade lines have been drawn utilizing the G.C.V. values derived from Proximate Overall analysis on 60%RH & 40°C.
- 9** The Resources have been categorized depth-wise, i.e. up to 300m, 300-600m & 600m and above depth zone.
- 10** Inter-seam dirt bands which have been excluded from effective seam thickness have been added to the seam overburden.
- 11** In some boreholes wherever the whole seam is deteriorated into carbshale, the “zone of deterioration” has been drawn by taking the midpoint of the distance between such boreholes with the surrounding boreholes. Whenever the seam has deteriorated with some remnants of coal/shcoal/carbsh low, as per band by band analysis, the thickness of the coal/shcoal/carbsh low is taken as effective thickness for the purpose of all other requirement viz, quality, parting, seam folio etc.

- 12** All the unworkable coal bands/coal sections occurring above the delineated roof of the seams and also occurring within the seam partings have been added to the overburden.
- 13** The reserves have been estimated for thickness ranges of 0.90m to 1.20m, 1.20m to 1.50m, 1.50m to 2.00m, 2.00m to 3.00m, 3.00m to 5.00m and >5.00m.

### 5.2.2 SEAMS AFFECTED BY INTRUSIVES

The intrusive has been encountered in the borehole number MSM-09 from 146.90m depth, however other area is devoid of any igneous activity as such the seam is free from pyrolitisation.

### 5.2.3 CATEGORISATION OF RESERVES

The reserves have been placed under 'Indicated' category as per Indian Standard Procedure. To obtain net insitu geological reserves, no deduction of have been made and the estimated gross resources have been presented in the chapter.

### 5.2.4 GRADE ESTIMATION

Coal quality of the non-coking coal of all the seams has been categorised into various grades on the basis of its Gross Calorific Value (GCV), which has been determined in few boreholes and the rest has been calculated by using Mazumder's formula:

Mazumder's formula for high moisture coal was used for calculation of Gross Calorific Value as described below:

$$\text{Gross CV in kcal/kg} = \frac{154 \times \{100 - (1.1 \times A + M)\} - 108 \times M}{1.8}$$

Where A = Ash% and M = Moisture % on equilibrated basis, determined as per IS 1350, Part I, 1984. The range of Useful Heat Value for the different grades is as follows:

**TABLE 5.2.4A: The range of GCV Value and average density (gm/cc) for the different grades is as follows:**

Sl. No.	Grade	GCV Kcal/kg		Density(gm/cc)
		Lower	Higher	
1	G1	>7000		1.36
2	G2	6701	7000	1.40
3	G3	6401	6700	1.43
4	G4	6101	6400	1.44
5	G5	5801	6100	1.47
6	G6	5501	5800	1.50
7	G7	5201	5500	1.53
8	G8	4901	5200	1.56
9	G9	4601	4900	1.58
10	G10	4301	4600	1.61
11	G11	4001	4300	1.65
12	G12	3701	4000	1.69
13	G13	3401	3700	1.73
14	G14	3101	3400	1.78
15	G15	2801	3100	1.81
16	G16	2501	2800	1.84
17	G17	2201	2500	1.87

## 5.3 PROCEDURE

### 5.3.1 BLOCK BOUNDARY

North : The block is bounded on north side by Pathora East, Arjuni West block

South : The south side bounded by Marwatola sector VII block.

East : The block is bounded on East side by Marwatola Sector III&IV block.

West : On the west the Ghunghuti block is located.

### **5.3.2 BARRIERS**

No barriers have been considered for resource estimation in the block.

### **5.3.3 SECTOR**

The block has been considered as a single sector.

### **5.3.4 MINE DATA CONSIDERED**

No mine exists within the block.

### **5.3.5 METHODOLOGY**

Effective thickness grids of 11 nos. of seams (IV, IIIA, III Top, III Middle, III Bottom, III Comb., II top, II Bottom, II Comb., IB and IA) are prepared through Grid Compute menu operations of MINEX 6.1.3 software. Likewise, GCV grids are prepared for 11 nos. of seams for different categories of grade (G1 to Ungraded) through same operations. For different grades of coal, different inputs of sp. gr. are fed to database of MINEX. Limit of thickness of seam 0.90 meter and above, limit of polygon of block boundary and limit of GCV values for different grades of coal from UNGR to Grade G1 were fed to database of MINEX through Seam Model / Detailed Resource Reporting operations menu of MINEX 6.1.3 software. Then by operating above menu of MINEX, grade-wise, seam-wise, coal area, coal volume, Gross insitu resources. Thus, gross indicated resources of representative coal seams occurring within the polygon of block boundary have been obtained through MINEX 6.1.3 software.

Similarly, depth grids for 11 seams are generated through grid arithmetic menu of MINEX by deducing flattened floor grids of respective Coal seams from TOPS grid (surface contour grid). Then by operating Detailed Resource Reporting operations menu of MINEX 6.1.3 software Depth-wise, Seam-wise and Grade-wise Gross insitu Coal resources occurring within the block boundary have been estimated (Annexure VIII) from MINEX generated depth, GCV and ET grids of respective seams. No deduction has been made for attaining Gross Indicated Resources.

### 5.3.6 SPECIFIC GRAVITY

Specific gravity of coal has been adopted as per ISP-2022 norms. The average density for G1-G17 grades of non coking coal has been mentioned in table 5.2.4A.

## 5.4 DESCRIPTION OF RESOURCES

### 5.4.1 SUMMARY

A total of 197.767 million tonnes of gross indicated resources have been estimated in the entire block. Details of the resources worked out for different seams are given in Annexure-VIII. The seam-wise proved resources are given in Table 5.4.1A.

**Table 5.4.1A:Seam-Wise ‘gross’ Indicated Geological resources In Marwatola Sector I&II block, Sohagpur Coalfield.**

**(Resources in million tonnes)**

Seam Name	Indicated Resources (mt)
IV	14.581
IIIA	8.229
IIIT	4.970
IIIM	3.601
IIIB	8.299
IIIC	31.583
IIT	32.314
IIB	15.635
II	22.321
IB	34.772
IA	21.460
<b>Grand Total</b>	<b>197.767</b>

The seam-wise, depth-wise and grade-wise gross Indicated resources are given in Table 5.4.1B.

**Table 5.4.1B:Seam-Wise and Grade-Wise ‘gross’Indicated Geological Resources in Marwatola Sector I&II block, Sohagpur Coalfield.**

**(Resources in million tonnes)**

Seam Name/Grade	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14	G15	Grand Total
IV	0.000	0.000	0.000	5.250	5.121	1.725	1.876	0.609	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<b>14.581</b>
IIIA	0.000	0.000	0.114	0.970	0.578	0.274	0.472	0.843	2.581	0.990	0.494	0.252	0.297	0.321	0.045	<b>8.229</b>
III Top	0.000	0.294	0.600	0.808	0.994	0.518	0.276	0.672	0.807	0.000	0.000	0.000	0.000	0.000	0.000	<b>4.970</b>
III Middle	0.000	0.000	0.000	0.000	2.691	0.910	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<b>3.601</b>
III Bottom	0.000	0.000	0.000	1.021	2.256	1.617	1.457	1.949	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<b>8.299</b>
III Comb.	0.000	0.000	0.270	1.507	3.839	4.933	3.432	6.354	5.797	2.259	0.796	0.766	1.378	0.240	0.012	<b>31.583</b>
II Top	0.000	0.181	3.684	5.918	10.120	5.117	7.147	0.146	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<b>32.314</b>
II Bottom	0.000	0.000	1.365	3.100	7.397	1.776	0.681	0.570	0.707	0.039	0.000	0.000	0.000	0.000	0.000	<b>15.635</b>
II Comb.	0.087	0.905	1.129	0.606	4.150	7.826	4.002	3.515	0.094	0.006	0.000	0.000	0.000	0.000	0.000	<b>22.321</b>
IB	0.000	0.962	1.184	6.282	8.578	10.762	5.639	1.364	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<b>34.772</b>
IA	0.000	0.087	1.686	6.067	5.792	2.036	2.107	2.139	0.807	0.329	0.272	0.137	0.003	0.000	0.000	<b>21.460</b>
<b>Grand Total</b>	<b>0.087</b>	<b>2.430</b>	<b>10.032</b>	<b>31.529</b>	<b>51.516</b>	<b>37.495</b>	<b>27.090</b>	<b>18.161</b>	<b>10.794</b>	<b>3.621</b>	<b>1.562</b>	<b>1.155</b>	<b>1.678</b>	<b>0.560</b>	<b>0.057</b>	<b>197.767</b>

The Grade-wise and Depth-wise gross Indicated geological resources are given in Table 5.4.1C.

**Table 5.4.1C:Grade-Wise and Depth-Wise ‘gross’ Indicated Geological Resources in Marwatola Sector-I&II block, Sohagpur Coalfield**

**(Resources in million tonnes)**

Depth Range/Grade	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14	G15	Total
<b>0-300</b>	0.000	0.669	2.306	13.154	23.816	19.316	15.565	14.029	9.506	3.342	1.387	1.083	1.675	0.560	0.057	<b>106.465</b>
<b>300-600</b>	0.087	1.400	7.361	17.762	27.110	16.891	10.399	3.029	0.931	0.279	0.176	0.071	0.003	0.000	0.000	<b>85.500</b>
<b>600 and above</b>	0.000	0.362	0.366	0.613	0.589	1.288	1.126	1.103	0.356	0.000	0.000	0.000	0.000	0.000	0.000	<b>5.802</b>
<b>Total</b>	<b>0.087</b>	<b>2.430</b>	<b>10.032</b>	<b>31.529</b>	<b>51.516</b>	<b>37.495</b>	<b>27.090</b>	<b>18.161</b>	<b>10.794</b>	<b>3.621</b>	<b>1.562</b>	<b>1.155</b>	<b>1.678</b>	<b>0.560</b>	<b>0.057</b>	<b>197.767</b>

The seam-wise and thickness-wise gross Indicated geological resources are given in Table 5.4.1D.

**Table 5.4.1D: Seam-Wise and Thickness-Wise 'gross' Indicated Geological Resources in Marwatola Sector I&II block, Sohagpur Coalfield**

**(Resources in million tonnes)**

Seam Name/ Thickness Range	0.90 to 1.20	1.20 to 1.50	1.50 to 2.00	2.00 to 3.00	3.00 to 5.00	5.00 and Above	Grand Total
IV	6.666	6.006	1.909	0.000	0.000	0.000	<b>14.581</b>
IIIA	2.252	1.735	1.543	2.699	0.000	0.000	<b>8.229</b>
III Top	0.934	3.976	0.061	0.000	0.000	0.000	<b>4.970</b>
III Middle	0.775	0.413	0.469	1.946	0.000	0.000	<b>3.601</b>
III Bottom	0.223	1.771	3.420	1.127	1.759	0.000	<b>8.299</b>
III Comb.	3.153	5.209	11.313	6.070	5.838	0.000	<b>31.583</b>
II Top	1.097	2.273	6.074	8.008	14.863	0.000	<b>32.314</b>
II Bottom	2.135	1.555	6.551	5.394	0.000	0.000	<b>15.635</b>
II Comb.	0.559	1.460	2.892	5.855	11.286	0.270	<b>22.321</b>
IB	4.106	12.802	16.900	0.964	0.000	0.000	<b>34.772</b>
IA	9.416	5.482	6.289	0.273	0.000	0.000	<b>21.460</b>
<b>Grand Total</b>	<b>31.315</b>	<b>42.682</b>	<b>57.419</b>	<b>32.336</b>	<b>33.745</b>	<b>0.270</b>	<b>197.767</b>

**Table 5.4.1E: Seam-Wise 'gross' Indicated Thin seam Resources in Marwatola Sector I&II block, Sohagpur Coalfield**

**(Resources in million tonnes)**

SEAM NAME	THICKNESS RANGE (0.5m-0.9m)
	Indicated Resources (mt)
IV	5.5888
IIIA	7.4631
IIIT	0.1253
IIIM	0.4016
IIIB	0.0000
IIIC	2.3605
IIT	0.7741
IIB	3.3211
II	0.8517
IB	1.3028
IA	7.3380
<b>Grand Total</b>	<b>29.5270</b>

## 5.5 UNITED NATIONS FRAMEWORK CLASSIFICATION

The UNFC consists of three dimensional system for assessing the reliability and degree of assurance, which is measured in terms of codes, viz, 1,2 and 3. The digit '1' indicates highest degree of assurance whereas the digit '3' indicates least degree of assurance. The reliability and degree of assurance of these codes are assessed with respect to three criteria. They are Economic viability, Feasibility Assessment like project report and Geological Assessment. With respect to the current report the Economic viability and Feasibility Assessment for the block is yet to be undertaken. Hence both these criteria are given least degree of assurance i.e the digit '3'. This report has dealt with the geological aspect completely. Thus the code '2' is given for Geological Assessment.

Thus the UNFC Code given for the resources 197.767mt of Marwatola Sector I&II block is **332**.

## **Chapter-6**

### **SPECIALIZED STUDIES**

#### **6.1 GENERAL**

Specialised studies in respect of following criteria were under taken and the same are detailed below.

- a. Physico Mechanical Studies
- b. Hydrological Studies
- c. Washability Studies
- d. REE Studies
- e. Trace Elements Studies

#### **6.2 PHYSICO-MECHANICAL PROPERTIES**

Indian Institute of Techonology Kharagpur received drill cores of borehole No.CSM I&II-12 of Marwatola Sector I&II Block, Sohagpur Coalfield, for determining its physico-mechanical properties. The core is from surface to a depth of 438.00m. The diameter of core received is about 47mm. The test data will be helpful during mine planning.

##### **6.2.1 EXPERIMENTAL DESIGN AND TEST PROCEDURE**

Depending upon the availability of suitable length of core samples for preparation of test specimen, most of the lithounit of the above borehole have been tested for the following properties:

1. Bulk Density
2. Unconfined compressive strength
3. Shear Strength
4. Tensile strength
5. Young's modulus of elasticity
6. Poisson's Ratio
7. Triaxial strength
8. Protodyaknow Strength Index (PSI)

## 6.2.2 TESTS

### 6.2.2.1 BULK DENSITY

Bulk density is defined as weight per unit volume of the rock specimen with natural voids. Due to non-availability of sufficient length of sample in each lithological section it is not possible to prepare regular shaped specimens for linear measurements, and therefore, to calculate volume of specimens, water displacement method is applied using electronic balance for quick measurements.

$$\text{Bulk density} = \frac{\text{Wt. of dry specimen}}{\text{Volume of specimen}}$$

### 6.2.2.2 UNCONFINED COMPRESSIVE STRENGTH

Rock cores free from surface irregularities are cut by a diamond saw using water as coolant. The specimens are finished as closely as possible to the tolerances specified by the Commission on Standardisation of Laboratory and Field Tests of the International Society for Rock Mechanics. To remove the moisture introduced at the time of cutting, specimens are kept at room temperature for about 5 days before testing.

Since it is difficult to obtain unbroken lengths of core suitable for test specimens of 2.5 to 3:1 length-diameter ratio, it is normally intended to have all specimens of 2:1 ratio only. However, some specimens of slightly less than 2:1 ratio are also tested in those cases where the number of specimens are too small. The following size-effect correction is employed such that the unconfined compressive strength calculated corresponds as closely as possible to the length-diameter ratio of 2:1.

$$C' = \frac{4.F}{\pi D^2}$$

$$C = \frac{0.889C'}{0.778+0.222(D/L)}$$

Where

C = Unconfined compressive strength for L/D = 2

C' = Unconfined compressive strength for L/D ≠ 2

F = Failure load

D = Diameter of specimen

L = Length of specimen

To determine the Failure load the specimens are loaded upto failure along the core axis in a hydraulic stiff compression testing machine at a stress rate of 0.2 to 0.5 MPa per second.

#### 6.2.2.3 SHEAR STRENGTH

The method employed is direct shear test using a double shear box. Cylindrical specimens of not less than 50mm length are loaded by a 20mm thick shear piece. Load is applied by the same stiff compression testing machine at a rate of around 0.2 to 0.5MPa per second. The following equation is used for determining shear strength S.

$$S = \frac{2F}{\pi D^2}$$

#### 6.2.2.4 TENSILE STRENGTH

The Brazilian disc method is employed for this test. The specimens with diameter to length ratio of 1:0.5 are tested on the same compression machine. Load is applied diametrically at a rate of around 0.2 to 0.5 MPa per second. The following equation is used for determining tensile strength T.

$$T = \frac{2F}{\pi l d}$$

Where F = Failure load  
 D = Diameter of specimen  
 L = Length of specimen

#### 6.2.2.5 YOUNG'S MODULUS OF ELASTICITY

For determining Young's modulus, the test specimens were similar to those prepared for uniaxial compressive strength test. Before readings of deformation under compressive load were taken the specimens are subjected to 3 cycles of loading and unloading to a level of about 50% of failure load as recommended by ISRM. Strain readings at equally loads were thereafter noted for axial deformation.

Young's modulus was then calculated as an average modulus of the linear segment of the axial stress-strain curve around 50% of the ultimate compressive strength.

#### 6.2.2.6 TRIAXIAL STRENGTH (COHESION AND ANGLE OF INTERNAL FRICTION)

Test specimens similar to those prepared for uniaxial compression test were subjected to confining pressures of 5 and 10 MPa in a triaxial cell. In a few cases, however, tests could be done for one confining pressure only due to limited number of test specimens available. The increase in confining pressure was carried out simultaneously with the application of axial stress by a compression testing machine until the desired confining pressure of 5 or 10 MPa was reached. Thereafter the axial stress was allowed to rise independently. The stress rate followed was about 0.5 MPa per second as specified by the ISRM.

Cohesion and angle of internal friction were obtained by first plotting the Mohr's diagram of stresses and the corresponding envelope. Cohesion was

estimated from the intercept of the envelope on the shear axis and the angle of internal friction from the slope of the envelope as a tangent to the Mohr's circles.

#### **6.2.2.7 PROTODYAKNOV STRENGTH INDEX**

Five test specimens weighing 50gm each comprising irregular pieces of samples in size range of 10 to 25 mm are pounded separately in a cylinder of 76mm internal diameter by a 2.4 kg drop weight falling through a height of 0.6m. Sample are subjected to 2 to 5 impacts depending upon the nature of samples crushing criteria.

Fines below 0.5mm size produced by the five specimens are combined and the total height of the fines column in a volumeter of 23mm internal diameter is measured.

The PSI is determined by the formula:

$$\text{Protodyakonov Strength Index} = \frac{20 \times \text{No. of drops}}{\text{Height of column of fines (mm)}}$$

#### **6.2.3 Test Results**

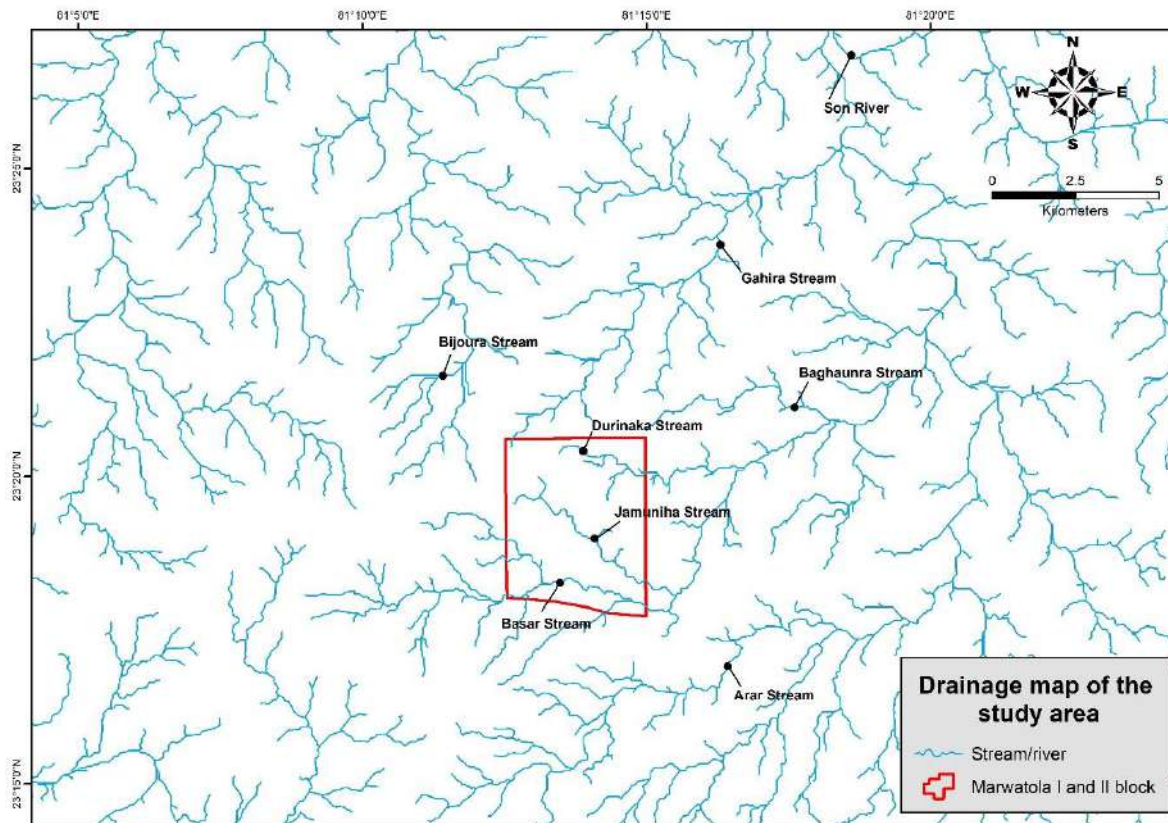
Summarised data results have been provided in Annexure-VII C.

### **6.3 HYDROGEOLOGICAL STUDIES**

#### **6.3.1 SURFACE HYDROLOGY**

Marwatola Sector I&II exploration block falls on the Son River watershed. Son River, the master drainage of the area flows at a distance of about 12 km from the northern boundary of the block. The surface drainage of this exploration block is dendritic in nature. Jamuniha stream flows through the central part of the block, ultimately joins the Basar stream flowing along the southern block boundary. Durinaka

stream flows towards east along the northern boundary of the block. The other streams flowing close to the block are Murdhoa, Bijoura, Gahira, Baghaunra, Koilarha, Bagaiha and Arar. The drainage map of the study area is shown in **figure 6.3.1**.



**Figure 6.3.1:** Drainage map of Marwatola Sector I&II exploration block, Sohagpur coalfield.

## 6.3.2 HYDROGEOLOGY

### 6.3.2.1 AQUIFER SYSTEM

The formations within the exploration block mainly belong to Gondwanas. The Barakar formation comprises of soil cover of variable thickness and sand stone of different grain sizes with shale beds and coal seams. The coal seams of the area are of lower Permian in age. These permeable sandstones are saturated and behave as aquifers. Whereas, shale and coal seams, the impermeable formations, act as aquicludes. Stratification and the presence of aquicludes lead to multi-layered aquifer system.

The formation, comprising mainly sandy soil, fine to coarse grained sandstone lying above the top coal seam (**Seam IV**) behaves as unconfined aquifer. The lower formations, consisting of compact and medium to coarse grained sandstone with secondary porosity, behave as semi-confined to confine in nature. In the unconfined aquifer, ground water moves laterally through the inter-granular pore spaces in the sandstone. Whereas, in lower aquifers the groundwater movement is restricted mainly through joints and fractures (i.e. secondary porosity). With intercalation of shale and carbonaceous shale beds and reduction in permeability with depths, the lower aquifers are very poor in potential.

The general hydrogeological units developed in the area are as follows

**TABLE 6.3.2.1: VARIOUS HYDROGEOLOGICAL UNITS DEVELOPED IN THE MARWATOLA SECTOR I & II EXPLORATION BLOCK, SOHAGPUR COALFIELD.**

Hydro geological unit	Lithology/ Formation	Generalized Thickness Range(m)	
		Min	Max
<b><i>Phreatic Aquifer</i></b>	<b><i>Sandy Soil, loosely cemented poorly consolidated weathered, fine to coarse grained sandstone</i></b>	99.64	225.97
Aquiclude	Coal Seam IV combined with carb shale, shaly coal and shale	0.10	2.50
<b><i>Aquifer A</i></b>	<b><i>Fine to Coarse grained Sandstone</i></b>	34.83	53.02
Aquiclude	Coal Seam IIIA combined with carb shale, shaly coal and shale	0.28	3.50
<b><i>Aquifer B</i></b>	<b><i>Fine to Very Coarse grained Sandstone</i></b>	5.97	16.30
Aquiclude	Coal Seam IIIT, IIIM, IIIB, IIIC combined with sandy shale, carb shale and shale	0.36	11.56
<b><i>Aquifer C</i></b>	<b><i>Fine to Coarse grained Sandstone</i></b>	18.67	56.93

Hydro geological unit	Lithology/ Formation	Generalized Thickness Range(m)	
		Min	Max
Aquiclude	Coal Seam IIT, IIB, IIC combined with carb shale, shaly coal and shale	0.03	8.66
<b>Aquifer D</b>	<b>Medium to Very Coarse grained Sandstone</b>	7.76	30.15
Aquiclude	Coal Seam IB combined with carb shale, shaly coal and shale	0.05	7.18
<b>Aquifer E</b>	<b>Fine to Medium grained Sandstone</b>	3.34	15.96
Aquiclude	Coal Seam IA combined with carb shale, shaly coal and shale	0.54	4.08
<b>Aquifer F</b>	<b>Fine to Coarse grained Sandstone</b>	5.48	22.22
Aquiclude	Coal Seam I-TOP, I-BOT combined with carb shale, shaly coal and shale	0.20	7.95

### 6.3.2.2 GENERAL AQUIFER PARAMETERS

The Hydrogeological investigation was carried out in Piparia UG mine of Umari district by CMPDIL. The aquifer parameters for the area have been reported in the "Report on Hydrogeological Conditions of Piparia U/G mine, Johilla Area, SECL" for pumping tests conducted by CMPDIL. The aquifer parameters computed by aquifer performance test in the Piparia Village are as follows: Transmissivity (**T**) = **140 m²/ day**; Hydraulic conductivity (**K**) = **3.52 m/day**.

The aquifer parameters for the Umari district were evaluated by CGWB (North Central Region) and were reported in the District Groundwater Information Booklet-2013. The following are interpreted from the exploratory Bore well results: Transmissivity (**T**) = **16 - 250 m²/ day**; Storativity (**S**) = **1.42×10⁻⁴**.

The aquifer parameters for the Pali UG mine (approx. 12 km west of Marwatola I and II block) of Umari district were evaluated by V. Pugazhendi (CGWA

Accredited Ground Water Consultant) and were reported in the report “Comprehensive Hydrogeological Report on Ground Water Conditions in both Core and Buffer Zones and Ground Water Modelling Studies for Pali UG mine, Johilla Area, SECL”. The aquifer characteristics interpreted are as follows: Transmissivity (**T**) = **3.1919 m²/ day**; Hydraulic conductivity (**K**) = **0.3755 m/day**; Storativity (**S**) = **2.0×10⁻⁴**.

Based on the ground water exploration carried out in the Umaria district by CMPDIL, CGWB and V. Pugazhendi (CGWA accredited Groundwater Consultant) the aquifer characteristics details are given in the Table 6.3.2.2 below.

**TABLE 6.3.2.2: AQUIFER CHARACTERISTICS OF THE UMARIA DISTRICT BY CMPDIL, CGWB AND V. PUGAZHENDI.**

Agency Name	K (m/day)	T (m ² /day)	S
CMPDIL	3.52	140	-
CGWB	-	16 - 250	1.42×10 ⁻⁴
V. Pugazhendi	0.3755	3.1919	2.0×10 ⁻⁴

### 6.3.3 GROUND WATER LEVELS

#### 6.3.3.1 GROUNDWATER LEVELS IN VILLAGES

The water levels in and around various villages of Marwatola Sector I&II block are under continuous monitoring. The monitored water levels in the block during the month of March (2022) is furnished below (Table 6.3.3.1).

**TABLE 6.3.3.1: DETAILS OF WELL INVENTORY & GROUNDWATER LEVELS, MARWATOLA SECTOR I&II EXPLORATION BLOCK, SOHAGPUR COALFIELD**

S. no	Location Name	Latitude (E)	Longitude (N)	Well Depth in m	Ground Water Level in m, March 2022
1	Jamuha	23° 25' 44.09"	81° 9' 45.464"	15.2	12.2
2	Ghunghuti	23° 21' 17.708"	81° 10' 47.914"	9.10	7.90
3	Sundadar	23° 15' 51.624"	81° 5' 42.313"	24.4	15.6
4	Mandri	23° 18' 39.629"	80° 57' 49.399"	31.7	3.00
5	Isanpura	23° 17' 12.638"	81° 0' 17.107"	7.00	4.30
6	Sastra	23° 23' 46.792"	80° 58' 12.976"	24.4	18.10
7	Changera	23° 16' 9.851"	81° 2' 2.375"	7.50	4.90

### 6.3.3.2 HISTORIC GROUND WATER LEVELS

The Central Groundwater Board, Bhopal (Madhya Pradesh) has been monitoring the Ground water levels in this region. The pre-monsoon and post-monsoon historical groundwater levels for the last 21 years (2000 to 2020) recorded at the nearest permanent water level monitoring stations (located at Amiliya and Ghunghuti/Ghunghuti New) were collected and are given below (Table 6.3.3.2):

**TABLE 6.3.3.2: GROUNDWATER LEVEL RECORDS, MARWATOLA SECTOR I&II EXPLORATION BLOCK, SOHAGPUR COALFIELD**

*(in meter, below ground level)*

Hydrogra ph Station / Year				Ghunghuti/Ghunghuti New			Rainfall (mm) Umari district
	Pre - Monsoon	Post - Monsoon	Fluctuati on	Pre - Monsoon	Post - Monsoon	Fluctuat ion	
2000	7.8	5.96	1.84	11.73	12.48	-0.75	891
2001	8.72	6.19	2.53	-	9.31	-	1129
2002	8.3	8	0.30	-	11.85	-	780
2003	8.8	2.9	5.90	-	7.23	-	1192
2004	7.65	6.4	1.25	10.38	10.33	0.05	968
2005	9.1	4	5.10	10.38	9.03	1.35	1341
2006	6.6	4.3	2.30	9.42	10.38	-0.96	1011
2007	6.3	5.15	1.15	10.38	7.93	2.45	775
2008	8.45	5.6	2.85	10.38	-	-	1017
2009	7.1	6.2	0.90	-	-	-	884

Hydrograph Station / Year				Ghunghuti/Ghunghuti New			Rainfall (mm) Umari district
	Pre - Monsoon	Post - Monsoon	Fluctuation	Pre - Monsoon	Post - Monsoon	Fluctuation	
2010	7.7	5.55	2.15	-	10.38	-	870
2011	12.24	4.05	8.19	11.15	-	-	1340
2012	10.2	4.7	5.50	10.38	7.53	2.85	964
2013	6.55	5.21	1.34	8.31	5.3	3.01	1220
2014	7.23	6.02	1.21	6.71	5.96	0.75	963
2015	7.65	8.5	-0.85	13.15	12.3	0.85	981
2016	-	5.96	-	13.4	10.92	2.48	1499
2017	7.71	7.37	0.34	11.9	11.52	0.38	811
2018	13.11	6.02	7.09	12.83	11.42	1.41	961
2019	6.76	10.3	-3.54	12.72	10.2	2.52	1211
2020	-	11.2	-	-	9.7	-	990
<b>Average</b>	<b>8.31</b>	<b>6.17</b>	<b>2.40</b>	<b>10.88</b>	<b>9.65</b>	<b>1.26</b>	<b>1038.09</b>

The above mentioned data show that the pre-monsoon water levels vary from minimum 6.30 m (2007, Amiliya) to maximum 13.40 m (2016, Ghunghuti) with an average of 9.60 m. Post monsoon water levels vary from 2.90 m (2003, Amiliya) to a maximum of 12.48 m (2000, Ghunghuti) with an average of 7.91 m. Whereas, the water level fluctuation varies from -3.54 m (2019, Amiliya) to 8.19 m (2011, Amiliya) with an average of 1.83 m in the study area.

### 6.3.3.3 WATER LEVEL TRENDS

The pre-monsoon and post-monsoon water level trends for the above monitoring stations are given in Figure. 6.3.3.3.1, 6.3.3.3.2, and Table 6.3.3.3.1, 6.3.3.3.2. As per the GEC-2015 guidelines, the trend of depth to water table below ground level during the various seasons in command area in centimeters per year can be calculated as:

$$a = (N \cdot S_4 - S_1 \cdot S_2 / N \cdot S_3 - S_1^2) \times 100$$

Where,

N = Number of pairs of data considered (Groundwater years)

S1 = Sum of monitoring years (Xi)

S2 = Sum of depth to water level below ground level in m during single season [Y (i)]

S3 = Sum of X (i)²

S4 = Sum of (Xi)* (Yi)

**TABLE 6.3.3.3.1: WATER LEVEL TREND DURING PRE-MONSOON AND POST-MONSOON SEASON AT AMILIYA (CGWB) STATION, SOHAGPUR COALFIELD.**

Sl. No.	GW Year	Year (x.i)	Depth to water table below GL in meters during		(x.i) ²	Pre-monsoon (x.i * y.i)	Post-monsoon (x.i * y.i)
			Pre-monsoon (y.i)	Post-monsoon (y.i)			
1	2000	1	7.8	5.96	1	7.80	5.96
2	2001	2	8.72	6.19	4	17.44	12.38
3	2002	3	8.3	8	9	24.90	24.00
4	2003	4	8.8	2.9	16	35.20	11.60
5	2004	5	7.65	6.4	25	38.25	32.00
6	2005	6	9.1	4	36	54.60	24.00
7	2006	7	6.6	4.3	49	46.20	30.10
8	2007	8	6.3	5.15	64	50.40	41.20
9	2008	9	8.45	5.6	81	76.05	50.40
10	2009	10	7.1	6.2	100	71.00	62.00
11	2010	11	7.7	5.55	121	84.70	61.05
12	2011	12	12.24	4.05	144	146.88	48.60
13	2012	13	10.2	4.7	169	132.60	61.10
14	2013	14	6.55	5.21	196	91.70	72.94
15	2014	15	7.23	6.02	225	108.45	90.30
16	2015	16	7.65	8.5	256	122.40	136.00
17	2016	17	7.65	5.96	289	130.05	101.32
18	2017	18	7.71	7.37	324	138.78	132.66
19	2018	19	13.11	6.02	361	249.09	114.38
20	2019	20	6.76	10.3	400	135.20	206.00
21	2020	21	6.76	11.2	441	141.96	235.20
	<b>N = 21</b>	<b>S1 = 231</b>	<b>S2 Pre = 172.38</b>	<b>S2 Post = 129.58</b>	<b>S3 = 3311</b>	<b>S4 Pre = 1903.65</b>	<b>S4 Post = 1553.19</b>

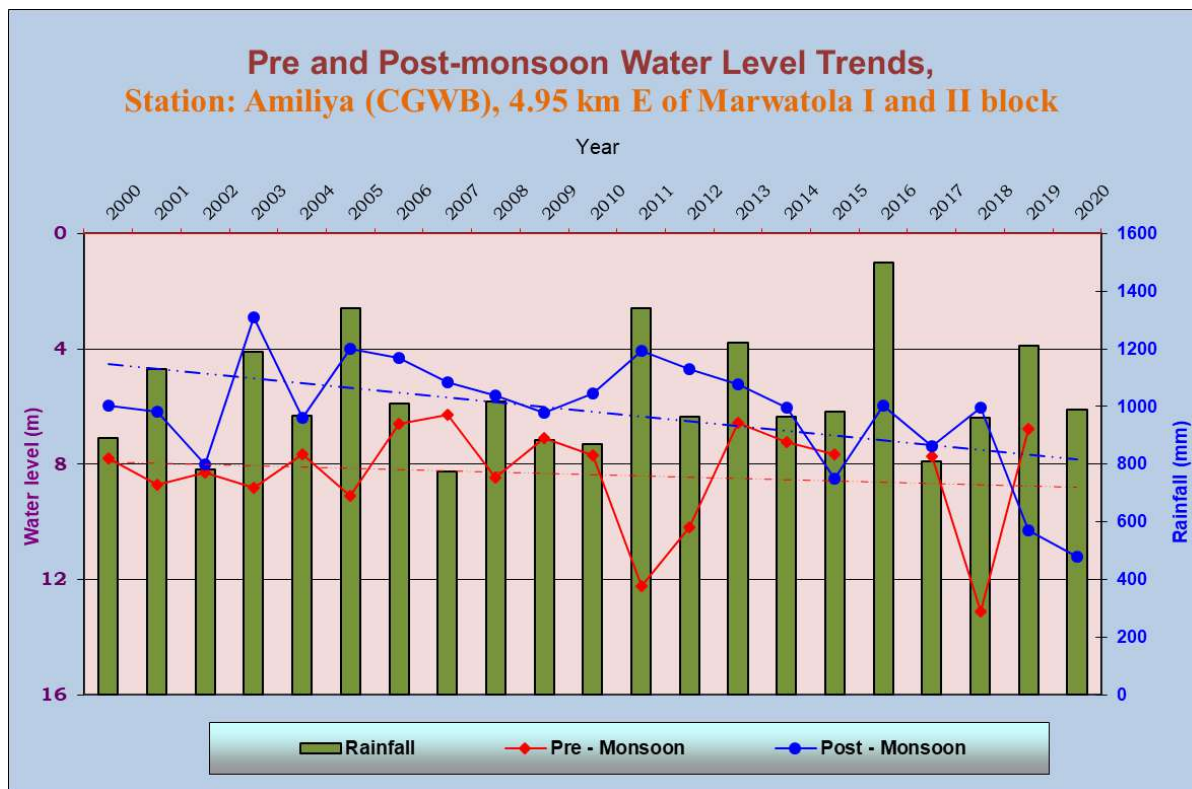
Thus,

Trend of water level during the pre-monsoon season (a) = 0.97 cm/year

Trend of water level during the post-monsoon season (a) = 16.59 cm/year

Therefore, at Amiliya station, the pre-monsoon water levels are showing neither rising nor falling trend while the post-monsoon water levels are showing

falling trend. Furthermore, the combination of Bar (Rainfall) and Line (water levels) plot prepared for Amiliya station is shown in Figure 6.3.3.3.1 and depicts that the pre-monsoon and post-monsoon water-levels are showing slight declining trend.



**Figure 6.3.3.3.1:** Pre and Post-monsoon groundwater level trends at Amiliya (CGWB) station.

Similarly, the trend of water level at Ghunghuti/Ghunghuti New (CGWB) station during pre-monsoon and post-monsoon seasons for the last 21 years is computed and provided in the Table 6.3.3.3.2.

**TABLE 6.3.3.3.2: WATER LEVEL TREND DURING PRE-MONSOON AND POST-MONSOON SEASON AT GHUNGHUTI/GHUNGHUTI NEW (CGWB) STATION, SOHAGPUR COALFIELD.**

Sl. No.	GW Year	Year (x.i)	Depth to water table below GL in meters during		(x.i) ²	Pre-monsoon (x.i * y.i)	Post-monsoon (x.i * y.i)
			Pre-monsoon (y.i)	Post-monsoon (y.i)			
1	2000	1	11.73	12.48	1	11.73	12.48
2	2001	2	11.73	9.31	4	23.46	18.62
3	2002	3	11.73	11.85	9	35.19	35.55
4	2003	4	11.73	7.23	16	46.92	28.92
5	2004	5	10.38	10.33	25	51.90	51.65
6	2005	6	10.38	9.03	36	62.28	54.18
7	2006	7	9.42	10.38	49	65.94	72.66
8	2007	8	10.38	7.93	64	83.04	63.44
9	2008	9	10.38	7.93	81	93.42	71.37
10	2009	10	10.38	7.93	100	103.80	79.3
11	2010	11	10.38	10.38	121	114.18	114.18
12	2011	12	11.15	10.38	144	133.80	124.56
13	2012	13	10.38	7.53	169	134.94	97.89
14	2013	14	8.31	5.3	196	116.34	74.2
15	2014	15	6.71	5.96	225	100.65	89.4
16	2015	16	13.15	12.3	256	210.40	196.8
17	2016	17	13.4	10.92	289	227.80	185.64
18	2017	18	11.9	11.52	324	214.20	207.36
19	2018	19	12.83	11.42	361	243.77	216.98
20	2019	20	12.72	10.2	400	254.40	204
21	2020	21	12.72	9.7	441	267.12	203.7
	<b>N =21</b>	<b>S1 = 231</b>	<b>S2 Pre = 231.89</b>	<b>S2 Post = 200.01</b>	<b>S3 = 3311</b>	<b>S4 Pre = 2595.28</b>	<b>S4 Post = 2202.88</b>

Thus,

Trend of water level during the pre-monsoon season (a) = 5.77 cm/year

Trend of water level during the post-monsoon season (a) = 0.35 cm/year

Therefore, at Ghunghuti/Ghunghuti New hydrograph station, the pre-monsoon and post-monsoon water levels show neither rising nor falling trend. Furthermore, the combination of Bar (Rainfall) and Line (water levels) plot prepared for Ghunghuti/Ghunghuti New station is shown in Figure 6.3.3.3.2 and depicts that the pre-monsoon water levels are slightly declining while the post-monsoon water level trends are neither rising nor falling.

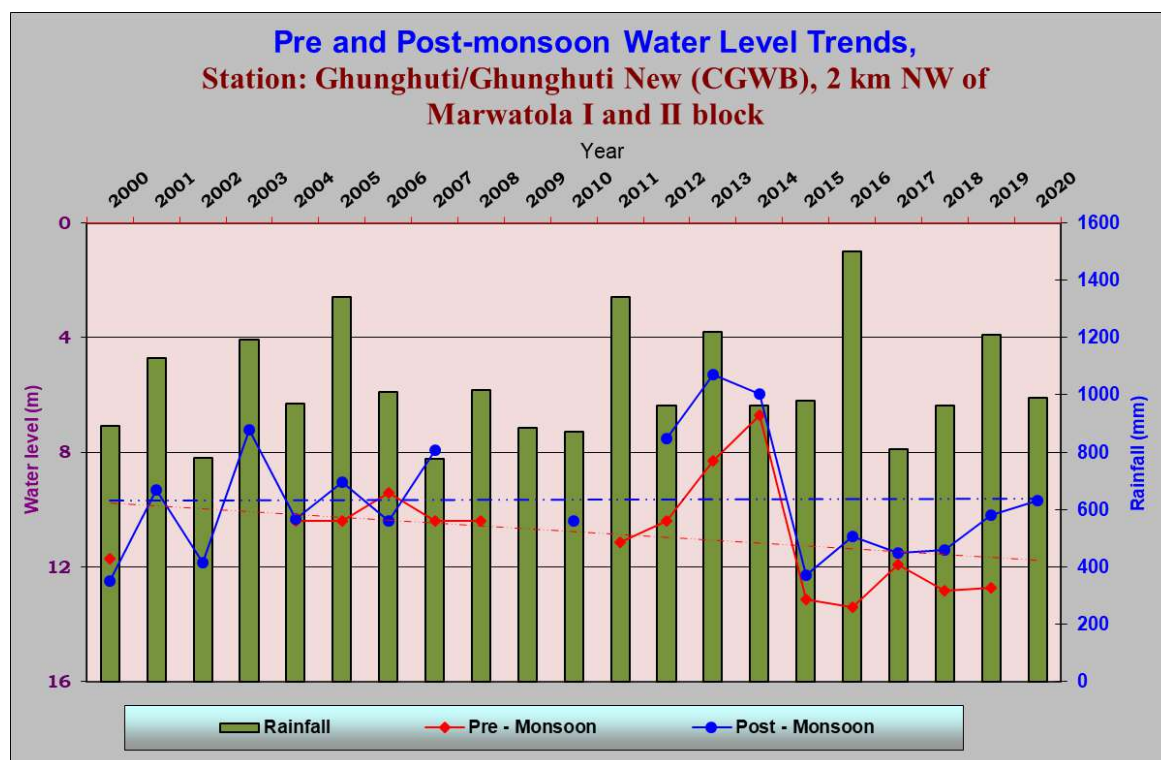


Figure 6.3.3.3.2: Pre and Post-monsoon groundwater level trends at Ghunghuti/Ghunghuti New (CGWB) station.

### 6.3.4 MINE DRAINAGE

In sedimentary rocks, due to stratification, the horizontal permeability ( $K_H$ ) is many folds higher than the vertical permeability ( $K_V$ ). The permeable beds act as individual hydrogeological units and develop a multi-layered aquifer system. Thereby, the groundwater velocity / flow along the bedding plane are higher than across the plane. A mine can be simulated as a large diameter well /sink and the mine inflow is contributing mainly from the saturated formation lying above the working seam. Thus, the mine inflow is directly proportional to the aquifer expose and progress of mine development.

If the block is selected for open cast method of mining, in the initial stage the mine inflow is in proportion to the rate of mine expansion. With the presence of low permeable formations, the inflow is significantly small. In due course, due to partial de-saturation of the aquifer, the inflow reduces and gets stabilized in time. With the development of secondary porosity due to blasting and continuity developed, the intermittent confined aquifer units get converted to unconfined near to the active mine face. With this, the unconfined aquifer close to the high wall face gets drained. So also, in the reclaimed area, with high porosity, about 40% of rainfall infiltrates and contributes to the groundwater system.

If the block is selected for underground method of mining, then the immediate aquifer present above the working coal seams will be affected by the mining. The inflow of the ground water within the mine takes places in two phases. During development phase continuous water inflow will take place from the active part of the mine and from some part of the older panels, whereas during depillaring stage immediate water inrush takes place from the depillared area and later the inflow reduces and gets stabilized in due course.

The mine discharge from the operating mines close to the Marwatola Sector I&II block are as follows (Table 6.3.4):

**TABLE 6.3.4: MINE DISCHARGE FROM THE OPERATING MINE CLOSE TO MARWATOLA SECTOR I & II BLOCK.**

Sl.No.	Mine	Average Mine discharge (m ³ /day)
1	P ali UG	1,063
2	Nowrozabad	3,153
3	Birsinghpur UG	622

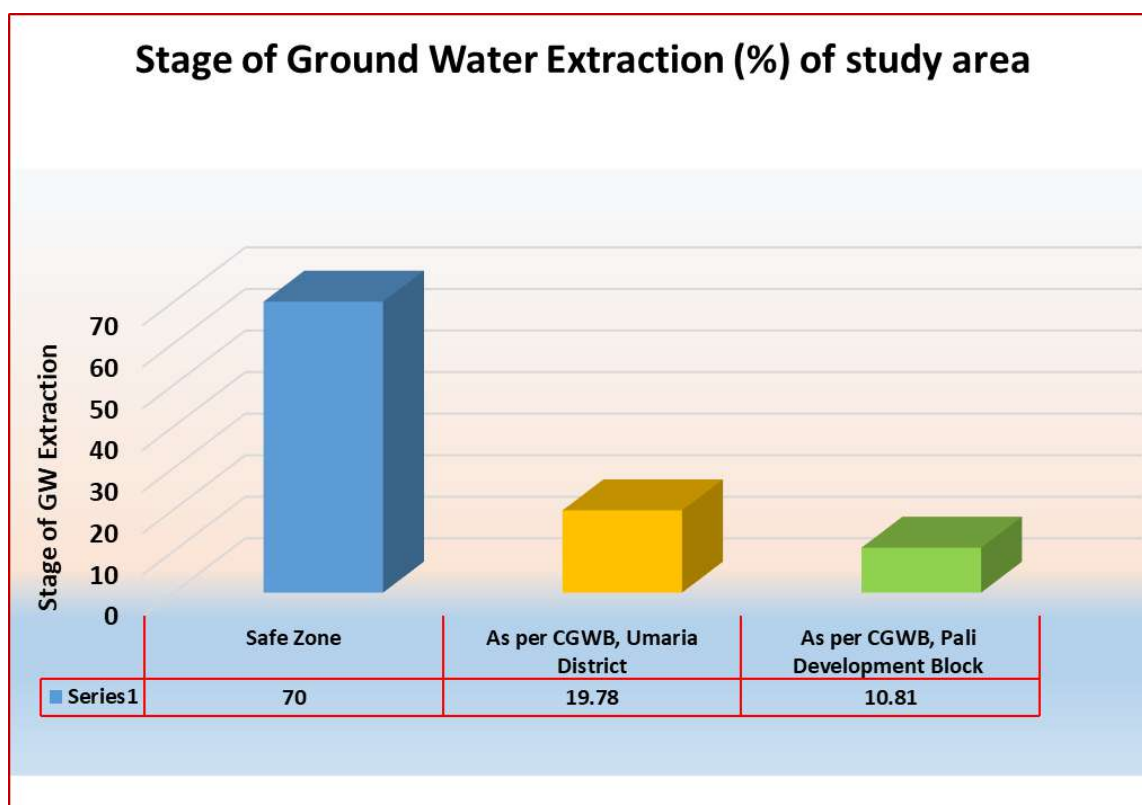
### 6.3.5 WATER QUALITY

Under routine environmental monitoring, the surface, groundwater and mine water quality for all the operating mines in the area have been monitored. Mine water and other effluents have been treated before disposal and usage.

In this context, the water quality data of the mine water effluent for the year 2021-22 of Pali UG, the drinking water/ groundwater quality data of Pre-monsoon (May 2021) and Post-monsoon (November 2021) of Pali UG mine have been provided in Table 6.3.5A to Table 6.3.5C.

### 6.3.6 GROUNDWATER STAGE EXTRACTION / DEVELOPMENT

Coal mining is the major industrial development activity in the area. The present block falls in Pali development block of Umaria district, Madhya Pradesh. The Central Ground Water Board, had assessed and reported 'Dynamic Ground Water Assessment', A.Y: 2021-2022 in the "INDIA-Groundwater Resource Estimation System (IN-GRES)" website that the stage of Groundwater extraction of **Pali Development Block** (where Marwatola Sector I&II block is located), as **10.81%** and identified the region within "**Safe**" category. The overall stage of ground water extraction in **Umaria District** was reported as **19.78%** and falls under the "**Safe**" category. These have been represented in the Figure 6.3.6 below:



**Figure 6.3.6:** Stage of Groundwater extraction in the study area

**TABLE 6.3.5A:**  
**MINE WATER EFFLUENT QUALITY DATA (2021-22), PALI UG**



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**cmpdi**  
*A Mini-Ratna Company*

**EFFLUENT WATER TEST REPORT**  
**For the year 2021- 2022**  
**JOHILLA AREA**

Name of the Customer	South Eastern Coalfields Ltd, Bilaspur		Date of Issue	19.01.2022
Name of the Project	Pali U.G.	Sample Reference No.	CMPDI/ENV/HSD/2021/755, Date: 16.12.2021	
Name of the Station	22. Mine discharge water	Date of Sampling	16.12.2021	Date of Analysis
Name of the Station		Date of Sampling	16.12.2021 to 11.01.2022	

Sl. No	Parameter	Method of Analysis	Observed Value		Lower Detection Limit	General Standards for Discharge of Environmental Pollution (Part A: Effluent) as per Schedule VI, Environment (Protection) Rules	Uncertainty of Measurement (at 95% C.I. & K= 1.96)
			22				
1	Colour & Odour	Qualitative Assessment	Acceptable		---	Acceptable	---
2	Temperature, °C	IS 3025 (Part 9):1984, (RA: 2002), Thermometric method	21.5		0.5°C	Shall not exceed 5°C above the receiving temp.	±0.14°C at 1.071°C
3	Total suspended Solids, mg/l	IS 3025 (Part 17):1984, (RA: 1996), Gravimetric Method	70		10.0	100.0	±0.445 mg/l at 24.429 mg/l
4	pH Value,	IS 3025 (Part 11):1983, (RA: 1996), Electrometric Method	6.95		1.00	5.5 to 9.0	±0.0884 at 4.025
5	Oil & Grease, mg/l	IS 3025 (Part 39):1991, (RA: 2003), Partition Gravimetric Method	BDL		4.0	10.0	±1.81 mg/l at 9.94 mg/l
6	Total Residual Chlorine, mg/l	APHA, 23rd Edition, 2017, DPD Method 4500G	0.03		0.02	1.0	±0.003 mg/l at 0.02 mg/l
7	Ammonical Nitrogen, mg/l	APHA, 23rd Edition, 2017 4500-NH ₃ F, Phenate Method	0.05		0.02	50.0	±0.0018 mg/l at 0.02 mg/l
8	Total Kjeldahl Nitrogen, mg/l	APHA, 23rd Edition, 2017 4500 B, Macro Kjeldahl Method	5.2		5.0	100.0	±0.7803 mg/l at 14.01 mg/l
9	Free Ammonia, mg/l	APHA, 23rd Edition, 2017 4500-NH ₃ F, Phenate Method	BDL		0.1	5.0	±0.0097 mg/l at 1.003 mg/l
10	B.O.D (3 days 27°C), mg/l	IS 3025 (Part 44):1993, (RA: 2003), 3 days incubation at 27°C	BDL		2.0	30.0	±11.7596682 mg/l at 202.4 mg/l
11	C.O.D, mg/l	APHA, 23rd Edition, 2017, 5220C, Closed Reflux, Titration Method	16		8.0	250.0	±6.209 mg/l at 247.427 mg/l
12	Arsenic, mg/l	IS 3025 (Part 37):1988, AAS- VGA Method	BDL		0.002	0.2	±0.0081 mg/l at 0.0184 mg/l
13	Lead, mg/l	APHA, 23rd Edition, 2017, 3113B, AAS-GTA Method	BDL		0.005	0.1	±0.000265 mg/l at 0.005098 mg/l
14	Hexavalent Chromium, mg/l	APHA, 23rd Edition, 2017, 3500 - Cr ⁶⁺ B Colorimetric Method	BDL		0.05	0.1	±0.001 mg/l at 0.098 mg/l
15	Total Chromium, mg/l	IS 3025 (Part 52):2003 AAS- Flame Method	BDL		0.05	2.0	±0.004 mg/l at 0.05 mg/l
16	Copper, mg/l	IS 3025 (Part 42): 1992, AAS-Flame Method	BDL		0.03	3.0	±0.131 mg/l at 4.895 mg/l
17	Zinc, mg/l	IS 3025 (Part 49): 1994, (RA: 2009), AAS-Flame Method	BDL		0.01	5.0	±0.0013 mg/l at 0.01 mg/l
18	Selenium, mg/l	IS 3025 (Part 56):2003,	BDL		0.001	0.05	±0.000938 mg/l at 0.001 mg/l
19	Nickel, mg/l	IS 3025 (Part 54):2003 AAS- Flame Method	BDL		0.1	3.0	±0.0106 mg/l at 0.1 mg/l
20	Fluoride, mg/l	APHA, 23rd Edition, 2017, 4500F- D SPADNS Method	0.60		0.01	2.0	±0.0014 mg/l at 0.976 mg/l
21	Dissolved Phosphate as (P), mg/l	APHA, 23rd Edition, 2017, 4500-P, C Vanadomolybdophosphoric Acid, Colorimetric Method	1.6		0.2	5.0	±0.125 mg/l at 1.230 mg/l
22	Sulphide, mg/l	APHA, 23rd Edition, 2017, 4500- S ²⁻ F. 4500D Iodometric Method	BDL		0.04	2.0	±0.0243 mg/l at 0.128 mg/l
23	Phenolic compounds, mg/l	APHA, 23rd Edition, 2017, 5530. C, Chloroform Extraction Method	BDL		0.001	1.0	±0.0204 mg/l at 0.1004 mg/l
24	Manganese, mg/l	IS 3025 (Part 59):2006 AAS- Flame Method	0.13		0.05	2.0	±0.026 mg/l at 2.442 mg/l
25	Iron, mg/l	IS 3025 (Part 53):2003 AAS- Flame Method	0.07		0.05	3.0	±0.0782 mg/l at 7.952 mg/l
26	Nitrate Nitrogen, mg/l	APHA, 23rd Edition, 2017, 4500 B, UV- Spectrophotometric Method	0.61		0.4	10.0	±0.088 mg/l at 0.4 mg/l
27	Cadmium, mg/l	APHA 23rd Edition, 2017, 3111B, AAS-GTA Method	BDL		0.001	2.0	±0.000232 mg/l at 0.00475 mg/l
28	Dissolved Solids, mg/l	IS 3025 (Part 16):1984 R: 2006, Gravimetric Method	874		30	2000	±4.473 mg/l at 592.0 mg/l

*Gopal Krishna*  
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*M. Reagan Singh*  
M. Reagan Singh  
Lab Co-ordinator

**Note:** 1) The results above relate to the samples tested as received.  
2) This report can not be reproduced in part or full without the written permission of the HOD (Env), CMPDI, RI-V.  
3) L.D.L. indicates Lower Detection Limit & B.D.L. indicates Below Detection Limit



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**EFFLUENT WATER TEST REPORT**  
**For the year 2021- 2022**  
**JOHILLA AREA**

Name of the Customer		South Eastern Coalfields Ltd, Bilaspur			Date of Issue		19.01.2022			
Name of the Project		Pali U.G.		Sample Reference No.		CMPDI/ENV/HSD/2021/755, Date: 16.12.2021				
Name of the Station		23. Gangara nala U/S water W.R.T. Pali mine discharge water			Date of Sampling		16.12.2021		Date of Analysis	
Name of the Station		24. Gangara nala U/S water W.R.T. Pali mine discharge water			Date of Sampling		16.12.2021		16.12.2021 to 11.01.2022	
Sl. No	Parameter	Method of Analysis	Observed Value		Lower Detection Limit	General Standards for Discharge of Environmental Pollution (Part A: Effluent) as per Schedule VI, Environment (Protection) Rules	Uncertainty of Measurement (at 95% C.L & K= 1.96)			
			23	24						
1	Colour & Odour	Qualitative Assessment	Acceptable	Acceptable	---	Acceptable	---			
2	Temperature, °C	IS 3025 (Part 9):1984, (RA:2002), Thermometric method	19.2	19.3	0.5°C	Shall not exceed 5°C above the receiving temp.	±0.14°C at 1.071°C			
3	Total suspended Solids, mg/l	IS 3025 (Part 17):1984, (RA: 1996), Gravimetric Method	11	10	10.0	100.0	±0.445 mg/l at 24.429 mg/l			
4	pH Value,	IS 3025 (Part 11):1983, (RA : 1996), Electrometric Method	8.04	7.86	1.00	5.5 to 9.0	±0.0884 at 4.025			
5	Oil & Grease, mg/l	IS 3025 (Part 39):1991 (RA : 2003), Partition Gravimetric Method	BDL	BDL	4.0	10.0	±1.81 mg/l at 9.94 mg/l			
6	Total Residual Chlorine, mg/l	APHA, 23rd Edition, 2017, DPD Method 4500G	0.02	0.04	0.02	1.0	±0.003 mg/l at 0.02 mg/l			
7	Ammonical Nitrogen, mg/l	APHA, 23rd Edition, 2017 4500-NH ₃ F, Phenate Method	0.06	0.07	0.02	50.0	±0.0018 mg/l at 0.02mg/l			
8	Total Kjeldahl Nitrogen, mg/l	APHA, 23rd Edition, 2017 4500 B, Macro Kjeldahl Method	2.8	3.1	5.0	100.0	±0.7803 mg/l at 14.01 mg/l			
9	Free Ammonia, mg/l	APHA, 23rd Edition, 2017 4500-NH ₃ F, Phenate Method	BDL	BDL	0.1	5.0	±0.0097 mg/l at 1.003 mg/l			
10	B.O.D(3 days 27°C), mg/l	IS 3025 (Part 44):1993, (RA : 2003), 3 days Incubation at 27°C	BDL	BDL	2.0	30.0	±11.7596682 mg/l at 202.4 mg/l			
11	C.O.D, mg/l	APHA, 23rd Edition, 2017, 5220C, Closed Reflux, Titration Method	8	BDL	8.0	250.0	±6.209 mg/l at 247.427 mg/l			
12	Arsenic, mg/l	IS 3025 (Part 37):1988, AAS- VGA Method	BDL	BDL	0.002	0.2	±0.0081 mg/l at 0.0184 mg/l			
13	Lead, mg/l	APHA, 23rd Edition, 2017, 3113B, AAS-GTA Method	BDL	BDL	0.005	0.1	±0.000266 mg/l at 0.005098 mg/l			
14	Hexavalent Chromium, mg/l	APHA, 23rd Edition, 2017, 3500 – Cr ⁶⁺ , 8 Colorimetric Method	BDL	BDL	0.05	0.1	±0.001 mg/l at 0.098mg/l			
15	Total Chromium, mg/l	IS 3025 (Part 52):2003 AAS-Flame Method	BDL	BDL	0.05	2.0	±0.004 mg/l at 0.05 mg/l			
16	Copper, mg/l	IS 3025 (Part 42) : 1992 , AAS-Flame Method	BDL	BDL	0.03	3.0	±0.131 mg/l at 4.895 mg/l			
17	Zinc, mg/l	IS 3025 ( Part 49 ) : 1994, (RA : 2009), AAS-Flame Method	0.04	0.08	0.01	5.0	±0.0013 mg/l at 0.01 mg/l			
18	Selenium, mg/l	IS 3025 (Part 56):2003,	BDL	BDL	0.001	0.05	±0.000938 mg/l at 0.001 mg/l			
19	Nickel, mg/l	IS 3025 (Part 54):2003 AAS- Flame Method	BDL	BDL	0.1	3.0	±0.0106 mg/l at 0.1 mg/l			
20	Fluoride, mg/l	APHA, 23rd Edition, 2017, 4500F-D SPADNS Method	0.52	0.32	0.01	2.0	±0.0014 mg/l at 0.976 mg/l			
21	Dissolved Phosphate as (P), mg/l	APHA, 23rd Edition, 2017, 4500-P, C Vanadomolybdophosphoric Acid, Colorimetric Method	1.4	0.7	0.2	5.0	±0.125 mg/l at 1.230 mg/l			
22	Sulphide, mg/l	APHA, 23rd Edition, 2017, 4500-S ² F. 4500D Iodometric Method	BDL	BDL	0.04	2.0	±0.0243 mg/l at 0.128 mg/l			
23	Phenolic compounds, mg/l	APHA, 23rd Edition, 2017, 5530. C, Chloroform Extraction Method	BDL	BDL	0.001	1.0	±0.0204 mg/l at 0.1004 mg/l			
24	Manganese, mg/l	IS 3025 (Part 59):2006 AAS- Flame Method	0.11	0.19	0.05	2.0	±0.026 mg/l at 2.442 mg/l			
25	Iron, mg/l	IS 3025 (Part 53):2003 AAS- Flame Method	0.06	0.12	0.05	3.0	±0.0782 mg/l at 7.952 mg/l			
26	Nitrate Nitrogen, mg/l	APHA, 23rd Edition, 2017, 4500 B, UV- Spectrophotometric Method	0.59	BDL	0.4	10.0	±0.088 mg/l at 0.4 mg/l			
27	Cadmium, mg/l	APHA 23rd Edition, 2017, 3111B, AAS-GTA Method	BDL	BDL	0.001	2.0	±0.000232 mg/l at 0.00475 mg/l			
28	Dissolved Solids, mg/l	IS 3025 (Part 16):1984 R : 2006, Gravimetric Method	223	221	30	2000	±4.473 mg/l at 592.0 mg/l			

*mle*  
 Gopal Krishna  
 Jr.Sc.Asst.

*M. Reagan Singh*  
 M. Reagan Singh  
 Lab Co-ordinator

**Note:** 1) The results above relate to the samples tested as received.  
 2) This report can not be reproduced in part or full without the written permission of the HOD (Env), CMPDI, RI-V.  
 3) L.D.L indicates Lower Detection Limit & B.D.L indicates Below Detection Limit

**TABLE 6.3.5B: DRINKING WATER/ GROUNDWATER QUALITY DURING PRE-MONSOON SEASON, PALI UG**



**CENTRAL MINE PLANNING AND DESIGN INSTITUTE LIMITED**

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**DRINKING WATER ANALYSIS REPORT**

CMPDI Complex, Songanga Colony

Bilaspur (C.G.)- 495 006

Phone: (07752) 246371

email: hk.gour@coalindia.in; mr.singh@coalindia.in

Month	MAY	2021	Area		Johilla		Report No.		MY21JH
Customer	South Eastern Coalfields Ltd (SECL), Bilaspur						Date of Issue		22-06-21 15:02
Project	Pali UGM			Sample Ref. No.		CMPDI/ENV/HSD/2021/313,Date-16/05/2021			
Sampling Stations	xx	Goraiya Village Drinking Water				Date of Sampling		15-May-2021	
	xxi	Pali Mine Premises Drinking Water				Date of Sampling		15-May-2021	
Sl. No.	Parameter	Method of Analysis	Date of Analysis		15-May-2021		to	8-Jun-2021	
			Observed Values		IS 10500: 2012		Uncertainty of Measurement (at 95% Confidence Level & K= 1.96)		
			xx	xxi	Acceptable Limit (Max)*	Permissible Limit in the Absence of Alternate Source (Max)			
1	Colour, Hazen LDL: 1.0 Hazen	APHA, 23rd Edition,2017, 2120. C. Spectrometric single wavelength method	1	BDL	5	15	±1.05+ Hazen at 49.86 Hazen		
2	Odour	IS 3025 (Part 5):1983, Physical (Qualitative)	Agreeable	Agreeable	Agreeable	Agreeable	None		
3	Phenolic compounds, mg/l LDL: 0.001 mg/l	APHA, 23rd Edition,2017, 5530. C, Chloroform Extraction Method	BDL	BDL	0.001	0.002	±0.0204 mg/l at 0.100 mg/l		
4	Turbidity, NTU LDL: 1.0 NTU	IS 3025 (Part 10):1984, R : 2006, Nephelometric Method	BDL	2	1	5	±0.855 NTU at 41.58 NTU		
5	pH LDL: 4.00	IS 3025 (Part 11):1983, R : 2012, Electrometric Method	6.7	7.71	6.5-8.5	No relaxation	±0.1272 at 7.01		
6	Alkalinity, mg/l as CaCO ₃ LDL: 5.0 mg/l	IS 3025(Part 23):1986,R 2003 Titration Method	285	170	200	600	±0.19696 mg/l at 10.0 mg/l		
7	Total Hardness, mg/l as CaCO ₃ LDL: 4.0 mg/l	IS 3025 (Part 21):2009, EDTA Method	584	372	200	600	±11.545 mg/l at 612.8 mg/l		
8	Iron, mg/l LDL: 0.05 mg/l	IS 3025 (Part 53) :2003, R:2009 AAS-Flame Method	BDL	BDL	0.3	No relaxation	±0.0782 mg/l at 7.95 mg/l		
9	Chlorides, mg/l LDL: 5.0 mg/l	IS 3025(Part 32):1988 , R : 2007, Argentometric Method	100.0	52.0	250	1000	±6.551 mg/l at 253.5 mg/l		
10	Residual Free Chlorine, mg/l LDL: 0.1 mg/l	APHA, 23rd Edition,2017, 4500G, DPD Colorimetric Method	BDL	BDL	0.2	1	±0.0082 mg/l at 0.1 mg/l		
11	Total Dissolved Solids, mg/l LDL: 30.0 mg/l	IS 3025 (Part 16):1984 R : 2006, Gravimetric Method	1001	645	500	2000	±4.473 mg/l at 592.0 mg/l		
12	Calcium, mg/l LDL: 5.0 mg/l	IS 3025 (Part 40): 1991, R : 2009, EDTA Method	124.8	96.8	75	200	±2.512 mg/l at 99.8 mg/l		
13	Copper, mg/l LDL: 0.03 mg/l	IS 3025 (Part 42) : 1992 R : 2009, AAS-Flame Method	BDL	BDL	0.05	1.5	±0.131 mg/l at 4.90 mg/l		
14	Manganese, mg/l LDL: 0.05 mg/l	IS 3025 (Part 59) : 2006, AAS-Flame Method	BDL	0.07	0.1	0.3	±0.026 mg/l at 2.44 mg/l		
15	Sulphate, mg/l LDL: 2.0 mg/l	APHA, 23rd Edition,2017, 4500- SO ₄ 2- C Turbidimetric Method	70	86	200	400	±0.640 mg/l at 19.88 mg/l		
16	Nitrate, mg/l LDL: 0.5 mg/l	APHA, 23rd Edition,2017, 4500, B UV-Spectrophotometric Method	46.64	5.40	45	No relaxation	±0.528 mg/l at 20.41 mg/l		
17	Fluoride, mg/l LDL: 0.1 mg/l	APHA, 23rd Edition,2017, 4500, F- D SPADNS Method	0.36	0.48	1	1.5	±0.014 mg/l at 0.98 mg/l		
18	Selenium, mg/l LDL: 0.001 mg/l	IS 3025 (Part 56):2003 AAS- VGA Method	BDL	BDL	0.01	No relaxation	±0.000938 mg/l at 0.001 mg/l		
19	Arsenic, mg/l LDL: 0.002 mg/l	IS 3025 (Part 37):1988,R 2003, AAS- VGA Method	BDL	BDL	0.01	0.05	±.0081 mg/l at 0.018 mg/l		
20	Lead, mg/l LDL: 0.005 mg/l	APHA, 23rd Edition,2017, 3113B, AAS-GTA Method	BDL	BDL	0.01	No relaxation	±0.000266 mg/l at 0.005 mg/l		
21	Zinc, mg/l LDL: 0.01 mg/l	IS 3025 (Part 49) : 1994, R : 2009, AAS-Flame Method	0.1	0.07	5	15	±0.0013 mg/l at 0.01 mg/l		
22	Total Chromium, mg/l LDL: 0.05 mg/l	IS 3025 (Part 52) : 2003, AAS-Flame Method	BDL	BDL	0.05	No relaxation	±0.004 mg/l at 0.05 mg/l		
23	Total Coliform, MPN/100 ml	APHA, 22nd Edition, 9221 Multiple Tube Fermentation Tech.	NIL	NIL	Nil	No relaxation	---		
24	Boron, mg/l LDL: 0.5 mg/l	APHA, 23rd Edition,2017, 4500 B, Carmine Method	BDL	BDL	0.5	1	±0.310 mg/l at 5.16 mg/l		

*Except Sl. No. 10 for which Acceptable Limit is Nil

LDL indicates Lower Detection Limit & BDL indicates Below Detection Limit

*Gopalakrishna*

Gopalakrishna  
Junior Scientific Asst

*M. Reagan Singh*

M. Reagan Singh  
Technical Manager

Note: The results above relate to the samples tested as received. This report cannot be reproduced in part or full without the written permission of the HOD (Env), CMPDI, RI-V. The Green, Yellow and Red color highlights in observed values indicate acceptable values, values exceeding acceptable limits but below permissible limits and values exceeding permissible limits respectively.

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# CENTRAL MINE PLANNING AND DESIGN INSTITUTE LIMITED

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**Environment Laboratory, Regional Institute-V**

## DRINKING WATER ANALYSIS REPORT

CMPDI Complex, Songanga Colony

Bilaspur (C.G.)- 495 006

Phone: (07752) 246371

email: hk.gour@coalindia.in; mr.singh@coalindia.in

Month	MAY	2021	Area	Johilla	Report No.	MY21JH	
Customer	South Eastern Coalfields Ltd (SECL), Bilaspur				Date of Issue	22-06-21 15:02	
Project	Pali UGM		Sample Ref. No.		CMPDI/ENV/HSD/2021/313,Date-16/05/2021		
Sampling Stations	xxii	Pali residential colony drinking water			Date of Sampling	15-May-2021	
			Date of Analysis		15-May-2021	to	8-Jun-2021
Sl. No.	Parameter	Method of Analysis	Observed Values		IS 10500: 2012		Uncertainty of Measurement (at 95% Confidence Level & K= 1.96)
			xxii		Acceptable Limit (Max)*	Permissible Limit in the Absence of Alternate Source (Max)	
1	Colour, Hazen LDL: 1.0 Hazen	APHA, 23rd Edition,2017, 2120. G, Spectrometric single wavelength method	BDL		5	15	±1.05+ Hazen at 49.86 Hazen
2	Odour	IS 3025 (Part 5):1983, Physical (Qualitative)	Agreeable		Agreeable	Agreeable	None
3	Phenolic compounds, mg/l LDL: 0.001 mg/l	APHA, 23rd Edition,2017, 5530. C, Chloroform Extraction Method	BDL		0.001	0.002	±0.0204 mg/l at 0.100 mg/l
4	Turbidity, NTU LDL: 1.0 NTU	IS 3025 (Part 10):1984, R : 2006, Nephelometric Method	2		1	5	±0.855 NTU at 41.58 NTU
5	pH LDL: 4.00	IS 3025 (Part 11):1983, R : 2012, Electrometric Method	7.35		6.5-8.5	No relaxation	±0.1272 at 7.01
6	Alkalinity, mg/l as CaCO ₃ LDL: 5.0 mg/l	IS 3025( Part 23):1986,R 2003 Titration Method	170		200	600	±0.19696 mg/l at 10.0 mg/l
7	Total Hardness, mg/l as CaCO ₃ LDL: 4.0 mg/l	IS 3025 (Part 21):2009, EDTA Method	376		200	600	±11.545 mg/l at 612.8 mg/l
8	Iron, mg/l LDL: 0.05 mg/l	IS 3025 (Part 53) :2003, R:2009 AAS-Flame Method	BDL		0.3	No relaxation	±0.0782 mg/l at 7.95 mg/l
9	Chlorides, mg/l LDL: 5.0 mg/l	IS 3025( Part 32):1988 , R : 2007, Argentometric Method	53.0		250	1000	±6.551 mg/l at 253.5 mg/l
10	Residual Free Chlorine, mg/l LDL: 0.1 mg/l	APHA, 23rd Edition,2017, 4500Cl ₂ DPD Colorimetric Method	BDL		0.2	1	±0.0082 mg/l at 0.1 mg/l
11	Total Dissolved Solids, mg/l LDL: 30.0 mg/l	IS 3025 (Part 16):1984, R : 2006, Gravimetric Method	656		500	2000	±4.473 mg/l at 592.0 mg/l
12	Calcium, mg/l LDL: 5.0 mg/l	IS 3025 (Part 40): 1991, R : 2009, EDTA Method	105.6		75	200	±2.512 mg/l at 99.8 mg/l
13	Copper, mg/l LDL: 0.03 mg/l	IS 3025 (Part 42) : 1992, R : 2009, AAS-Flame Method	BDL		0.05	1.5	±0.131 mg/l at 4.90 mg/l
14	Manganese, mg/l LDL: 0.05 mg/l	IS 3025 (Part 59) : 2006, AAS-Flame Method	0.06		0.1	0.3	±0.026 mg/l at 2.44 mg/l
15	Sulphate, mg/l LDL: 2.0 mg/l	APHA, 23rd Edition,2017, 4500- SO ₄ ²⁻ E Turbidimetric Method	90		200	400	±0.640 mg/l at 19.88 mg/l
16	Nitrate, mg/l LDL: 0.5 mg/l	APHA, 23rd Edition,2017, 4500, B UV-Spectrophotometric Method	4.94		45	No relaxation	±0.528 mg/l at 20.41 mg/l
17	Fluoride, mg/l LDL: 0.1 mg/l	APHA, 23rd Edition,2017, 4500, F- D SPADNS Method	0.29		1	1.5	±0.014 mg/l at 0.98 mg/l
18	Selenium, mg/l LDL: 0.001 mg/l	IS 3025 (Part 56):2003, AAS- VGA Method	BDL		0.01	No relaxation	±0.000938 mg/l at 0.001 mg/l
19	Arsenic, mg/l LDL: 0.002 mg/l	IS 3025 (Part 37):1988,R 2003, AAS- VGA Method	BDL		0.01	0.05	±.0081 mg/l at 0.018 mg/l
20	Lead, mg/l LDL: 0.005 mg/l	APHA, 23rd Edition,2017, 3113B, AAS-GTA Method	BDL		0.01	No relaxation	±0.000266 mg/l at 0.005 mg/l
21	Zinc, mg/l LDL: 0.01 mg/l	IS 3025 ( Part 49 ) : 1994, R : 2009, AAS-Flame Method	0.02		5	15	±0.0013 mg/l at 0.01 mg/l
22	Total Chromium, mg/l LDL: 0.05 mg/l	IS 3025 ( Part 52 ) : 2003, AAS-Flame Method	BDL		0.05	No relaxation	±0.004 mg/l at 0.05 mg/l
23	Total Coliform, MPN/100 ml	APHA, 22nd Edition, 9221 Multiple Tube Fermentation Tech.	NIL		Nil	No relaxation	---
24	Boron, mg/l LDL: 0.5 mg/l	APHA, 23rd Edition,2017, 4500-B, Carmine Method	BDL		0.5	1	±0.310 mg/l at 5.16 mg/l

*Except Sl. No. 10 for which Acceptable Limit is Min

LDL indicates Lower Detection Limit & BDL indicates Below Detection Limit

*(Signature)*

Gopalakrishna  
Junior Scientific Asst

*(Signature)*

M. Reagan Singh  
Technical Manager

Note: The results above relate to the samples tested as received. This report cannot be reproduced in part or full without the written permission of the HOD (Env), CMPDI, RI-V. The Green, Yellow and Red color highlights in observed values indicate acceptable values, values exceeding acceptable limits but below permissible limits and values exceeding permissible limits respectively.

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**TABLE 6.3.5C: DRINKING WATER/ GROUNDWATER QUALITY DURING POST-MONSOON SEASON, PALI UG**



**CENTRAL MINE PLANNING AND DESIGN INSTITUTE LIMITED**

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**DRINKING WATER ANALYSIS REPORT**

CMPDI Complex, Songanga Colony  
Bilaspur (C.G.)- 495 006  
Phone: (07752) 246371  
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Month	November	2021	Area		Johilla		Report No.		NV21JH			
Customer	South Eastern Coalfields Ltd (SECL), Bilaspur						Date of Issue		20-12-21 10:38			
Project	Pali UGM				Sample Ref. No.		CMPDI/ENV/HSD/2021/695,Date- 14/11/2021					
Sampling Stations	xx	Goraiya Village Drinking Water					Date of Sampling		13-Nov-2021			
	xxi	Pali Mine Premises Drinking Water					Date of Sampling		13-Nov-2021			
					Date of Analysis		13-Nov-2021		to		14-Dec-2021	
					Observed Values		IS 10500: 2012					
Sl. No.	Parameter		Method of Analysis		xx	xxi	Acceptable Limit (Max)*	Permissible Limit in the Absence of Alternate Source (Max)		Uncertainty of Measurement (at 95% Confidence Level & K= 1.96)		
1	Colour, Hazen LDL: 1.0 Hazen		APHA, 23rd Edition,2017, 2120. C. Spectrometric single wavelength method		BDL	2	5	15		±1.05+ Hazen at 49.86 Hazen		
2	Odour		IS 3025 (Part 5):1983, Physical (Qualitative)		Agreeable	Agreeable	Agreeable	Agreeable		None		
3	Phenolic compounds, mg/l LDL: 0.001 mg/l		APHA, 23rd Edition,2017, 5530. C, Chloroform Extraction Method		BDL	BDL	0.001	0.002		±0.0204 mg/l at 0.100 mg/l		
4	Turbidity, NTU LDL: 1.0 NTU		IS 3025 (Part 10):1984. R : 2006, Nephelometric Method		1	2	1	5		±0.855 NTU at 41.58 NTU		
5	pH LDL: 4.00		IS 3025 (Part 11):1983, R : 2012, Electrometric Method		6.8	7.31	6.5-8.5	No relaxation		±0.1272 at 7.01		
6	Alkalinity, mg/l as CaCO ₃ LDL: 5.0 mg/l		IS 3025(Part 23):1986,R 2003 Titration Method		360	150	200	600		±0.19696 mg/l at 10.0 mg/l		
7	Total Hardness, mg/l as CaCO ₃ LDL: 4.0 mg/l		IS 3025 (Part 21):2009, EDTA Method		566	384	200	600		±11.545 mg/l at 612.8 mg/l		
8	Iron, mg/l LDL: 0.05 mg/l		IS 3025 (Part 53) :2003, R:2009 AAS-Flame Method		BDL	BDL	0.3	No relaxation		±0.0782 mg/l at 7.95 mg/l		
9	Chlorides, mg/l LDL: 5.0 mg/l		IS 3025(Part 32):1988, R : 2007, Argentometric Method		103.5	47.0	250	1000		±6.551 mg/l at 253.5 mg/l		
10	Residual Free Chlorine, mg/l LDL: 0.1 mg/l		APHA, 23rd Edition,2017, 4500G, DPD Colorimetric Method		BDL	BDL	0.2	1		±0.0082 mg/l at 0.1 mg/l		
11	Total Dissolved Solids, mg/l LDL: 30.0 mg/l		IS 3025 (Part 16):1984 R : 2006, Gravimetric Method		1012	600	500	2000		±4.473 mg/l at 592.0 mg/l		
12	Calcium, mg/l LDL: 5.0 mg/l		IS 3025 (Part 40): 1991, R : 2009, EDTA Method		84.0	90.4	75	200		±2.512 mg/l at 99.8 mg/l		
13	Copper, mg/l LDL: 0.03 mg/l		IS 3025 (Part 42) : 1992 R : 2009, AAS-Flame Method		BDL	BDL	0.05	1.5		±0.131 mg/l at 4.90 mg/l		
14	Manganese, mg/l LDL: 0.05 mg/l		IS 3025 (Part 59) : 2006, AAS-Flame Method		0.12	BDL	0.1	0.3		±0.026 mg/l at 2.44 mg/l		
15	Sulphate, mg/l LDL: 2.0 mg/l		APHA, 23rd Edition,2017, 4500- SO4 ²⁻ E Turbidimetric Method		86	86	200	400		±0.640 mg/l at 19.88 mg/l		
16	Nitrate, mg/l LDL: 0.5 mg/l		APHA, 23rd Edition,2017, 4500, B UV-Spectrophotometric Method		49.40	6.43	45	No relaxation		±0.528 mg/l at 20.41 mg/l		
17	Fluoride, mg/l LDL: 0.1 mg/l		APHA, 23rd Edition,2017, 4500, F- D SPADNS Method		0.21	0.31	1	1.5		±0.014 mg/l at 0.98 mg/l		
18	Selenium, mg/l LDL: 0.001 mg/l		IS 3025 (Part 56):2003 AAS- VGA Method		BDL	BDL	0.01	No relaxation		±0.000938 mg/l at 0.001 mg/l		
19	Arsenic, mg/l LDL: 0.002 mg/l		IS 3025 (Part 37):1988,R 2003, AAS- VGA Method		BDL	BDL	0.01	0.05		±.0081 mg/l at 0.018 mg/l		
20	Lead, mg/l LDL: 0.005 mg/l		APHA, 23rd Edition,2017, 3113B, AAS-GTA Method		BDL	BDL	0.01	No relaxation		±0.000266 mg/l at 0.005 mg/l		
21	Zinc, mg/l LDL: 0.01 mg/l		IS 3025 (Part 49) : 1994, R : 2009, AAS-Flame Method		0.01	0.08	5	15		±0.0013 mg/l at 0.01 mg/l		
22	Total Chromium, mg/l LDL: 0.05 mg/l		IS 3025 (Part 52) : 2003, AAS-Flame Method		BDL	BDL	0.05	No relaxation		±0.004 mg/l at 0.05 mg/l		
23	Total Coliform, MPN/100 ml		APHA, 22nd Edition, 9221 Multiple Tube Fermentation Tech.		NIL	NIL	Nil	No relaxation		---		
24	Boron, mg/l LDL: 0.5 mg/l		APHA, 23rd Edition,2017, 4500-B, Carmine Method		BDL	BDL	0.5	1		±0.310 mg/l at 5.16 mg/l		

*Except Sl. No. 10 for which Acceptable Limit is Min

LDL indicates Lower Detection Limit & BDL indicates Below Detection Limit

*mllb*  
Gopalakrishna  
Junior Scientific Asst

*[Signature]*  
M. Reagan Singh  
Technical Manager

Note: The results above relate to the samples tested as received. This report cannot be reproduced in part or full without the written permission of the HOD (Env), CMPDI, RI-V. The Green, Yellow and Red color highlights in observed values indicate acceptable values, values exceeding acceptable limits but below permissible limits and values exceeding permissible limits respectively.

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# CENTRAL MINE PLANNING AND DESIGN INSTITUTE LIMITED

**cmpdi**  
A Mini-Ratna Company

**Environment Laboratory, Regional Institute-V**

## DRINKING WATER ANALYSIS REPORT

CMPDI Complex, Songanga Coloy

Bilaspur (C.G.)- 495 006

Phone: (07752) 246371

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Month	November	2021	Area	Johilla	Report No.	NV21JH	
Customer	South Eastern Coalfields Ltd (SECL), Bilaspur				Date of Issue	20-12-21 10:38	
Project	Pali UGM		Sample Ref. No.		CMPDI/ENV/HSD/2021/695,Date- 14/11/2021		
Sampling Stations	xxii	Pali residential colony drinking water			Date of Sampling	13-Nov-2021	
Sl. No.	Parameter	Method of Analysis	Date of Analysis		13-Nov-2021	to	14-Dec-2021
			Observed Values		IS 10500: 2012		Uncertainty of Measurement (at 95% Confidence Level & K= 1.96)
			xxii		Acceptable Limit (Max)*	Permissible Limit in the Absence of Alternate Source (Max)	
1	Colour, Hazen LDL: 1.0 Hazen	APHA, 23rd Edition,2017, 2120. C, Spectrometric single wavelength method	1		5	15	±1.05+ Hazen at 49.86 Hazen
2	Odour	IS 3025 (Part 5):1983, Physical (Qualitative)	Agreeable		Agreeable	Agreeable	None
3	Phenolic compounds, mg/l LDL: 0.001 mg/l	APHA, 23rd Edition,2017, 5530. C, Chloroform Extraction Method	BDL		0.001	0.002	±0.0204 mg/l at 0.100 mg/l
4	Turbidity, NTU LDL: 1.0 NTU	IS 3025 (Part 10):1984, R : 2006, Nephelometric Method	3		1	5	±0.855 NTU at 41.58 NTU
5	pH LDL: 4.00	IS 3025 (Part 11):1983, R : 2012, Electrometric Method	7.48		6.5-8.5	No relaxation	±0.1272 at 7.01
6	Alkalinity, mg/l as CaCO ₃ LDL: 5.0 mg/l	IS 3025( Part 23):1986,R 2003 Titration Method	140		200	600	±0.19696 mg/l at 10.0 mg/l
7	Total Hardness, mg/l as CaCO ₃ LDL: 4.0 mg/l	IS 3025 (Part 21):2009, EDTA Method	358		200	600	±11.545 mg/l at 612.8 mg/l
8	Iron, mg/l LDL: 0.05 mg/l	IS 3025 (Part 53) :2003, R:2009 AAS-Flame Method	0.06		0.3	No relaxation	±0.0782 mg/l at 7.95 mg/l
9	Chlorides, mg/l LDL: 5.0 mg/l	IS 3025( Part 32):1988 , R : 2007, Argentometric Method	50.0		250	1000	±6.551 mg/l at 253.5 mg/l
10	Residual Free Chlorine, mg/l LDL: 0.1 mg/l	APHA, 23rd Edition,2017, 4500G, DPD Colorimetric Method	0.10		0.2	1	±0.0082 mg/l at 0.1 mg/l
11	Total Dissolved Solids, mg/l LDL: 30.0 mg/l	IS 3025 (Part 16):1984 R : 2006, Gravimetric Method	566		500	2000	±4.473 mg/l at 592.0 mg/l
12	Calcium, mg/l LDL: 5.0 mg/l	IS 3025 (Part 40): 1991, R : 2009, EDTA Method	100.0		75	200	±2.512 mg/l at 99.8 mg/l
13	Copper, mg/l LDL: 0.03 mg/l	IS 3025 (Part 42) : 1992 R : 2009, AAS-Flame Method	BDL		0.05	1.5	±0.131 mg/l at 4.90 mg/l
14	Manganese, mg/l LDL: 0.05 mg/l	IS 3025 (Part 59) : 2006, AAS-Flame Method	BDL		0.1	0.3	±0.026 mg/l at 2.44 mg/l
15	Sulphate, mg/l LDL: 2.0 mg/l	APHA, 23rd Edition,2017, 4500- SO42- E Turbidimetric Method	91		200	400	±0.640 mg/l at 19.88 mg/l
16	Nitrate, mg/l LDL: 0.5 mg/l	APHA, 23rd Edition,2017, 4500. B UV-Spectrophotometric Method	6.65		45	No relaxation	±0.528 mg/l at 20.41 mg/l
17	Fluoride, mg/l LDL: 0.1 mg/l	APHA, 23rd Edition,2017, 4500. F- D SPADNS Method	0.40		1	1.5	±0.014 mg/l at 0.98 mg/l
18	Selenium, mg/l LDL: 0.001 mg/l	IS 3025 (Part 56):2003 AAS- VGA Method	BDL		0.01	No relaxation	±0.000938 mg/l at 0.001 mg/l
19	Arsenic, mg/l LDL: 0.002 mg/l	IS 3025 (Part 37):1988,R 2003, AAS- VGA Method	BDL		0.01	0.05	±0.0081 mg/l at 0.018 mg/l
20	Lead, mg/l LDL: 0.005 mg/l	APHA, 23rd Edition,2017, 3113B, AAS- GTA Method	BDL		0.01	No relaxation	±0.000266 mg/l at 0.005 mg/l
21	Zinc, mg/l LDL: 0.01 mg/l	IS 3025 ( Part 49 ) : 1994, R : 2009, AAS-Flame Method	0.1		5	15	±0.0013 mg/l at 0.01 mg/l
22	Total Chromium, mg/l LDL: 0.05 mg/l	IS 3025 ( Part 32 ) : 2003, AAS-Flame Method	BDL		0.05	No relaxation	±0.004 mg/l at 0.05 mg/l
23	Total Coliform, MPN/100 ml	APHA, 22nd Edition, 9221 Multiple Tube Fermentation Tech.	NIL		Nil	No relaxation	---
24	Boron, mg/l LDL: 0.5 mg/l	APHA, 23rd Edition,2017, 4500- B, Carmine Method	BDL		0.5	1	±0.310 mg/l at 5.16 mg/l

*Except Sl. No. 10 for which Acceptable Limit is Min

LDL Indicates Lower Detection Limit & BDL Indicates Below Detection Limit

Gopalakrishna  
Junior Scientific Asst

M. Reagan Singh  
Technical Manager

Note: The results above relate to the samples tested as received. This report cannot be reproduced in part or full without the written permission of the HOD (Env), CMPDI, RI-V. The Green, Yellow and Red color highlights in observed values indicate acceptable values, values exceeding acceptable limits but below permissible limits and values exceeding permissible limits respectively.

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#### **6.4 WASHABILITY STUDIES**

The washability test has not been carried out in the block.

#### **6.5 REE STUDIES**

The Rare earth element test has been carried out for borehole No. CSM I&II-11, the summarised data results have been provided in Annexure-VII D.

#### **6.6 TRACE ELEMENT STUDIES**

The Trace element test has been carried out for borehole No. CSM I&II-11, the summarised data results have been provided in Annexure-VII E.

## Chapter – 7

### MINING & UTILISATION

#### 7.1 GENERAL

The coal bearing barakars are mostly covered by soil, as a result the area is mostly devoid of any coal exposures. The sub-surface data have been established the existence of 16 seams including splits and local seams well defined coal seams of Barakar Formation namely seam V, IV, L2, IIIA, III Top, III Middle, III Bottom, III Comb, II Top, II Bottom, II Comb, IB, IA, I Top, I Bottom and I Comb in descending order have been proved in Marwatola Sector I&II block of Sohagpur Coalfield, in which of 11seams IV, IIIA, III Top, III Middle, III Bottom, III Comb, II Top, II Bottom, II Comb, IB and IA have been found to attain workable seam thickness in the major part of the block. All the Seams incrop within the block area. As discussed with the mine planners, all the Seams are to exploited by underground method.

#### 7.2 UNDERGROUND RESERVES

##### 7.2.1 SEAM DETAILS

The seam-wise and parting variation as deciphered from borehole intersections are given below.

Seams / Parting	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		No s. of Bh
	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	
SOIL	0.00	0.00	0.20	6.00	0.20	6.00	470.08	542.58			35
			CSM I&II-06	MSM-23	CSM I&II-06	MSM-23		MSM-27			
WM	0.20	6.00	5.20	46.00	5.20	46.00					35
	CSM I&II-06	MSM-23	CSM I&II-01	MSM-55	CSM I&II-01	MSM-55					
Parting	5.20	46.00	120.00	201.73	107.56	179.17					03
	CSM I&II-01	MSM-55	CSM I&II-02	MSM-23	CSM I&II-02	MSM-55					
V	120.00	201.73	120.10	201.83	0.05	0.10	316.56	378.64	316.46	378.54	03
	CSM I&II-02	MSM-23	CSM I&II-02	MSM-23	CSM I&II-13	CSM I&II-02	MSM-23	CSM I&II-02	MSM-23	CSM I&II-02	
Parting	120.00	201.73	136.20	217.81	15.98	58.49					03

Seams / Parting	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		No s. of Bh
	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	
	CSM I&II-02	MSM- 23	CSM I&II-02	MSM- 23	MSM- 23	CSM I&II-13					
IV	99.64	480	100.24	480.65	0.10	1.80	43.28	405.79	42.63	405.1 9	27
	CSM I&II -10	CSM I&II-03	CSM I&II-10	CSM I&II-03	MSM- 51	MSM- 24	CSM I&II-03	CSM I&II-10	CSM I&II-03	CSM I&II-10	
Parting	100.24	226.73	135.84	258.50	24.73	52.20					17
	CSM I&II -10	MSM- 28	CSM I&II-10	MSM- 24	CSM I&II-12	MSM- 52					
L2	135.84	258.50	135.94	258.50	0.05	1.35	249.57	369.59	249.13	369.4 9	19
	CSM I&II-10	MSM- 24	CSM I&II-10	MSM- 24	CSM I&II-06	CSM I&II-05	MSM- 54	CSM I&II-10	MSM- 54	CSM I&II-10	
Parting	135.94	258.50	144.60	595.50	6.51	20.10					13
	CSM I&II-10	MSM- 24	CSM- I&II-10	CSM- I&II-15	CSM I&II-02	MSM- 28					
IIIA	144.60	595.50	147.00	597.00	0.10	2.70	-82.86	360.83	-84.36	358.4 3	21
	CSM- I&II-10	CSM- I&II-15	CSM- I&II-10	CSM I&II-15	MSM- 18	MSM- 28	CSM- I&II-15	CSM- I&II-10	CSM- I&II-15	CSM- I&II-10	
Parting	147.00	597.00	164.40	619.10	6.17	25.73	-	-	-	-	21
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-04	CSM I&II-03					
III Top	251.45	272.30	252.93	273.60	0.30	1.48	227.94	257.96	227.64	256.8 2	04
	CSM- I&II-07	MSM- 55	CSM- I&II-07	MSM- 55	MSM- 51	CSM- I&II-07	MSM- 51	CSM- I&II-12	MSM- 51	CSM- I&II-12	
Parting	252.93	273.60	253.93	277.00	0.62	3.40					04
	CSM- I&II-07	MSM- 55	CSM- I&II-07	MSM- 55	MSM- 51	MSM- 55					
III Middle	253.93	277.00	254.23	278.15	0.30	3.22	227.02	255.10	226.72	251.8 8	04
	CSM- I&II-07	MSM- 55	CSM- I&II-07	MSM- 55	CSM- I&II-07	CSM- I&II-12	MSM- 51	CSM- I&II-12	MSM- 51	CSM- I&II-12	
Parting	254.23	278.15	256.78	280.45	1.08	2.55					04
	CSM- I&II-07	MSM- 55	CSM I&II-07	MSM- 55	MSM- 51	CSM I&II-07					
III Bottom	256.78	280.45	258.66	281.90	1.30	3.42	225.64	249.82	224.34	246.4 0	04
	CSM I&II-07	MSM- 55	CSM I&II-07	MSM- 55	MSM- 51	CSM I&II-12	MSM- 51	CSM I&II-12	MSM- 51	CSM I&II-12	

Seams / Parting	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		No s. of Bh
	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	
<b>III Comb.</b>	164.40	619.10	166.30	619.20	0.10	5.06	-106.46	341.03	-106.56	339.13	<b>26</b>
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-15	CSM I&II-05	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-10	
Parting	166.30	619.20	199.30	650.37	18.67	60.96					<b>28</b>
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	MSM-27	MSM-26					
<b>II Top</b>	237.85	328.65	238.75	330.13	0.40	4.80	188.53	269.52	188.13	268.99	<b>15</b>
	CSM I&II-02	MSM-24	CSM I&II-02	MSM-24	MSM-52	MSM-55	MSM-52	MSM-27	MSM-52	MSM-27	
Parting	238.75	330.13	239.22	332.15	0.47	4.10					<b>15</b>
	CSM I&II-02	MSM-24	CSM I&II-02	MSM-24	CSM I&II-02	MSM-21					
<b>II Bottom</b>	239.22	332.15	241.15	334.40	0.36	2.30	186.88	268.30	185.38	267.94	<b>15</b>
	CSM I&II-02	MSM-24	CSM I&II-02	MSM-24	MSM-27	MSM-21	MSM-52	MSM-27	MSM-52	MSM-27	
<b>II Comb.</b>	199.30	650.37	202.30	650.87	0.03	5.60	-137.73	306.13	-138.23	303.13	<b>14</b>
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-13	MSM-17	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-10	
Parting	202.30	650.87	215.90	684.70	7.99	33.83					<b>26</b>
	CSM I&II-10	CSM I&II-15	CSM I&II-10	CSM I&II-15	CSM I&II-04	CSM I&II-15					
<b>I B</b>	20	684.7	20.98	685.3	0.05	2.47	-172.06	467.99	-172.66	467.01	<b>28</b>
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	MSM-27	CSM I&II-10	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	
Parting	20.98	685.30	29.72	691.98	3.90	17.25					<b>28</b>
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	CSM I&II-12	CSM I&II-08					
<b>I A</b>	29.72	691.98	29.83	693.88	0.11	2.11	-179.34	458.27	-181.24	458.16	<b>29</b>

Seams / Parting	ROOF DEPTH		FLOOR DEPTH		THICKNESS		RRL		FRL		No s. of Bh
	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	MIN (M)	MAX (M)	
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	MSM-20	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	
Parting	233.10	362.85	243.38	369.40	4.11	11.77					14
	CSM I&II-10	MSM-23	MSM I&II-10	MSM-23	MSM-19	MSM-34					
I Top	243.38	369.40	243.58	369.70	0.10	0.75	148.20	262.05	147.70	261.85	14
	CSM I&II-10	MSM-23	CSM I&II-10	MSM-23	MSM-19	CSM I&II-02	MSM-34	CSM I&II-10	MSM-34	CSM I&II-10	
Parting	243.58	369.70	246.24	371.70	0.40	6.15					14
	CSM I&II-10	MSM-23	CSM I&II-10	MSM-23	CSM I&II-13	MSM-21					
I Bottom	246.24	371.70	247.60	372.25	0.20	1.36	144	259.19	143.80	257.83	14
	CSM I&II-10	MSM-23	CSM I&II-10	MSM-23	CSM I&II-13, MSM-34	CSM I&II-10	MSM-34	CSM I&II-10	MSM-34	CSM I&II-10	
Parting	247.60	372.25	-	-	-	-					14
	CSM I&II-10	MSM-23									
I Comb.	51.80	700.05	52.00	702.10	0.20	2.05	-187.41	436.19	-189.46	435.99	13
	MSM-22	CSM I&II-15	MSM-22	CSM I&II-15	MSM-17, MSM-22	CSM I&II-15	CSM I&II-15	MSM-22	CSM I&II-15	MSM-22	

## 7.3 COAL UTILISATION

### 7.3.1 COAL CHARACTERISTICS

The coals of Marwatola Sector I&II Block are generally low to medium moisture, medium to high ash, low rank, non-coking coals of both superior and inferior quality. The seam-wise quality is given in Table 7.3.1A.

**Table 7.3.1A**  
**Quality of Coal Seams (I₃₀ basis)**

Coal Seam/ Section	At 60% RH & 40°C				GCV (K. cal./ kg.)		Grade
	M%		A%				
	Min.	Max.	Min.	Max.	Min.	Max.	
IV	4.7	11.6	12.6	25.9	5088	6368	G8-G4
IIIA	2.3	4.8	20.1	55.1	2934	6439	G15-G3
III Top	3.7	5.6	15.4	33.7	4715	6571	G9-G3
III Middle	3.4	3.4	24.0	24.0	5805	5805	G5
III Bottom	3.2	6.7	15.2	32.8	5005	6192	G8-G4
III Comb.	2.0	10.0	13.4	49.6	3394	6366	G14-G4
II Top	2.8	7.6	11.6	30.2	5192	6637	G8-G3
II Bottom	2.5	6.7	13.4	35.1	4643	6402	G9-G3
II Comb	1.9	7.4	12.0	33.8	4961	7025	G8-G1
IB	1.5	7.9	9.6	32.8	4918	6910	G8-G2
IA	2.7	7.6	11.3	42.2	4324	6737	G10-G2

### 7.3.2 SUITABILITY FOR USE

The coals of medium to high ash, non-coking coals of inferior grade pertaining to the seams IV, IIIA, III Top, III Middle, III Bottom, III Comb, II Top, II Bottom, II Comb, IB and IA can be utilised in the power generation, cement, fertilizer, refractory, ceramic and cement industries.

## Chapter – 8

### ENVIRONMENTAL BASE LINE DATA

#### 8.1 GENERAL

The Marwatola Sector I& II Block is situated in Sohagpur Coalfields near to Pali UG & Birsinghpur UG of Johilla Area, Umaria district, Madhya Pradesh. The details of environmental data of nearest mining projects of block are being discussed below.

#### 8.2 METEOROLOGICAL FEATURES

The area is characterized by tropical climate. May is the hottest month when the temperature rises to as high as 46°C and during winter temperature falls up to 0°C. The annual average rainfall recorded was 1076 mm. Meteorological trends are based IMD, Umaria, and M.P data.

#### 8.3 AIR QUALITY

The air quality data observed from April to June 2022 at different environmental monitoring stations of Pali UG & Birsinghpur UG are given in Table 8.3. This indicates that the PM100, PM10, SO₂, & NO₂ values in respect of different category are within the permissible limit of NAAQS/Coal mine standards.

**Table 8.3 Air quality observed at different stations of Pali UG & Birsinghpur UG**

S. No.	Location Name	PM100	PM10	SO ₂	NO _x
	Industrial zone (GSR 742(E) dt 25.9.2000 A-O/A-N	600	300	120	120
		500	250	120	120
	Residential zone (B)	----	100	80	80
1	PALI COLONY	162-187	80-94	15-28	25-38
2	PALI VILLAGE	156-182	75-89	12-26	21-35
3	KUMARDA VILLAGE	140-164	68-78	7-23	17-31
4	MANDARIA VILLAGE	144-178	71-84	11-24	21-34
5	PALI PIT OFFICE	503-592	248-286	39-47	49-57
6	CHHAJA VILLAGE	154-180	77-88	10-25	20-34
7	GORAIYA VILLAGE	152-171	73-86	9-24	19-33
8	BIRSINGHPUR PIT OFFICE	289-342	143-166	20-31	30-41

## 8.4 WATER QUALITY

The water quality data as obtained from April to June 2022 at different environmental monitoring stations of Pali UG & Birsinghpur UG is given in Table 8.4.

**Table 8.4: Water quality observed at different stations of Pali UG & Birsinghpur UG**

Sr.No	Parameter	IS 10500:2012		Observed values	
		Acceptable Limit	Permissible Limit	Min	Max
1	Alkalinity, mg/l as CaCO ₃	200	1000	170	435.75
2	Arsenic, mg/l	0.01	0.05	BDL	BDL
3	Boron, mg/l	0.5	1	BDL	BDL
4	Calcium, mg/l	75	200	52.8	150.7
5	Chlorides, mg/l	250	1000	22.83	130.5
6	Colour, Hazen	5	15	1	21
7	Copper, mg/l	0.05	1.5	BDL	BDL
8	Fluoride, mg/l	1	1.5	0.27	1.5
9	Iron, mg/l	0.3	No Relaxation	0.06	0.13
10	Lead, mg/l	0.01	No relaxation	BDL	BDL
11	Manganese, mg/l	0.1	0.3	0.06	0.24
12	Nitrate mg/l	45	No Relaxation	1.02	64.82
13	Odour	Agreeable	Agreeable	Agreeable	Agreeable
14	pH Value	6.5 - 8.5	No Relaxation	6.8	7.33
15	Phenolic compounds, mg/l	0.001	0.002	BDL	BDL
16	Residual Free Chlorine, mg/l	0.2	1	0.1	0.13
17	Selenium, mg/l	0.01	No Relaxation	BDL	BDL
18	Sulphate, mg/l	200	400	7	176
19	Total Chromium, mg/l	0.05	No relaxation	BDL	BDL
20	Total Dissolved Solids, mg/l	500	2000	396	1122
21	Total Hardness, mg/l as CaCO ₃	200	600	234	644
22	Turbidity, NTU	1	5	BDL	4
23	Zinc, mg/l	5	15	0.02	0.22
24	Total Coliform, MPN/100 ml	Nil	No Relaxation	Nil	Nil

## 8.5 NOISE LEVEL

The Noise quality data as obtained from April to June 2022 at different environmental monitoring stations of Pali UG & Birsinghpur UG are given in Table 8.5.

**Table 8.5: Noise quality observed at different stations of Pali UG & Birsinghpur UG**

Sl. No.	Name of Location	Category	Noise Level in dB(A)			
			Day		Night	
			Min.	Max.	Min.	Max.
	<b>Industrial area</b>	<b>A</b>	<b>75</b>		<b>70</b>	
	<b>Commercial area</b>	<b>B</b>	<b>65</b>		<b>55</b>	
	<b>Residential area</b>	<b>C</b>	<b>55</b>		<b>45</b>	
	<b>Silence zone</b>	<b>D</b>	<b>50</b>		<b>40</b>	
1	PALI COLONY	A	52.6	53.5	41.3	43.9
2	PALI VILLAGE	C	50.8	51.6	40.0	42.6
3	KUMARDA VILLAGE	C	41.2	51.5	36.5	38.9
4	MANDARIA VILLAGE	C	49.5	55.7	39.4	42.6
5	PALI PIT OFFICE	A	64.0	68.5	57.8	62.1
6	CHHAJA VILLAGE	C	50.4	55.6	40.1	42.4
7	GORAIYA VILLAGE	C	48.7	52.2	39.0	40.9
8	BIRSINGHPUR PIT OFFICE	A	50.9	56.4	46.4	49.4

## 8.6 FLORA & FAUNA

As per the baseline data of Khairaha UG - Sohagpur area, this area has no eco-sensitive zones. A detailed flora and fauna study will be carried while preparing EIA/EMP.

## 8.7 SOCIO ECONOMIC STATUS

In respect of basic and civic amenities, it has been found that there is a fair network of infrastructure facilities and civic amenities available within 10 km radius of Marwatola Sector I&II coal Block. A detailed Socio-economic study will be carried while preparing EIA/EMP.

## Chapter – 9

### CONCLUSION & RECOMMENDATION

#### 9.1 CONCLUSION

##### 9.1.1 GENERAL INFORMATION ABOUT THE BLOCK

The Marwatola sector I&II Block, with an area of 21.03 sq.km, is situated in the western part of the Sohagpur coalfield. The block is located in the northern west part of Sohagpur coalfield.

The regional exploration in Marwatola block, Sohagpur Coalfield has been done initially by MECL covering an area of around 78 sq.km. Later the block has been sub-divided into Marwatola Sector I&II, Marwatola Sector III&IV, Marwatola Sector VI and Marwatola Sector VII respectively. The Marwatola Sector VI and Marwatola Sector VII block have been explored in detail by CMPDIL (excluding forest area). In order to increase the borehole density to 2bhs/sq.km the Marwatola Sector I&II and Marwatola Sector III&IV have been proposed for G2 stage of exploration through NMET. In the present Marwatola Sector I&II block, the drilling was undertaken, to prove the structure and existence of the coal seams V to Seam I during the period from 22-12-2021 to 17-06-2022.

Marwatola Sector I&II block is located in the district of Umaria, Madhya Pradesh state. It lies in the western part of Sohagpur coalfield and is about 12 Km. North-West of Shahdol town, the district headquarter of Shahdol, Madhya Pradesh. The block is located at a distance of 75 Km. from Umaria, the district headquarter of Umaria, Madhya Pradesh.

The block measures 21.03 sq.km. It is bounded by:

North	:The block is bounded on north side by Pathora East, Arjuni West block
South	: The south side bounded by Marwatola sector VII block.
East	:The block is bounded on East side by Marwatola Sector III&IV block.
West	: On the west the Ghunghuti block is located.

**Table: 9.1.1 A**  
**Cardinal Points of Block Boundary of Marwatola Sector I&II Block, Sohagpur**  
**Coalfield in (Coal grid and UTM).**

CARDINAL POINTS	UTM COORDINATE SYSTEM		COAL GRID COORDINATE	
S No	EASTING	NORTHING	X	Y
1	525316.549	2581637.035	2718942.695	1040098.846
2	525331.130	2576290.313	2718847.772	1034754.006
3	525096.743	2576307.516	2718613.833	1034776.001
4	524221.384	2576373.162	2717740.177	1034859.546
5	523523.379	2576565.217	2717046.391	1035065.815
6	522576.445	2576729.579	2716103.209	1035249.499
7	521933.799	2576782.859	2715461.916	1035315.916
8	521174.687	2576838.767	2714704.258	1035387.344
9	521163.071	2578026.391	2714716.963	1036574.723
10	521156.016	2578026.270	2714709.909	1036574.746
11	521154.760	2578618.761	2714720.785	1037167.023
12	521134.090	2581063.874	2714750.190	1039611.569
13	521135.210	2581138.048	2714752.828	1039685.691
14	521132.991	2581472.141	2714757.451	1040019.694
15	521135.001	2581610.436	2714762.293	1040157.892
16	521235.834	2581611.789	2714863.113	1040157.180
17	524038.593	2581631.913	2717665.152	1040119.899

The area in and around Marwatola Sector I&II Block is virgin with no mining activities. Dhanpuri UG and burdhar Mines of SECL is about 40 km (Aerial distance) and by road 45km. southeast of the block (by road). Nowrozabad Mine of SECL is about 24 km south west of the block. For Malachua mine PR is under developing stages.

In the block regional exploration was initiated by MECL during 09/04/2004 to 13/05/2007 and CMPDI through outsourcing drilling agency 15 boreholes drilled during 22/12/2021 to 17/06/2022 in the Marwatola Sector I&II Block. However, for establishing the structure and for proving the coal seam incrops these boreholes were drilled in the block. A total of 35 No. of boreholes have been drilled in the area. The details are as follows:

S. No.	Series of Boreholes	Nos. of boreholes drilled	Period
1	MWS	20	April.2004 to May.2007
2	CSM I&II	15	Dec.2021 to June 2022

Geologically the formations housing the coal seams belong to Barakar Formations. The strata including coal seams exhibit a broad NE-SW trend in the south-western part of the block exhibits a rolling strike. The strata including coal seams show a broad NE-SW trend. The strata in general show a northwesterly dip. The general dip of the strata varies from 2°-3° but near and in between the faults, the dip of the strata varies from 6°-8°.

Six faults have been interpreted in the block. These faults are numbered as F₁-F₁ to F₆-F₆ from north to south and fault strike is almost E-W.

The existence of 16 seams including splits and local seams well defined coal seams of Barakar Formation namely seam V, IV, L2, IIIA, III Top, III Middle, III Bottom, III Comb, II Top, II Bottom, II Comb, IB, IA, I Top, I Bottom and I Comb in descending order have been proved in Marwatola Sector I&II block of Sohagpur Coalfield, in which of 11 seams IV, IIIA, III Top, III Middle, III Bottom, III Comb, II Top, II Bottom, II Comb, IB and IA have been found to attain workable seam thickness in the major part of the block. (In few patch of Raniganj seam encounter) The focus of documentation/ interpretation has been confined only within the workable zone of the block. All the seams are non-coking type.

A total of 197.767 million tonnes of gross indicated resources have been estimated in the entire block. Details of the resources worked out for different seams are given in Annexure-VIII. The seam-wise proved resources are given in Table 9.1.1B.

**Table 9.1.1B: Seam-Wise 'gross' Indicated Geological resources In Marwatola Sector I&II block, Sohagpur Coalfield.**

**(Resources in million tonnes)**

<b>Seam Name</b>	<b>Indicated Resources (mt)</b>
IV	14.581
IIIA	8.229
III Top	4.970
III Middle	3.601
III Bottom	8.299
III Comb	31.583
II Top	32.314
II Bottom	15.635
II Comb.	22.321
IB	34.772
IA	21.460
<b>Grand Total</b>	<b>197.767</b>

## 9.2 RECOMMENDATIONS

1. Marwatola Sector I&II block is a virgin block. G2 stage of exploration has proved that the block has underground reserves of 197.767 million tonnes of gross indicated resources mainly of grade G15 to G1 (Dominant grade G9 to G3).

2. The chemical analysis of the coal seams indicate that these are mainly high moisture, medium to high ash, low rank non coking coals suitable for power generation, cement, refractory & ceramic industries.

3. All the workable coal seams incrops in the block.

4. The block may be taken up for detailed exploration to prove the coal seams and to upgrade the resource in the block to proved category.

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